EECS 16A Designing Information Devices and Systems I
Homework 0

Submission Format
Your homework submission should consist of **one** file.

- `hw0.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned).

Submit the file to the appropriate assignment on Gradescope.

1. **Reading Assignment**

   For this homework, please read Note 0 and Note 1 until Section 1.6. This will provide an overview of linear equations and augmented matrices. You are always welcome and encouraged to read ahead beyond this as well. Write a paragraph about how this relates to what you have learned before and what is new.

2. **Study Groups** To complete this part of the HW, you only have to fill out the following survey and indicate in your submitted answer that you filled it out. Nothing else is required.

   (a) We highly encourage working with your peers to complete homework assignments and work through course content together and are offering a platform for you to find study groups! Study groups are optional, but we strongly recommend them. In addition, we are offering a 1 unit P/NP class, EE194-3, in which you will meet and work with your assigned group. If interested, please fill out the following group matching form:
   https://forms.gle/toUttRRnHoP5ZyiS7

3. **Syllabus**

   Read the course syllabus and answer the following questions. The syllabus can be found here: https://eecs16a.org/policies.html.

   (a) When is homework 0 due? When is homework 0's self-grade due? In general, what day of the week is the homework due and at what time? In general, what day of the week are the self-grades due and at what time?

   (b) When are homework parties? Homework parties are where groups of students can get together to work on the homework together.

   (c) How many homework drops do you get? Reminder, the homework drop is for extenuating circumstance such as getting sick, family emergencies etc. You should plan on completing and submitting all homeworks and self-grades.

   (d) What is the penalty if you turn in your self-grades up to one week late?

   (e) What score will you get on a homework if you do not submit your self-grades?

   (f) How would you opt in for discussion attendance to count to your grade? Can you change it after your decision? How will the grade for discussion attendance be calculated?

(g) Fill in the blank: You should attend one discussion section on ______ and one discussion section on ______ each week. These sections need not be taught by the same instructor.

(h) Provide a complete list of everything you must do in order to receive credit for your homework assignments.

(i) Read the following guide: www.tinyurl.com/ee16a-gradescope. What are the five steps in the submission process for a PDF on Gradescope? Please note that if you do not select pages for each question/subquestion we cannot grade your homework and we will be forced to give you a 0.

(j) If you submit your homework but forget to select pages, can you reselect pages?

(k) What percentage do you need to get on a homework assignment for you to get full credit for the assignment?

(l) Will the exams in this class be in person? How many cheatsheet can you have in each exam?

(m) Fill in the blank:
If you miss ___ or more labs you will fail the class.

(n) Fill in the blank:

During buffer lab periods, you may get checked off for at most _______ missed lab that occurred during that lab module by attending your _______ section.

4. Homework resources

If you need help on a homework problem or have a question about the material, what are some of the resources you might be able to use?

(i) Homework party
(ii) TA office hours
(iii) Professor office hours
(iv) Asking a friend taking 16A
(v) Posting on ED
(vi) Going to discussion
(vii) All of the above

5. Counting Solutions

Learning Goal: *(This problem is meant to illustrate the different types of systems of equations. Some have a unique solution and others have no solutions or infinitely many solutions. We will learn in this class how to systematically figure out which of the three above cases holds.)*

Directions: For each of the following systems of linear equations, determine if there is a unique solution, no solution, or an infinite number of solutions. If there is a unique solution, find it. If there is an infinite number of solutions, describe the set of solutions. If there is no solution, explain why. **Show your work**.

Example: We will provide an example to show a way of solving systems of linear equations.

$$\begin{array}{rcrl} 2x & + & 3y & = 5 \\ x & + & y & = 2 \end{array}$$

$$2x + 3y = 5 \tag{1}$$
$$x + y = 2 \tag{2}$$

Subtract: (1) - 2*(2)

$$y = 1 \tag{3}$$

Now we plug in (3) into (2) and solve for x

$$x + 1 = 2$$
$$\rightarrow \quad x = 1 \tag{4}$$

From (3) and (4), we get the unique solution:

$$x = 1$$
$$y = 1$$

(a)

$$\begin{array}{rcrcrl} x & + & y & + & z & = 3 \\ 2x & + & 2y & + & 2z & = 5 \end{array}$$

(b)

$$\begin{array}{rcrcrl} & - & y & + & 2z & = 1 \\ 2x & & & + & z & = 2 \end{array}$$

(c)

$$\begin{array}{rcrl} x & + & 2y & = 3 \\ 2x & - & y & = 1 \\ 3x & + & y & = 4 \end{array}$$

(d)

$$\begin{array}{rcrl} x & + & 2y & = 3 \\ 2x & - & y & = 1 \\ x & - & 3y & = -5 \end{array}$$

EECS 16A Designing Information Devices and Systems I
Homework 1

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned) as well as your IPython notebook saved as a PDF.

If you do not attach a PDF "printout" of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

1. **Reading Assignment**

   For this homework, please read Notes 1A, 1B and 2A. The notes will provide an overview of systems of linear equations, vectors, and matrices. You are always welcome and encouraged to read beyond this as well, in particular, a quick look at Note 3 will help you.

   - Describe how Gaussian Elimination can help you understand if there are no solutions to a particular system of equations?

   - What about a unique solution?

   - Does a row of zeros always mean there are infinite solutions?

2. **Gaussian Elimination**

   (a) In this problem we will investigate the relationship between Gaussian elimination and the geometric interpretation of linear equations. You are welcome to draw plots by hand or using software. Please be sure to label your equations with a legend on the plot.

      i. Draw the following set of linear equations in the x-y plane. If the lines intersect, write down the point or points of intersection.

$$x + 2y = 4 \tag{1}$$
$$2x - 4y = 4 \tag{2}$$
$$3x - 2y = 8 \tag{3}$$

      ii. Write the above set of linear equations in augmented matrix form and do the first step of Gaussian elimination to eliminate the x variable from equation 2. Now, the second row of the augmented matrix has changed. Plot the corresponding new equation created in this step on the same graph as above. What do you notice about the new line you draw?

      iii. Complete all of the steps of Gaussian elimination including back substitution. Now plot the new equations represented by the rows of the augmented matrix in the last step (after completing back substitution) on the same graph as above. What do you notice about the new line you draw?

   (b) Write the following set of linear equations in augmented matrix form and use Gaussian elimination to determine if there are no solutions, infinite solutions, or a unique solution. If any solutions exist, determine what they are. You may do this problem by hand or use a computer. We encourage you to try it by hand to ensure you understand Gaussian elimination.

$$\begin{aligned} x+2y+5z &= 3 \\ x+12y+6z &= 1 \\ 2y+z &= 4 \\ 3x+16y+16z &= 7 \end{aligned}$$

(c) Consider the following system of equations:

$$\begin{aligned} x+2y+5z &= 6 \\ 3x+9y+6z &= 3 \end{aligned}$$

You are given a set S of candidate solutions,

$$S = \left\{ \vec{v} \;\middle|\; \vec{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 16 \\ -5 \\ 0 \end{bmatrix} + \begin{bmatrix} -11 \\ 3 \\ 1 \end{bmatrix} t \quad , \quad t \in \mathbb{R} \right\}$$

[Note: $\in$ is a mathematical symbol meaning "is an element of".]

This vector notation can be expressed in terms of its components:

$$\vec{v} = \begin{bmatrix} 16-11t \\ -5+3t \\ t \end{bmatrix} \quad \text{means} \quad \begin{aligned} x &= 16-11t \\ y &= -5+3t \\ z &= t \end{aligned}$$

Show, by substitution, that any $\vec{v} \in S$ is a solution to the system of equations given above. Note that this means that the candidate solution must satisfy the system of equations for all $t \in \mathbb{R}$.

(d) Consider the following system:

$$\begin{aligned} 4x+4y+4z+w+v &= 1 \\ x+y+2z+4w+v &= 2 \\ 5x+5y+5z+w+v &= 0 \end{aligned}$$

If you were to write the above equations in augmented matrix form and use Gaussian elimination to solve the system, you would get the following (for extra practice, you can try and do this yourself):

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 3 & 16 \\ 0 & 0 & 1 & 0 & -3 & -17 \\ 0 & 0 & 0 & 1 & 1 & 5 \end{array}\right]$$

How many variables are free variables? Determine the solutions to the set of equations.

3. Filtering Out The Troll

Learning Goal: *The goal of this problem is to represent a practical scenario using a simple model of directional microphones. You will tackle the problem of sound reconstruction through solving a system of linear equations.*

You attended a very important public speech and recorded it using a recording device that had two directional microphones[1]. However, there was a person in the audience who was trolling around, adding interference to the recording. When you went back home to listen to the recording, you realized that the two recordings were dominated by the troll's interference and you could not hear the speech. Fortunately, since your recording device contained two microphones, you realized there is a way to combine the two individual microphone recordings so that the troll's interference is removed. You remembered the locations of the speaker and the troll and created the diagram shown in Figure 1. You (and your two microphones) are located at the origin.

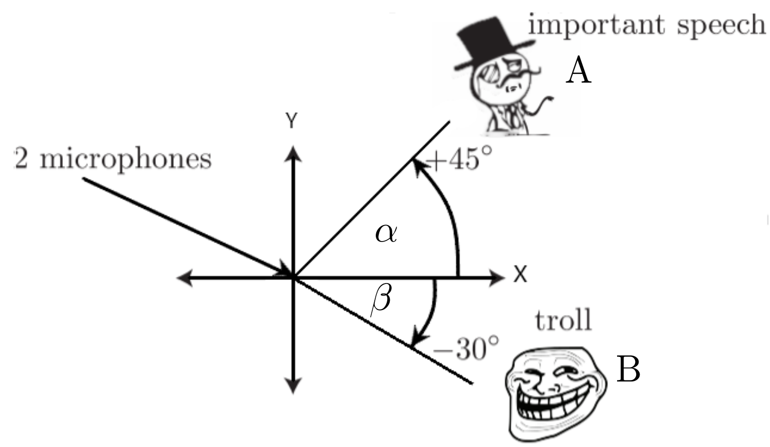


Figure 1: Locations of the speaker and the troll.

Each directional microphone records signals differently based on their angles of arrival. The first microphone weights (multiplies) a signal, coming from an angle θ with respect to the x-axis, by the factor $f_1(\theta) = \cos(\theta)$. If two signals are simultaneously playing (as is the case with the speech and the troll interference), then a linear combination (i.e. a weighted sum) of both the signals is recorded, each weighted by the respective $f_1(\theta)$ for their angles. The second microphone weights a signal, coming from an angle θ with respect to the x-axis, by the factor $f_2(\theta) = \sin(\theta)$. Again, if there are two simultaneous signals, then the second microphone also records the weighted sum of both the signals.

For example, an audio source that lies on the x axis will be recorded by the first microphone with weight equal to 1 (since $\cos(0) = 1$), but will not be recorded up by the second microphone (since $\sin(0) = 0$). Note that the weights can also be negative.

Let us represent the speech sample at a particular time-instant by the variable a and the interference caused by the troll at the same time-instant by the variable b. Remember, we do not know either a or b. The recording of the first microphone at that time instant is given by m_1:

$$m_1 = f_1(\alpha) \cdot a + f_1(\beta) \cdot b,$$

and the second microphone recorded the signal

$$m_2 = f_2(\alpha) \cdot a + f_2(\beta) \cdot b.$$

[1] If you want to learn more about directional microphones, click on this **link**

where α and β are the angles at which the public speaker A and the troll B respectively are located with respect to the x-axis, and vectors a and b are the audio signals produced by the public speaker A and the troll B respectively.

(As a side note, we could represent the entire speech with a vector $\vec{a}$ by stacking all the speech samples on top of each other, and this is what we would typically do in a real-world speech processing example. However, here we consider just one time instant of the speech for simplicity.)

(a) Plug in the values of α and β to write the recordings of the two microphones m_1 and m_2 as a linear combination (i.e. a weighted sum) of a and b.

(b) Solve the system you wrote out on the earlier part to recover the important speech a, as a weighted combination of m_1 and m_2. In other words, write $a = u \cdot m_1 + v \cdot m_2$ (where u and v are scalars). What are the values of u and v?

(c) Partial IPython code can be found in `prob1.ipynb`, which you can access through the datahub link associated with this assignment on the course website. Complete the code to get the signal of the important speech. Write out what the speaker says. (Optional: Where is the speech taken from?)

Note: You may have noticed that the recordings of the two microphones sound remarkably similar. This means that you could recover the real speech from two "trolled" recordings that sound almost identical! Leave out the fact that the recordings are actually different, and have some fun with your friends who aren't lucky enough to be taking EECS16A.

4. Anish's Optimal Tea

Learning Goal: *Recognize a problem that can be cast as a system of linear equations.*

Anish's Optimal Tea has a unique way of serving its customers. To ensure the best customer experience, each customer gets a combination drink personalized to their tastes. Anish knows that a lot of customers don't know what they want, so when customers walk up to the counter, they are asked to taste four standard combination drinks that each contain a different mixture of the available pure teas.

Each combination drink (Classic, Roasted, Mountain, and Okinawa) is made of a mixture of pure teas (Black, Oolong, Green, and Earl Grey), with the total amount of pure tea in each combination drink always the same, and equal to one cup. The table below shows the quantity of each pure tea contained in each of the four standard combination drinks.

Tea [cups]	Classic	Roasted	Mountain	Okinawa
Black	$\frac{1}{3}$	$\frac{1}{3}$	0	$\frac{2}{3}$
Oolong	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{2}{5}$	$\frac{1}{3}$
Green	0	$\frac{1}{3}$	$\frac{3}{5}$	0
Earl Grey	$\frac{1}{3}$	0	0	0

Initially, the customer's ratings for each of the pure teas are unknown. Anish's goal is to determine how much the customer likes each of the pure teas, so that an optimal combination drink can then be made. By letting the customer taste and score each of the four standard combination drinks, Anish can use linear algebra to determine the customer's initially unknown ratings for each of the pure teas. After a customer gives a score (all of the scores are real numbers) for each of the four standard combination drinks, Anish then calculates how much the customer likes each pure tea and mixes up a special combination drink that will maximize the customer's score.

The score that a customer gives for a combination drink is a linear combination of the ratings of the constituent pure teas, based on their proportion. For example, if a customer's rating for black tea is 6 and oolong tea is 3, then the total score for the Okinawa Tea drink would be $6 \cdot \frac{2}{3} + 3 \cdot \frac{1}{3} = 5$ because Okinawa has $\frac{2}{3}$ black tea and $\frac{1}{3}$ Oolong tea.

Professor Ranade was thirsty after giving the first lecture, so Professor Ranade decided to take a drink break at Anish's Optimal Tea. Professor Ranade walked in and gave the following ratings:

Combination Drink	**Score**
Classic	7
Roasted	7
Mountain	$\frac{37}{5}$
Okinawa	$\frac{19}{3}$

(a) What were Professor Ranade's ratings for each tea? **Show your work for this problem, step by step. You may use a calculator to do algebra.**

(b) What mystery tea combination could Anish put in Professor Ranade's personalized drink to maximize the customer's score? If there is more than one correct answer, state that there are many answers, and give one such combination. What score would Professor Ranade give for the answer you wrote down? Assume the total amount of tea must be one cup.

5. Fountain Codes

Learning Goal: *Linear algebra shows up in many important engineering applications. Wireless communication and information theory heavily rely on principles of linear algebra. This problem illustrates some of the techniques used in wireless communication.*

Vivian wants to send a message to her friend Kanav. Vivian sends her message $\vec{m}$ across a wireless channel in the form of a transmission vector $\vec{w}$. Kanav receives a vector of symbols denoted as $\vec{r}$.

Vivian knows some of the symbols in the transmission vector that she sends may be corrupted, so she needs a way to protect her message from the corruptions. (Transmission corruptions occur commonly in real wireless communication systems, for instance, when your laptop connects to a WiFi router, or your cellphone connects to the nearest cell tower.) Ideally, Vivian can come up with a transmission vector such that if some of the symbols get corrupted, Kanav can still figure out what Vivian is trying to say!

One way to accomplish this goal is to use fountain codes, which are part of a broader family of codes called error correcting codes. The basic principle is that instead of sending the exact message, Vivian sends a modified longer version of the message so that even if some parts are corrupted Kanav can recover what she meant. Fountain codes are based on principles of linear algebra, and were actually developed right here at Berkeley! The company that commercialized them, Digital Fountain, (started by a **Berkeley grad, Mike Luby**), was later acquired by Qualcomm. In this problem, we will explore some of the underlying principles that make fountain codes work in a very simplified setting.

The message that Vivian wants to send to Kanav are the three numbers a, b, and c. The message vector representing these numbers is $\vec{m} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$. Figure 2 shows how Vivian's message is encoded in a transmission vector and how Kanav's received vector may have some corrupted symbols.

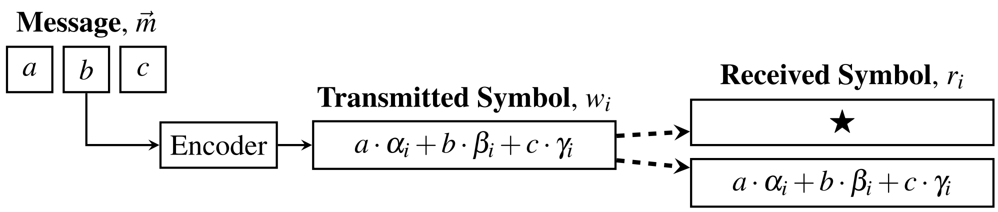


Figure 2: Each symbol in a transmission vector $\vec{w}$ is a linear combination of a, b, and c. Each transmitted symbol, w_i, is either received exactly as it was sent, or it is corrupted. A corrupted symbol is denoted by ★. The ith row of a symbol generating matrix G determines the values of α_i, β_i, and γ_i.

[Note: You may find it helpful to familiarize yourself with some Greek letters, to be able to effectively refer to them in this course! See **Greek Alphabet**.]

(a) Since Vivian has three numbers she wishes to send, she could transmit six symbols in her transmission vector for redundancy. This transmission strategy is called the **repetition code**.

If Vivian uses the repetition code, her transmission vector is $\vec{w} = \begin{bmatrix} a \\ b \\ c \\ a \\ b \\ c \end{bmatrix}$.

As depicted in Figure 2, the received vector may have corrupted symbols. For example, suppose only the first symbol was corrupted, then Kanav would receive the vector $\vec{r} = \begin{bmatrix} \bigstar \\ b \\ c \\ a \\ b \\ c \end{bmatrix}$, where the ★ symbol represents a corrupted symbol.

Using the repetition code scheme, give an example of a received vector $\vec{r}$ with only two corrupted symbols such that a is unrecoverable but b and c are still recoverable.

(b) Vivian can generate $\vec{w}$ by multiplying her message $\vec{m} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ by a matrix. Write a matrix-vector multiplication that Vivian can use to generate $\vec{w}$ according to the repetition code scheme. Specifically, find a "generating" matrix G_R such that $G_R\vec{m} = \vec{w}$, where $\vec{w} = \begin{bmatrix} a \\ b \\ c \\ a \\ b \\ c \end{bmatrix}$.

(c) Instead of a repetition code, it is also possible to use other codes (e.g. fountain codes). Vivian and Kanav can choose any symbol generating matrix, as long as they agree upon it in advance. Each different matrix represents a different "code." Vivian's TA recommends using the symbol generating matrix G_F:

$$G_F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Vivian then uses the symbol generating matrix G_F to produce a new transmission vector: $G_F\vec{m} = \vec{w}$.

Suppose Kanav receives the vector $\vec{r} = \begin{bmatrix} 7 \\ \bigstar \\ \bigstar \\ 3 \\ 4 \\ \bigstar \\ \bigstar \end{bmatrix}$, which is a corrupted version of $\vec{w}$.

Write a system of linear equations that Kanav can use to recover the message vector $\vec{m}$. Solve it to recover the three numbers that Vivian sent.

Hint: Consider the rows of G_F that correspond to the uncorrupted symbols in $\vec{r}$.

(d) We explore one example case where Vivian and Kanav agree to use G_F (i.e. $G_F\vec{m} = \vec{w}$) and there are three corruptions, so Kanav receives four uncorrupted symbols.

Suppose Kanav receives $\vec{r} = \begin{bmatrix} 1 \\ \bigstar \\ 3 \\ \bigstar \\ 4 \\ \bigstar \\ 9 \end{bmatrix}$.

Can you determine the message $\vec{m}$ that Vivian sent? If so, determine $\vec{m}$.

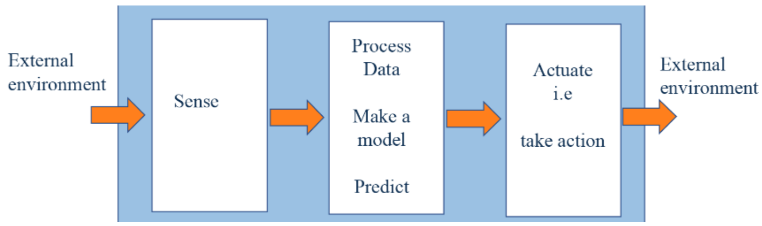
Note: it can be shown that receiving *any* four uncorrupted symbols when Vivian is using G_F is enough to recover Vivian's message. On the other hand, we showed in part (a) that receiving *any* four uncorrupted symbols using G_R does not guarantee we can recover Vivian's message. This is why, in practice, we would prefer to use the fountain code G_F instead of the repetition code G_R — it is a more reliable way to send messages.

6. Group Problem: Explore an Interesting Application of EECS

This problem is meant to be solved in your study groups and start a discussion. The conversation you have is more important to us than the answers you write down to this problem. If you chose to not have a study group, you may do this on your own, or discuss with other students during homework party.

As a group, identify an exciting technology you would like to build. In this problem, you will go through the design process for this technology by identifying the required sensing, processing and actuation components. Some examples of technologies you could consider are the step counters on phones, GPS, self-driving cars, a robotic drummer accompanist, a fitbit, an advanced pacemaker, a hearing aid, touchscreens, an automated fruit picker that picks ripe fruit, augmented reality for emotional awareness, a car that can apologize to defuse road rage, a friend bot, a pollution cleaning-bot etc.

This task is intended to be a fun exploration, so responses for each part need not exceed a paragraph in length. You are also encouraged to come to office hours to connect with your classmates and share ideas!



(a) What kind of information does your technology need to collect from the external environment? How is this collected? Are any sensors used? If so, what physical quantities do they measure?

(b) How do you propose to process the data obtained from the sensor measurements? What would you need to model? Would you have to make any predictions from the data? What techniques would you use?

(c) What actions would you want to take in the real world using your prediction or processed data? Describe the actuation mechanisms, if applicable, in this technology.

(d) What are the potential social implications of this technology? Good? Bad? Transformative? Redistributive?

7. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

EECS 16A Designing Information Devices and Systems I
Homework 2

Submission Format
Your homework submission should consist of **one** file.

- `hw2.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned) as well as your IPython notebook saved as a PDF.

  If you do not attach a PDF "printout" of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

Submit each file to its respective assignment on Gradescope.

1. Reading Assignment

For this homework, please read Notes 2B section 2.2, 3, 4, and 11A up to and including 11.3. These notes will give you an overview of linear transformations, linear independence, span, basic circuit elements, IV curves, and an introduction to thinking about and writing proofs. You are always welcome and encouraged to read beyond this as well. Write a paragraph about how you can use the strategies in the notes to tackle proof questions.

2. Easing into Proofs

(Contributors: Urmita Sikder, Gireeja Ranade)

Learning Objectives: *This is an opportunity to practice your proof development skills.*

(a) **Show that if the system of linear equations, $\mathbf{A}\vec{x} = \vec{b}$, has infinitely many solutions, then columns of A are linearly dependent.** Let us use the structure delineated in **Note 4** to approach this proof. This problem has 4 sub-parts and the following is a chart showing the sequential steps we are going to take to approach this proof.

In a text book you might see the steps in a proof written out in the order in the middle column of the table. But when you are building a proof you usually want to go in another order — this is the order of the subparts in this problem.

Proof steps		**Corresponding problem sub-parts**
1	Write what is known	Sub-part (i)
2	Manipulate what is known	Sub-part (iii)
3	Connecting it up	Sub-part (iv)
4	What is to be shown	Sub-part (ii)

(i) **First Step: write what you know**
Think about the *information we already know* from the problem statement. We know that system of equations, $\mathbf{A}\vec{x} = \vec{b}$, has infinitely many solutions. Infinitely many solutions are hard to work with, but perhaps we can simplify to something that we can work with. If the system has infinite number of solutions, it must have at least ___ distinct solutions (Fill in the blank).

So let us assume that $\vec{u}$ and $\vec{v}$ are two different vectors, both of which are solutions to $\mathbf{A}\vec{x} = \vec{b}$. Express the sentence above in a mathematical form (Just writing the equations will suffice; no need to do further mathematical manipulation).

(ii) **What we want to show:**
Now consider *what we need to show*. We have to show that the columns of $\mathbf{A}$ are linearly dependent. Let us assume that $\mathbf{A}$ has columns $\vec{c_1}$, $\vec{c_2}$, ..., and $\vec{c_n}$, i.e. $\mathbf{A} = \begin{bmatrix} | & | & \cdots & | \\ \vec{c_1} & \vec{c_2} & \cdots & \vec{c_n} \\ | & | & \cdots & | \end{bmatrix}$. Using the definition of linear dependence from **Note 3 Subsection 3.1.1**, write a mathematical equation that conveys linear dependence of $\vec{c_1}$, $\vec{c_2}$, ..., and $\vec{c_n}$.

(iii) **Manipulating what we know:**
Now let us try to start from the **First step: equations from (i)**, make mathematically logical steps and reach the **What we want to show: equations from (ii)**. Since your answer to (ii) is expressed in terms of the column vectors of $\mathbf{A}$, let us try to express the mathematical equations from (i), in terms of the the column vectors too. For example, we can write

$$\mathbf{A}\vec{x} = \vec{b}$$
$$\implies \begin{bmatrix} | & | & \cdots & | \\ \vec{c_1} & \vec{c_2} & \cdots & \vec{c_n} \\ | & | & \cdots & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdots \\ x_n \end{bmatrix} = \vec{b}$$
$$\implies x_1\vec{c_1} + x_2\vec{c_2} + \ldots + x_n\vec{c_n} = \vec{b}$$

Notice that x_1,... x_n etc are scalars. Now use your answer to part (i) to repeat the above formulation for distinct solutions $\vec{u}$ and $\vec{v}$. Note that this is proceeding slightly differently from how we did this proof in lecture. This is fine — there are often many correct ways to do a proof.

(iv) **Connecting it up:**
Now think about how you can mathematically manipulate your answer from part (iii) (**Manipulating what we know**) to **match the pattern** of your answer from part (ii) (**What we want to show**).

(b) **[PRACTICE]** Now try this proof on your own. Similar proofs will also be covered in your discussion section 3B. Given some set of vectors $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$, show the following:

$$\text{span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\} = \text{span}\{\vec{v}_1 + \vec{v}_2, \vec{v}_2, \ldots, \vec{v}_n\}$$

In order to show this, you have to prove the two following statements:

- If a vector $\vec{q}$ belongs in $\text{span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$, then it must also belong in $\text{span}\{\vec{v}_1 + \vec{v}_2, \vec{v}_2, \ldots, \vec{v}_n\}$.
- If a vector $\vec{r}$ belong is $\text{span}\{\vec{v}_1 + \vec{v}_2, \vec{v}_2, \ldots, \vec{v}_n\}$, then it must also belong in $\text{span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$.

In summary, you have to prove the problem statement from both directions. Now use the method developed in part (a) to prove these statements.

3. Image Stitching

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Learning Objective: *This problem is similar to one that students might experience in an upper division computer vision course. Our goal is to give students a flavor of the power of tools from fundamental linear algebra and their wide range of applications.*

Often, when people take pictures of a large object, they are constrained by the field of vision of the camera. This means that they have two options to capture the entire object:

- Stand as far away as they need to include the entire object in the camera's field of view (clearly, we do not want to do this as it reduces the amount of detail in the image)
- (This is more exciting) Take several pictures of different parts of the object and stitch them together like a jigsaw puzzle.

We are going to explore the second option in this problem. Daniel, who is a professional photographer, wants to construct an image by using "image stitching". Unfortunately, Daniel took some of the pictures from different angles as well as from different positions and distances from the object. While processing these pictures, Daniel lost information about the positions and orientations from which the pictures were taken. Luckily, you and your friend Marcela, with your wealth of newly acquired knowledge about vectors and matrices, can help him!

You and Marcela are designing an iPhone app that stitches photographs together into one larger image. Marcela has already written an algorithm that finds common points in overlapping images. **It's your job to figure out how to stitch the images together using Marcela's common points to reconstruct the larger image.**

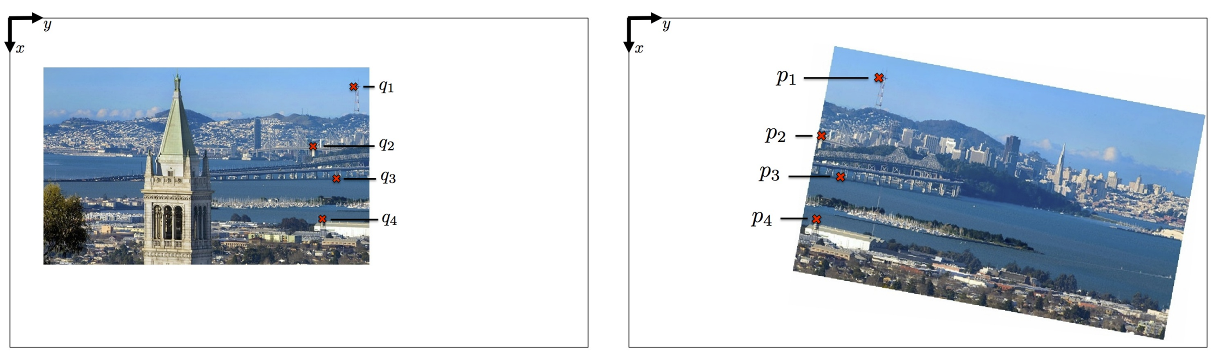


Figure 1: Two images to be stitched together with pairs of matching points labeled.

We will use vectors to represent the common points which are related by a linear transformation. Your idea is to find this linear transformation. For this you will use a single matrix, $\mathbf{R}$, and a vector, $\vec{t}$, that transforms every common point in one image to their corresponding point in the other image. Once you find $\mathbf{R}$ and $\vec{t}$ you will be able to transform one image so that it lines up with the other image.

Suppose $\vec{p} = \begin{bmatrix} p_x \\ p_y \end{bmatrix}$ is a point in one image, which is transformed to $\vec{q} = \begin{bmatrix} q_x \\ q_y \end{bmatrix}$, the corresponding point in the other image (i.e., they represent the same object in the scene). For example, Fig. 1 shows how the points $\vec{p}_1$, $\vec{p}_2$ ... in the right image are transformed to points $\vec{q}_1$, $\vec{q}_2$ ... on the left image. You write down the following relationship between $\vec{p}$ and $\vec{q}$.

$$\begin{bmatrix} q_x \\ q_y \end{bmatrix} = \underbrace{\begin{bmatrix} r_{xx} & r_{xy} \\ r_{yx} & r_{yy} \end{bmatrix}}_{\mathbf{R}} \begin{bmatrix} p_x \\ p_y \end{bmatrix} + \underbrace{\begin{bmatrix} t_x \\ t_y \end{bmatrix}}_{\vec{t}} \tag{1}$$

This problem focuses on finding the unknowns (i.e. the components of $\mathbf{R}$ and $\vec{t}$), so that you will be able to stitch the image together.

(a) To understand how the matrix $\mathbf{R}$ and vector $\vec{t}$ transforms any vector representing a point on a image, Consider this equation similar to Equation (1),

$$\vec{v} = \begin{bmatrix} 2 & 2 \\ -2 & 2 \end{bmatrix} \vec{u} + \vec{w} = \vec{v}_1 + \vec{w}. \tag{2}$$

Use $\vec{w} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\vec{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ for this part.

We want to find out what geometric transformation(s) can be applied on $\vec{u}$ to give $\vec{v}$.

Step 1: Find out how $\begin{bmatrix} 2 & 2 \\ -2 & 2 \end{bmatrix}$ is transforming $\vec{u}$. Evaluate $\vec{v}_1 = \begin{bmatrix} 2 & 2 \\ -2 & 2 \end{bmatrix} \vec{u}$.

What **geometric transformation(s)** might be applied to $\vec{u}$ to get $\vec{v}_1$? Choose the option that answers the question and explain your choice.

(i) Rotation
(ii) Scaling
(iii) Shifting/Translation
(iv) Rotation and Scaling

Drawing the vectors $\vec{u}$, and $\vec{v}_1$ in two dimensions on a single plot might help you to visualize the transformations. You can also look into the corresponding demo in the IPython notebook `prob2.ipynb`.

Step 2: Find out $\vec{v} = \vec{v}_1 + \vec{w}$. Find out how **addition of $\vec{w}$ is geometrically transforming** $\vec{v}_1$. Choose the option that answers the question and explain your choice.

(i) Rotation
(ii) Scaling
(iii) Shifting/ Translation.

Drawing the vectors $\vec{v}$, $\vec{w}$, and $\vec{v}_1$ in two dimensions on a single plot might help you to visualize the transformations. You can also look into the corresponding demo in the IPython notebook `prob2.ipynb`.

(b) Multiply Equation (1) out into **two scalar linear equations**.

(i) What are the known values and what are the unknowns in each equation?
(ii) How many unknowns are there?
(iii) How many independent equations do you need to solve for all the unknowns?
(iv) How many pairs of common points $\vec{p}$ and $\vec{q}$ will you need in order to write down a system of equations that you can use to solve for the unknowns? ***Hint:*** *Remember that each pair of $\vec{p}$ and $\vec{q}$ will give you* ***two*** *different linear equations.*

(c) Write out a system of linear equations that you can use to solve for $\vec{\alpha} = \begin{bmatrix} r_{xx} \\ r_{xy} \\ r_{yx} \\ r_{yy} \\ t_x \\ t_y \end{bmatrix}$. Assume that all four pairs of points from Fig. 1 are labeled as:

$$\vec{q}_1 = \begin{bmatrix} q_{1x} \\ q_{1y} \end{bmatrix}, \vec{p}_1 = \begin{bmatrix} p_{1x} \\ p_{1y} \end{bmatrix} \quad \vec{q}_2 = \begin{bmatrix} q_{2x} \\ q_{2y} \end{bmatrix}, \vec{p}_2 = \begin{bmatrix} p_{2x} \\ p_{2y} \end{bmatrix} \quad \vec{q}_3 = \begin{bmatrix} q_{3x} \\ q_{3y} \end{bmatrix}, \vec{p}_3 = \begin{bmatrix} p_{3x} \\ p_{3y} \end{bmatrix} \quad \vec{q}_4 = \begin{bmatrix} q_{4x} \\ q_{4y} \end{bmatrix}, \vec{p}_4 = \begin{bmatrix} p_{4x} \\ p_{4y} \end{bmatrix}.$$

Now think of your answer to Part b(iv). How many pairs of these points would you need to solve for $\vec{\alpha}$. Choose **just enough** equations required to solve for $\vec{\alpha}$ and express these linear equations using **matrix-vector form**.

(d) In the IPython notebook `prob2.ipynb`, you will have a chance to test out your solution. Plug in the values that you are given for p_x, p_y, q_x, and q_y for each pair of points into your system of equations to solve for the matrix, $\mathbf{R}$, and vector, $\vec{t}$. The notebook will solve the system of equations, apply your transformation to the second image, and show you if your stitching algorithm works. **You are NOT responsible for understanding the image stitching code or Marcela's algorithm.**

4. Kinematic Model for a Simple Car

Learning Objective: *Many real world systems are not actually linear and have more complex behaviors. However, linear models can approximate non-linear systems under certain conditions.*

Building a self-driving car first requires understanding the basic motions of a car. In this problem, we will explore how to model the motion of a car.

There are several models that we can use to model the motion of a car. Assume we use a kinematic model, described in the following four equations and Figure 2.

$$x[k+1] = x[k] + v[k]\cos(\theta[k])\Delta t \tag{3}$$
$$y[k+1] = y[k] + v[k]\sin(\theta[k])\Delta t \tag{4}$$
$$\theta[k+1] = \theta[k] + \frac{v[k]}{L}\tan(\phi[k])\Delta t \tag{5}$$
$$v[k+1] = v[k] + a[k]\Delta t \tag{6}$$

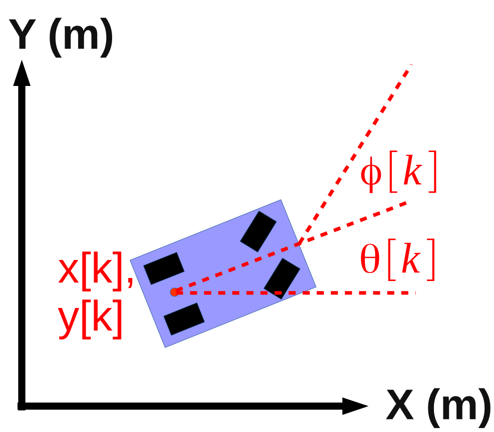


Figure 2: Vehicle Kinematic Model

where

- k, a nonnegative integer, indicates the time step at which we measure each variable (e.g. $v[k]$ is the speed at time step k and $v[k+1]$ is the speed at the following time step)
- $x[k]$ and $y[k]$ denote the coordinates of the vehicle (meters)
- $\theta[k]$ denotes the heading of the vehicle, or the angle with respect to the x-axis (radians)

- $v[k]$ is the speed of the car (meters per second)
- $a[k]$ is the acceleration of the car (meters per second squared)
- $\phi[k]$ is the steering angle input we command (radians)
- Δt is a constant measuring the time difference (in seconds) between time steps $k+1$ and k
- L is a constant and is the length of the car (in meters)

For this problem, let L be 1.0 meter and Δt be 0.1 seconds.

The variables $x[k], y[k], \theta[k], v[k]$ describe the **state** of the car at time step k. The state captures all the information needed to fully determine the current position, speed, and heading of the car. The **inputs** at time step k are $a[k]$ and $\phi[k]$. These are provided by the driver. The current value of these inputs, along with the current state of the vehicle, will determine the state of the vehicle at the next time step.

We note that the problem is nonlinear, due to the sine, cosine and tangent functions, as well as terms including the product of states and inputs.

The purpose of this problem is to show that we can approximate a nonlinear model with a simple linear model and do reasonably well. This is why, despite many systems being nonlinear, linear algebra tools are widely used in practice.

For Parts (b) - (d), fill out the corresponding sections in prob2.ipynb.

(a) We assume that the car has a small heading, θ, which is a **very small but nonzero** value, and that the steering angle ϕ is also **very small but nonzero**. In this case, we could use the following approximations:

$$\sin(\alpha) \approx 0,$$
$$\cos(\alpha) \approx 1,$$
$$\tan(\alpha) \approx 0.$$

where α is the small angle of interest, and $\approx$ means "approximately equal to". Here, we use a very simple approximation for small angles; in later classes, you may learn better approximations.

Draw, by hand, graphs of $\sin(\alpha)$ and $\cos(\alpha)$, for α ranging from $-\pi$ to π. Using these graphs, can you justify the approximation we are making for small values of α?

(b) Applying the approximation described in the previous part, write down a system of linear equations that approximates the nonlinear vehicle model given above in Equations (3) to (6). In particular, find the 4 x 4 matrix **A** and 4 x 2 matrix **B** that satisfy the equation given below.

$$\begin{bmatrix} x[k+1] \\ y[k+1] \\ \theta[k+1] \\ v[k+1] \end{bmatrix} = \mathbf{A} \begin{bmatrix} x[k] \\ y[k] \\ \theta[k] \\ v[k] \end{bmatrix} + \mathbf{B} \begin{bmatrix} a[k] \\ \phi[k] \end{bmatrix}$$

Hint: *Start with simplifying Equations* (3) *to* (6)*, using the* $\sin(\alpha)$*,* $\cos(\alpha)$*, and* $\tan(\alpha)$ *approximations from part (a) only to approximate nonlinear terms. Do NOT replace* ϕ*,* θ *with 0 unless it is for the* $\sin(\alpha)$*,* $\cos(\alpha)$*, and* $\tan(\alpha)$ *approximations.*

(c) Suppose we drive the car so that the direction of travel is aligned with the x-axis, and we are driving nearly straight, i.e. the steering angle is $\phi[k] = 0.0001$ radians. (Driving exactly straight would have the steering angle $\phi[k] = 0$ radians.) The initial state and input are:

$$\begin{bmatrix} x[0] \\ y[0] \\ \theta[0] \\ v[0] \end{bmatrix} = \begin{bmatrix} 5.0 \\ 10.0 \\ 0.0 \\ 2.0 \end{bmatrix}$$

$$\begin{bmatrix} a[k] \\ \phi[k] \end{bmatrix} = \begin{bmatrix} 1.0 \\ 0.0001 \end{bmatrix}$$

You can use these values in the IPython notebook to compare how the nonlinear system evolves in comparison to the linear approximation that you made. The IPython notebook simulates the car for ten time steps. Are the trajectories similar or very different? Why?

(d) Now suppose we drive the vehicle from the same starting state, but we turn left instead of going straight, i.e. the steering angle is $\phi[k] = 0.5$ radians. The initial state and input are:

$$\begin{bmatrix} x[0] \\ y[0] \\ \theta[0] \\ v[0] \end{bmatrix} = \begin{bmatrix} 5.0 \\ 10.0 \\ 0.0 \\ 2.0 \end{bmatrix}$$

$$\begin{bmatrix} a[k] \\ \phi[k] \end{bmatrix} = \begin{bmatrix} 1.0 \\ 0.5 \end{bmatrix}$$

You can use these values in the IPython notebook to compare how the nonlinear system evolves in comparison to the linear approximation that you made. The IPython notebook simulates the car for ten time steps. Are the trajectories similar or very different? Why?

5. Basic Circuit Components

Learning Objectives: *Review basics of cucuit components and current-voltage relationships*

In the laboratory, you are tasked with identifying a single unknown component within a circuit. You can use a piece of electrical equipment called a multimeter to measure either voltage (voltmeter) across or the current (ammeter) through the component (you can measure both quantities simultaneously using two multimeters).

For each part of the problem, deduce the most likely type of circuit component based on the provided voltage and current measurements. Also draw the component circuit symbol and sketch the IV curve.

Hint: You are told the possible choices are ***short circuit (wire)****,* ***open circuit****,* ***resistor****,* ***voltage source****, and* ***current source****. Moreover, each part in this problem has a unique component, there are no repeats.*

(a) First, to familiarize yourself with common quantities of *voltage*, *current*, and *resistance*, fill in the unit name and unit symbol for each:

Quantity	**Symbol**	**Unit Name**	**Unit Symbol**
Voltage	V		
Current	I		
Resistance	R		

(b) You take one measurement and find the voltage is $V = 10$ V and current is $I = 1$ A. After changing a part of the circuit (not the part you are measuring), you take another measurement and find $V = 10$ V and $I = 2$ A. What is the most likely component type? Draw the component symbol and sketch the IV curve.

(c) You take one measurement and find the voltage is $V = 0$ V and current is $I = 1$ A. What is the most likely component type? Draw the component symbol and sketch the IV curve.

(d) You take one measurement and find the voltage is $V = 5$ V and current is $I = 0$ A. What is the most likely component type? Draw the component symbol and sketch the IV curve.

(e) You take one measurement and find the voltage is $V = 10$ V and current is $I = 2$ A. After changing a part of the circuit (not the part you are measuring), you take another measurement and find $V = -5$ V and $I = -1$ A. What is the most likely component type? Draw the component symbol and sketch the IV curve.

(f) You take one measurement and find the voltage is $V = 10$ V and current is $I = 2$ A. After changing a part of the circuit (not the part you are measuring), you take another measurement and find $V = -3$ V and $I = 2$ A. What is the most likely component type? Draw the component symbol and sketch the IV curve.

6. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

EECS 16A Designing Information Devices and Systems I
Homework 3

Submission Format
Your homework submission should consist of **one** file.

- `hw3.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned) as well as your IPython notebook saved as a PDF.

  If you do not attach a PDF "printout" of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

Submit each file to its respective assignment on Gradescope.

1. Reading Assignment

For this homework, please read Notes 4 through 6. Note 4 will discuss mathematical derivation techniques useful for proofs. The notes 5 and 6 will provide an overview of multiplication of matrices with vectors, by considering the example of water reservoirs and water pumps, and matrix inversion.

(a) Please summarize (in one or two paragraphs) the main learnings in Notes 4-6.

(b) You have seen in Note 5 that the pump system can be represented by a state transition matrix. What constraint must this matrix satisfy in order for the pump system to obey water conservation?

(c) Repeat the proof of Theorem 6.4 in Note 6. That is, prove that **If a matrix A is invertible, its columns are linearly independent.**

2. Study Groups Tasks

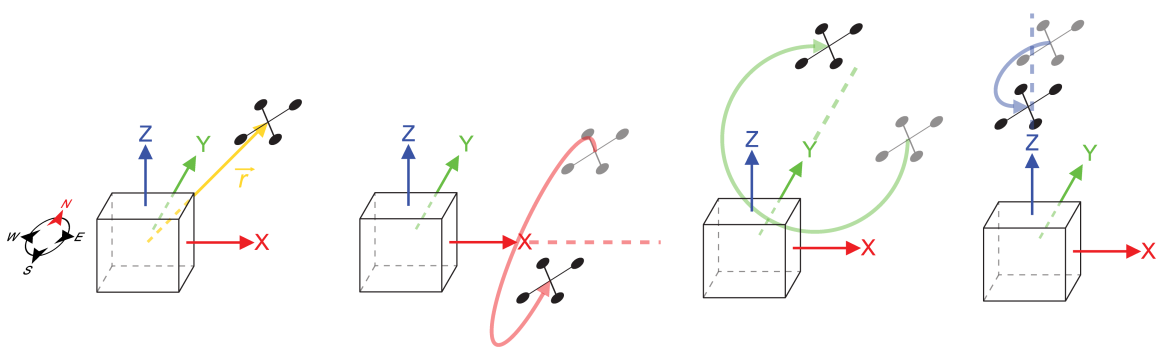
(a) We recommend you use some time during your study group to develop a plan for studying for the first midterm, which is coming soon. Discuss how you can support each other in your studying. Do you want to have an extra meeting to discuss some harder concepts? If you are not working with a group, you may make your plan alone. Make a list of what you want to review — this should include lectures, discussion problems, homework problems, notes.

(b) We hope your study groups have been productive and fun so far! In case you missed the first study group matching form and have not connected with a group yet, or you would like to try a new study group, we are conducting a second matching process. Please use the link below to make your request.
`https://forms.gle/Man8tZskVJyne7Pu8`
To get full credit for this question, you must (1) fill out the survey (it will record your email) and (2) indicate in your HW submission that you filled out the survey.

3. Quadcopter Transformations

Learning Objectives: *Linear algebra is often used to represent transformations in robotics. This problem introduces some of the basic uses of transformations.*

Sunash and his colleagues are interested in developing a communication link using a laser to control the location of a quadrotor. Consider a vector $\vec{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3$ representing the location of the quadcopter relative to the origin. The quadcopter is only capable of three different maneuvers relative to the origin. The maneuvers are rotations about the x, y, and z axes. For perspective, the positive x-axis points east, the positive y-axis points north, and the positive z-axis points up towards the sky. The figures below illustrate the quadcopter and these maneuvers.



We can represent each of these rotations, that are linear transformations, as matrices that operate on the location vector of the quadcopter, $\vec{r}$, to position it at its new location. The matrices $\mathbf{R_x}(\theta)$, $\mathbf{R_y}(\psi)$, and $\mathbf{R_z}(\phi)$ represent rotations about the x-axis, y-axis, and z-axis, respectively. The matrices are:

$$\mathbf{R_x}(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}, \mathbf{R_y}(\psi) = \begin{bmatrix} \cos\psi & 0 & \sin\psi \\ 0 & 1 & 0 \\ -\sin\psi & 0 & \cos\psi \end{bmatrix}, \mathbf{R_z}(\phi) = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

.

(a) Sunash wants to make the quadcopter rotate first by 30° about the x-axis, and then by 60° about the z-axis. Use $\mathbf{R_x}(\theta)$, $\mathbf{R_y}(\psi)$, and $\mathbf{R_z}(\phi)$ to construct **a single matrix that performs the operations in the specified order**. Show the matrix operations and calculations by hand.

(b) Sunash accidentally punched in the two rotation commands in reverse. The 60° rotation about the z-axis occurred before the 30° rotation about the x-axis. Use $\mathbf{R_x}(\theta)$, $\mathbf{R_y}(\psi)$, and $\mathbf{R_z}(\phi)$ to construct **a single matrix that performs the operations in the accidentally reversed order**. Show the matrix operations and calculations by hand.

(c) Say the quadcopter was initially positioned at $\vec{r} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$.

  i. Where did Sunash intend for the quadcopter to end up? Use your result from part (a) to find this out.

  ii. Where did the quadcopter actually end up with the accidentally reversed order of the rotations? Use your result from part (b) to find this out.

  iii. Did the quadcopter end up where it was supposed to go?

4. Mechanical Inverses

***Learning Objectives**: Matrices represent linear transformations, and their inverses represent the opposite transformation. Here we practice inversion, but are also looking to develop an intuition. Visualizing the transformations might help develop this intuition.*

For each of the following values of matrix A:

i Find the inverse, $\mathbf{A}^{-1}$, if it exists. If you found that the inverse does not exist, mention how you decided that. Solve this by hand.

ii **For parts (a)-(d)**, in addition to finding the inverse (if it exists), describe how the matrix, $\mathbf{A}$ transforms an arbitrary vector, $\begin{bmatrix} x \\ y \end{bmatrix}$.

For example, if $\mathbf{A}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$, then $\mathbf{A}$ could scale $\begin{bmatrix} x \\ y \end{bmatrix}$ by 2 to get $\begin{bmatrix} 2x \\ 2y \end{bmatrix}$. If $\mathbf{A}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ -y \end{bmatrix}$, then $\mathbf{A}$ could reflect $\begin{bmatrix} x \\ y \end{bmatrix}$ across the x axis, etc. *Hint:* It may help to plot a few examples to recognize the pattern.

iii **For parts (a)-(d)**, if we use $\mathbf{A}$ to geometrically transform $\begin{bmatrix} x \\ y \end{bmatrix}$ to get $\begin{bmatrix} u \\ v \end{bmatrix} = \mathbf{A}\begin{bmatrix} x \\ y \end{bmatrix}$, **is it possible to reverse the transformation geometrically,** i.e. is it possible to retrieve $\begin{bmatrix} x \\ y \end{bmatrix}$ from $\begin{bmatrix} u \\ v \end{bmatrix}$ geometrically?

(a) $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(b) $\mathbf{A} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

(c) **(PRACTICE / NOT GRADED)**

$\mathbf{A} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

(d) **(PRACTICE / NOT GRADED)**

$\mathbf{A} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

Assume $\cos\theta \neq 0$. *Hint:* $\cos^2\theta + \sin^2\theta = 1$.

(e) $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 0 \end{bmatrix}$

(f) $\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 2 \\ 1 & 4 & 4 \end{bmatrix}$

(g) **(PRACTICE / NOT GRADED)**

$\mathbf{A} = \begin{bmatrix} -1 & 1 & -\frac{1}{2} \\ 1 & 1 & -\frac{1}{2} \\ 0 & 1 & 1 \end{bmatrix}$

(h) **(PRACTICE / NOT GRADED)**

$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$

(i) **(PRACTICE / NOT GRADED)**

$$\mathbf{A} = \begin{bmatrix} 3 & 0 & -2 & 1 \\ 0 & 2 & 1 & 3 \\ 3 & 1 & 0 & 4 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Hint 1: What do the linear (in)dependence of the rows and columns tell us about the invertibility of a matrix? Hint 2: We're reasonable people!

5. Properties of Pump Systems

Learning Objectives: *This problem illustrates how matrices and vectors can be used to represent linear transformations.*

Throughout this problem, we will consider a system of reservoirs connected to each other through pumps. An example system is shown below in Figure 1, represented as a graph. Each node in the graph is marked with a letter and **represents a reservoir**. Each arrow in the graph represents a pump which moves a fraction of the water from one reservoir to the next at every time step. The **fraction of water** is written on top of the arrow.

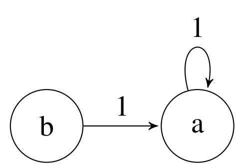


Figure 1: Pump system

(a) Consider the system of pumps shown above in Figure 1. Let $x_a[n]$ and $x_b[n]$ represent the amount of water in reservoir a and b, respectively, at time step n. Find a system of equations that represents $x_a[n+1]$ and $x_b[n+1]$ in terms of $x_a[n]$, $x_b[n]$ etc.

(b) For the system shown in Figure 1, find the associated state transition matrix. In other words, find the matrix $\mathbf{A}$ such that:

$$\vec{x}[n+1] = \mathbf{A}\vec{x}[n], \text{ where } \vec{x}[n] = \begin{bmatrix} x_a[n] \\ x_b[n] \end{bmatrix}$$

(c) Let us assume that at time step 0, the reservoirs are initialized to the following water levels: $x_a[0] = 0.5, x_b[0] = 0.5$. In a completely alternate universe, the reservoirs are initialized to the following water levels: $x_a[0] = 0.3, x_b[0] = 0.7$. For both initial states, what are the water levels at timestep 1 ($\vec{x}[1]$)? Use your answer from part (b) to compute your solution.

(d) If you observe the reservoirs at timestep 1, i.e if you know $\vec{x}[1]$, can you figure out what the initial ($\vec{x}[0]$) water levels were? Why or why not?

(e) Now let us generalize what we observed. Say there is a transition matrix $\mathbf{A}$ representing a pump system. Say there exist two distinct initial state vectors/ water levels: $\vec{x_u}[0]$ and $\vec{x_v}[0]$, that lead to the same state vector $\vec{x}[1]$ after $\mathbf{A}$ acts on them.You do not know which of the two initial state vectors you started in. Can you decide which initial state you started in by observing $\vec{x}[1]$? What does this say about the matrix $\mathbf{A}$?

(f) Now, we want to prove the following theorem in a step-by-step fashion.

Theorem: Consider a system consisting of k reservoirs such that the entries of each column in the system's state transition matrix sum to one. If s is the total amount of water in the system at timestep n, then total amount of water at timestep $n+1$ will also be s.

i. Rewrite the theorem statement for a graph with only two reservoirs.

ii. Since the problem does not specify the transition matrix, let us consider the transition matrix $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and the state vector $\vec{x}[n] = \begin{bmatrix} x_1[n] \\ x_2[n] \end{bmatrix}$. In general, it is helpful to write as much out mathematically as you can in proofs. It can also be helpful to draw the transition graph. Write out what is "known", i.e. ALL the information that is given to you in the theorem statement *in mathematical form*.

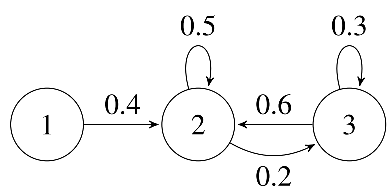
iii. Now write out what is to be proved *in mathematical form*.

iv. Prove the statement i.e. the ans for your last part for the case of two reservoirs.

v. Now use what you learned to generalize to the case of k reservoirs. *Hint:* Think about $\mathbf{A}$ in terms of its columns, since you have information about sum of each column.

(g) Set up the state transition matrix $\mathbf{A}$ for the system of pumps shown below. Compute the sum of the entries of each column of the state transition matrix. Are the sums greater than/less than/equal to 1? Explain what this $\mathbf{A}$ matrix physically implies about how the total amount of water in this system changes over time.

Note: Arrows not shown assumed to be 0.



6. Segway Tours

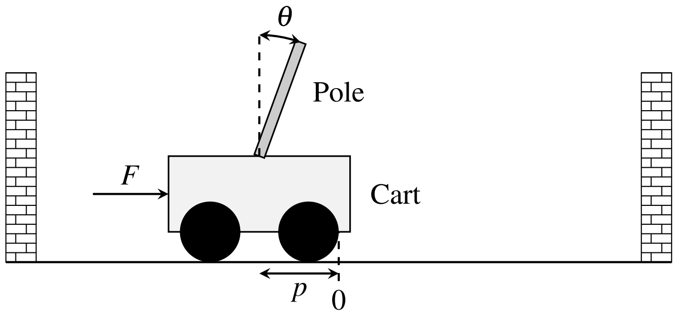
Learning Objective: *The learning objective of this problem is to see how the concept of span can be applied to control problems. If a desired state vector of a linear control problem is in a span of a particular set of vectors, then the system may be steered to reach that particular vector using the available inputs.*

Your friend has decided to start a new SF tour business, and you suggest they use segways.

They become intrigued by your idea and asks you how a segway works. A segway is essentially a stand on two wheels.

The segway works by applying a force (through the spinning wheels) to the base of the segway. This controls both the position on the segway and the angle of the stand. As the driver pushes on the stand, the segway tries to bring itself back to the upright position, and it can only do this by moving the base.

Is it possible for the segway to be brought upright and to a stop from any initial configuration? There is only one input (force) used to control two outputs (position and angle). You both talk to a third friend who is GSIing EE128, and she tells you that a segway can be modeled as a cart-pole system.



A cart-pole system can be fully described by its position p, velocity $\dot{p}$, angle θ, and angular velocity $\dot{\theta}$. We write this as a "state vector", $\vec{x}$:

$$\vec{x} = \begin{bmatrix} p \\ \dot{p} \\ \theta \\ \dot{\theta} \end{bmatrix}.$$

The input to this system is a scalar quantity $u[n]$ at time n, which is the force F applied to the cart (or base of the segway).[1]

The cart-pole system can be represented by a linear model:

$$\vec{x}[n+1] = \mathbf{A}\vec{x}[n] + \vec{b}u[n], \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{4\times4}$ and $\vec{b} \in \mathbb{R}^{4\times1}$.

The control $u[n]$ allows us to move the state ($\vec{x}$) in the direction of $\vec{b}$. So, if $u[n] = 2$, we move the state by $2\vec{b}$ at time n, and so on. We can choose different controls at different times.

The model tells us how the the state vector, $\vec{x}$, will evolve over time as a function of the current state vector and control inputs.

You look at this general linear system and try to answer the following question: Starting from some initial state $\vec{x}_0$, can we reach a final desired state, $\vec{x}_f$, in N steps?

The challenge seems to be that the state is four-dimensional and keeps evolving and that we can only apply a one-dimensional (scalar) control at each time. Typically, to set the values of four variables to desired quantities, you would need four inputs. Can you do this with just one input?

We will solve this problem by walking through several steps.

(a) Express $\vec{x}[1]$ in terms of $\vec{x}[0]$ and the input $u[0]$. (*Hint:* This is not meant to be complicated.)

(b) i. Express $\vec{x}[2]$ in terms of *only* $\vec{x}[0]$ and the inputs, $u[0]$ and $u[1]$.

ii. Then express $\vec{x}[3]$ in terms of *only* $\vec{x}[0]$ and the inputs, $u[0]$, $u[1]$, and $u[2]$.

[1] You might note that velocity and angular velocity are derivatives of position and angle respectively. Differential equations are used to describe continuous time systems, which you will learn more about in EECS 16B. But even without these techniques, we can still approximate the solution to be a continuous time system by modeling it as a discrete time system where we take very small steps in time. We think about applying a force constantly for a given finite duration and we see how the system responds after that finite duration.

iii. Finally express $\vec{x}[4]$ in terms of *only* $\vec{x}[0]$ and the inputs, $u[0]$, $u[1]$, $u[2]$, and $u[3]$. **Your expressions can have other relevant variables (e.g. A, $\vec{b}$ etc) and mathematical operators.**

(c) Now, generalize the pattern you saw in the earlier part to write an expression for $\vec{x}[N]$ in terms of $\vec{x}[0]$ and the inputs from $u[0], \ldots, u[N-1]$. **Your expression can have other relevant variables (e.g. A, $\vec{b}$ etc) and mathematical operators.**

For the next four parts of the problem, you are given the matrix $\mathbf{A}$ and the vector $\vec{b}$:

$$\mathbf{A} = \begin{bmatrix} 1 & 0.05 & -0.01 & 0 \\ 0 & 0.22 & -0.17 & -0.01 \\ 0 & 0.10 & 1.14 & 0.10 \\ 0 & 1.66 & 2.85 & 1.14 \end{bmatrix}$$

$$\vec{b} = \begin{bmatrix} 0.01 \\ 0.21 \\ -0.03 \\ -0.44 \end{bmatrix}$$

Assume the cart-pole starts in an initial state $\vec{x}[0] = \begin{bmatrix} -0.3853493 \\ 6.1032227 \\ 0.8120005 \\ -14 \end{bmatrix}$, and you want to reach the desired state $\vec{x}_f = \vec{0}$ using the control inputs $u[0], u[1], \ldots$ etc. The state vector $\vec{x}_f = \vec{0}$ corresponds to the cart-pole (or segway) being upright and stopped at the origin. **Reaching $\vec{x}_f = \vec{0}$ in N steps means that, given that we start at $\vec{x}[0]$, we can find control inputs ($u[0], u[1], \ldots$ etc), such that we get $\vec{x}[N]$ (i.e. state vector at Nth time step) equal to $\vec{x}_f = \vec{0}$.**

Note: Please use the Jupyter notebook to solve parts (d) - (g) of the problem. You may use the function we provided `gauss_elim(matrix)` to help you find the **upper triangular form** of matrices. An example of Gaussian Elimination using (`gauss_elim(matrix)`) is provide in the Jupyter notebook under section **Example Usage of gauss_elim**. You may also use the function (`np.linalg.solve`) to solve the equations.

(d) Can you reach $\vec{x}_f$ in *two* time steps? Show work to justify your answer. You should manipulate the equations on paper, but then use the Jupyter notebook for numerical computations.
(*Hint: Express $\vec{x}[2] - \mathbf{A}^2\vec{x}[0]$ in terms of the inputs $u[0]$ and $u[1]$. Then determine if the system of equations can be solved to obtain $u[0]$ and $u[1]$. If we obtain valid solutions for $u[0]$ and $u[1]$, then we can say we will reach $\vec{x}_f$ in two time steps. Use the notebook to see if the system of equations can be solved.*)

(e) Can you reach $\vec{x}_f$ in *three* time steps? Show work to justify your answer. You should manipulate the equations on paper, but then use the Jupyter notebook for numerical computations.
(*Hint: Similar to the last part, express $\vec{x}[3] - \mathbf{A}^3\vec{x}[0]$ in terms of the inputs $u[0]$, $u[1]$ and $u[2]$. Then determine if we can obtain valid solutions for $u[0]$, $u[1]$ and $u[2]$.*)

(f) Can you reach $\vec{x}_f$ in *four* time steps? Show work to justify your answer. You should manipulate the equations on paper, but then use the Jupyter notebook for numerical computations. (*Use the hints from the last two parts.*)

(g) If you have found that you can get to the final state in 4 time steps, find the required correct control inputs, i.e. $u[0]$, $u[1]$, $u[2]$ and $u[3]$, using Jupyter and verify the answer by entering these control inputs into the **Plug in your controller** section of the code in the Jupyter notebook. You need to just show

that you reached the desired final state $\vec{x_f}$ by plugging in the control inputs. The code has been already written to simulate this system.
Suggestion: See what happens if you enter all four control inputs equal to 0. This gives you an idea of how the system naturally evolves!

(h) Let us reflect on what we just did. Recall the system we have:

$$\vec{x}[n+1] = \mathbf{A}\vec{x}[n] + \vec{b}u[n].$$

The control allows us to move the state at time step $n+1$ by $u[n]$ in direction $\vec{b}$, remember $u[n] \in \mathbb{R}$ is just a scalar. We know from part (c) that:

$$\vec{x}[2] = \mathbf{A}^2\vec{x}[0] + \mathbf{A}\vec{b}u[0] + \vec{b}u[1].$$

Again, here $u[0], u[1] \in \mathbb{R}$ can be thought of as arbitrary scalars, and $\mathbf{A}\vec{b}u[0] + \vec{b}u[1]$ can be thought of as the set of all linear combinations of the vectors $\vec{b}$ and $\mathbf{A}\vec{b}$. Using this observation, can you express the possible states you can arrive at in two time steps using the span of exactly *two* vectors plus a vector offset?

(i) Let's try to generalize the idea in the previous part. Express the states you can reach in N timesteps as a span of some vectors plus a vector offset. (Hint: Consider the direction that each control input $u[0], \ldots, u[N-1]$ can move $\vec{x}[N]$ by.)

(j) **(CHALLENGE, OPTIONAL)** Now say you wanted to reach anywhere in $\mathbb{R}^4$, i.e. $\vec{x}_f$ is an unspecified vector in $\mathbb{R}^4$. Under what conditions can you guarantee that you can "reach" $\vec{x}_f$ from any $\vec{x}_0$ in N time steps?
Wouldn't this be cool?

7. Discussion Opt-In/Opt-Out Form

Here is the form for indicating whether you'd like to opt-in or opt-out of discussion participation. For those who've already filled it out, great! For those who haven't, the deadline is now extended to September 22nd at 11:00PM PT. If you've already filled out the form but would like to change your decision, you may edit your response before the deadline. After the deadline passes, your decision is binding, so please read the section "Participation in Discussion" carefully before deciding. If you do not fill it out, we will assume that you are opting-out.

8. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

EECS 16A Designing Information Devices and Systems I
Homework 4

Submission Format
Your homework submission should consist of **one** file.

- `hw4.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned) as well as your IPython notebook saved as a PDF.

  If you do not attach a PDF "printout" of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

Submit the file to the appropriate assignment on Gradescope.

1. Reading Assignment

For this homework, please read Note 7 through 9. These notes will give you an overview of matrix subspaces and eigenvalues/eigenvectors. You are always welcome and encouraged to read beyond this as well. Write a paragraph about how this relates to what you have learned before and what is new for you.

2. Mechanical Basis

(a) Let vectors $\vec{v}_1, \vec{v}_2, \vec{v}_3 \in \mathbb{R}^4$:

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \vec{v}_3 = \begin{bmatrix} 2 \\ 0 \\ 3 \\ 0 \end{bmatrix}.$$

Can the set of vectors $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ form a basis for the vector space $\mathbb{R}^4$? Justify your answer.

(b) Let $\vec{x} = \begin{bmatrix} 5 \\ 3 \\ 6 \end{bmatrix}$. Given a new set of vectors in $\mathbb{R}^3$:

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \vec{v}_3 = \begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix}, \vec{v}_4 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \vec{v}_5 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}.$$

Can the set of vectors $\{\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4, \vec{v}_5\}$ be a basis for $\mathbb{R}^3$? If so, express $\vec{x}$ as a linear combination of these basis vectors.

If the set of five vectors cannot form a basis for $\mathbb{R}^3$, choose a new basis including $\vec{v}_1, \vec{v}_2$ and any number of additional vectors from the set. Describe in words how you might write $\vec{x}$ as a linear combination of the entries in the basis (no need to do any computation).

3. Introduction to Eigenvalues and Eigenvectors

(Contributors: Adhyyan Narang, Ava Tan, Avi Pandey, Christos Adamopoulos, Gireeja Ranade, Michael Kellman, Panos Zarkos, Titan Yuan, Urmita Sikder, Wahid Rahman)

Learning Goal: *Practice calculating eigenvalues and eigenvectors. The importance of eigenvalues and eigenvectors will become clear in the following problems.*

For each of the following matrices, find their eigenvalues and the corresponding eigenvectors. For simple matrices, you may do this by inspection if you prefer.

(a) $\mathbf{A} = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$.

(b) $\mathbf{A} = \begin{bmatrix} 22 & 6 \\ 6 & 13 \end{bmatrix}$.

(c) $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$.

(d) Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be a general square matrix. Show that the set of eigenvectors corresponding to a particular eigenvalue of $\mathbf{A}$ is a subspace of $\mathbb{R}^n$. In other words, show that

$$\{\vec{x} \in \mathbb{R}^n : \mathbf{A}\vec{x} = \lambda\vec{x}, \lambda \in \mathbb{R}\}$$

is a subspace. You have to show that all three properties of a subspace (as mentioned in Note 8) hold.

4. Noisy Images

(Contributors: Ava Tan, Avi Pandey, Gireeja Ranade, Grace Kuo, Matthew McPhail, Michael Kellman, Panos Zarkos, Richard Liou, Titan Yuan, Urmita Sikder, Wahid Rahman)

Learning Goal: *The imaging lab uses the eigenvalues of the masking matrix to understand which masks are better than others for image reconstruction in the presence of additive noise. This problem explores the underlying mathematics.*

In lab, we used a single pixel camera to capture many measurements of an image $\vec{i}$. A single scalar measurement s_i is captured using a mask $\vec{h}_i$ such that $s_i = \vec{h}_i^T\vec{i}$. Many measurements can be expressed as a matrix-vector multiplication of the masks with the image, where the masks lie along the rows of the matrix.

$$\begin{bmatrix} s_1 \\ \vdots \\ s_N \end{bmatrix} = \begin{bmatrix} \vec{h}_1^T \\ \vdots \\ \vec{h}_N^T \end{bmatrix} \vec{i} \tag{1}$$

$$\vec{s} = \mathbf{H}\vec{i} \tag{2}$$

In the real world, noise, $\vec{w}$, creeps into our measurements, so instead we have,

$$\vec{s} = \mathbf{H}\vec{i} + \vec{w}. \tag{3}$$

(a) Express $\vec{i}$ in terms of $\mathbf{H}$ (or its inverse), $\vec{s}$, and $\vec{w}$. Assume $\mathbf{H}$ is invertible. (*Hint:* Think about what you did in the imaging lab.)

(b) It turns out that the eigenvalues of $\mathbf{H}$ and $\mathbf{H}^{-1}$ impact how well we can reconstruct the image from the measurements $\vec{s}$. We will see this in subsequent parts of the problem. First, let us compute the eigenvalues of $\mathbf{H}^{-1}$. The eigenvalues of $\mathbf{H}^{-1}$ are actually related to the eigenvalues of $\mathbf{H}$! Show that if λ is an eigenvalue of a matrix $\mathbf{H}$, then $\frac{1}{\lambda}$ is an eigenvalue of the matrix $\mathbf{H}^{-1}$.
Hint: Start with an eigenvalue λ and one corresponding eigenvector $\vec{v}$, such that they satisfy $\mathbf{H}\vec{v} = \lambda\vec{v}$.

(c) We are going to try different $\mathbf{H}$ matrices in this problem and compare how they deal with noise. Run all of the cells in the attached IPython notebook. Observe the **plots and the printed results.** Which matrix performs best in reconstructing the original image and why? What do you observe regarding the eigenvalues of matrices $\mathbf{H_1}, \mathbf{H_2}$ and $\mathbf{H_3}$? What special matrix is $\mathbf{H}_1$? (*Notice that each plot in the iPython notebook returns the result of trying to image a noisy image as well as the minimum absolute value of the eigenvalue of each matrix.*) Comment on the effect of small eigenvalues on the noise in the image.

(d) Now, because there is noise in our measurements, there will be noise in our recovered image. However, the noise is scaled. From the results of part (a), you know that: $\vec{i} = \mathbf{H}^{-1}\vec{s} - \mathbf{H}^{-1}\vec{w}$, so the impact of the noise on the image $\vec{i}$ is given by $\mathbf{H}^{-1}\vec{w}$.
Let us call this quantity $\hat{\vec{w}}$, often called "w-hat".

$$\hat{\vec{w}} = \mathbf{H}^{-1}\vec{w} \tag{4}$$

To analyze how this transformation alters $\vec{w}$, we represent $\vec{w}$ as a linear combination of the eigenvectors of $\mathbf{H}^{-1}$,

$$\vec{w} = \alpha_1\vec{b}_1 + \ldots + \alpha_N\vec{b}_N, \tag{5}$$

where, $\vec{b}_i$ is $\mathbf{H}^{-1}$'s eigenvector corresponding to eigenvalue $\frac{1}{\lambda_i}$.

Show that we can express the noise in the recovered image as the following linear combination of the vectors $\vec{b}_i$:

$$\hat{\vec{w}} = \mathbf{H}^{-1}\vec{w} = \alpha_1\frac{1}{\lambda_1}\vec{b}_1 + \ldots + \alpha_N\frac{1}{\lambda_N}\vec{b}_N. \tag{6}$$

Now, if λ_i is very large, will the coefficient of $\vec{b}_i$ be large or small in $\hat{\vec{w}}$? If we want $\hat{\vec{w}}$ to be as small as possible, do we prefer large λ_i's or small λ_i's?

5. The Dynamics of Romeo and Juliet's Love Affair

(Contributors: Ava Tan, Avi Pandey, Karina Chang, Vijay Govindarajan, Gireeja Ranade)

Learning Goal: *Eigenvalues and eigenvectors of state transition matrices tend to reveal useful information about the dynamical systems they model. This problem serves as an example of extracting useful information through analysis of the eigenvalues of the state transition matrix of a dynamical system.*

In this problem, we will study a discrete-time model of the dynamics of Romeo and Juliet's love affair—adapted from Steven H. Strogatz's original paper, *Love Affairs and Differential Equations*, Mathematics Magazine, 61(1), p.35, 1988, which describes a continuous-time model.

Let $R[n]$ denote Romeo's feelings about Juliet on day n, and let $J[n]$ denote Juliet's feelings about Romeo on day n, where $R[n]$ and $J[n]$ are **scalars**. The **sign** of $R[n]$ (or $J[n]$) indicates like or dislike. For example, if $R[n] > 0$, it means Romeo likes Juliet. On the other hand, $R[n] < 0$ indicates that Romeo dislikes Juliet. $R[n] = 0$ indicates that Romeo has a neutral stance towards Juliet.

The **magnitude** (i.e. absolute value) of $R[n]$ (or $J[n]$) represents the intensity of that feeling. For example, a larger magnitude of $R[n]$ means that Romeo has a stronger emotion towards Juliet (strong love if $R[n] > 0$ or strong hatred if $R[n] < 0$). Similar interpretations hold for $J[n]$.

We model the dynamics of Romeo and Juliet's relationship using the following linear system:

$$R[n+1] = aR[n] + bJ[n], \quad n = 0,1,2,\dots$$

and

$$J[n+1] = cR[n] + dJ[n], \quad n = 0,1,2,\dots,$$

which we can rewrite as

$$\vec{s}[n+1] = \mathbf{A}\,\vec{s}[n],$$

where $\vec{s}[n] = \begin{bmatrix} R[n] \\ J[n] \end{bmatrix}$ denotes the state vector and $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ denotes the state transition matrix for our dynamic system model.

The selection of the parameters a,b,c,d results in different dynamic scenarios. The fate of Romeo and Juliet's relationship depends on these model parameters (i.e. a,b,c,d) in the state transition matrix and the initial state ($\vec{s}[0]$). In this problem, we'll explore some of these possibilities.

(a) Consider the case where $a+b = c+d$ in the state-transition matrix

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Show that

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

is an eigenvector of $\mathbf{A}$, and determine its corresponding eigenvalue λ_1.
Show that

$$\vec{v}_2 = \begin{bmatrix} b \\ -c \end{bmatrix}$$

is an eigenvector of $\mathbf{A}$, and determine its corresponding eigenvalue λ_2.
Now, express the first and second eigenvalues and their eigenspaces in terms of the parameters $a,b,c,$ and d.
Hint: *Consider* $\mathbf{A}\vec{v}_1$. *Is it equal to a scalar multiple of* $\vec{v}_1$*? Repeat a similar process for* $\vec{v}_2$.

For parts (b) - (e), consider the following state-transition matrix:

$$\mathbf{A} = \begin{bmatrix} 0.75 & 0.25 \\ 0.25 & 0.75 \end{bmatrix}.$$

(b) Determine the eigenvalues and corresponding eigenvectors (i.e. $\lambda_1, \vec{v}_1$ and $\lambda_2, \vec{v}_2$) for this system. Note that this matrix is a special case of the matrix explored in part (a), so you can use results from that part to help you.

(c) Determine all of the non-zero *steady states* of the system. That is, find all possible state vectors $\vec{s}_*$ such that if Romeo and Juliet start at, or enter, any of those state vectors, their states will stay in place forever: $\{\vec{s}_* \mid \mathbf{A}\vec{s}_* = \vec{s}_*\}$.

(d) Suppose Romeo and Juliet start from an initial state $\vec{s}[0] \in \operatorname{span}\left\{\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right\}$. What happens to their relationship over time? Specifically, what is $\vec{s}[n]$ as $n \to \infty$?

(e) Suppose the initial state is $\vec{s}[0] = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$. What happens to their relationship over time? Specifically, what is $\vec{s}[n]$ as $n \to \infty$?
Hint: *Can you use what you learned about the eigenvectors of* $\mathbf{A}$ *(in parts c and d) to help you solve this problem? You can represent the starting state as a linear combination of eigenvectors* $\vec{v}_1$ *and* $\vec{v}_2$*.*

Now suppose we have the following state-transition matrix:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Use this state-transition matrix for parts (f) - (h).

(f) Determine the eigenvalues and corresponding eigenvectors (i.e. $\lambda_1, \vec{v}_1$ and $\lambda_2, \vec{v}_2$) for this system. Note that this matrix is **a special case** of the matrix explored in part (a), so you can use results from that part to help you.

(g) Suppose Romeo and Juliet start from an initial state $\vec{s}[0] \in \operatorname{span}\left\{\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right\}$. What happens to their relationship over time? Specifically, what is $\vec{s}[n]$ as $n \to \infty$?

(h) Now suppose that Romeo and Juliet start from an initial state $\vec{s}[0] \in \operatorname{span}\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$. What happens to their relationship over time? Specifically, what is $\vec{s}[n]$ as $n \to \infty$?

Finally, we consider the case where we have the following state-transition matrix:

$$\mathbf{A} = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}.$$

Use this state-transition matrix for parts (i) - (k).

(i) Determine the eigenvalues and corresponding eigenvectors (i.e. $\lambda_1, \vec{v}_1$ and $\lambda_2, \vec{v}_2$) for this system. Note that this matrix is **a special case** of the matrix explored in part (a), so you can use results from that part to help you.

(j) Suppose Romeo and Juliet start from an initial state $\vec{s}[0] = \begin{bmatrix} R[0] \\ J[0] \end{bmatrix}$, where $\vec{s}[0] \in \operatorname{span}\left\{\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right\} = \operatorname{span}\left\{\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right\}$. What happens to their relationship over time if $R[0] > 0$ and $J[0] < 0$? What about if $R[0] < 0$ and $J[0] > 0$? Specifically, what is $\vec{s}[n]$ as $n \to \infty$?

(k) Now suppose that Romeo and Juliet start from an initial state $\vec{s}[0] \in \operatorname{span}\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$. What happens to their relationship over time? Specifically, what is $\vec{s}[n]$ as $n \to \infty$?

6. Traffic Flows

(Contributors: Adhyyan Narang, Ava Tan, Avi Pandey, Gireeja Ranade, Grace Kuo, Laura Brink, Panos Zarkos, Raghav Anand, Richard Liou, Titan Yuan, Vijay Govindarajan)

Learning Objective: *The learning objective of this problem is to see how the concept of nullspaces can be applied to flow problems.*

Your goal is to measure the flow rates of vehicles along roads in a town. It is prohibitively (too) expensive to place a traffic sensor along every road. You realize, however, that the number of cars flowing into an intersection must equal the number of cars flowing out. You can use this "flow conservation" to determine the traffic along all roads in a network by measuring the flow along only some roads. In this problem, we will explore this concept.

(a) Let's begin with a network with three intersections, A, B and C. Define the flow t_1 as the rate of cars (cars/hour) on the road between B and A, flow t_2 as the rate on the road between C and B, and flow t_3 as the rate on the road between C and A.

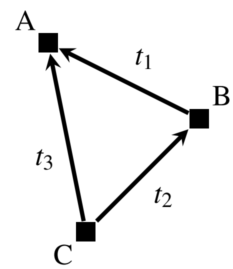


Figure 1: A simple road network.

(Note: The directions of the arrows in the figure are the way that we define positive flow by convention. For example, if there were 100 cars per hour traveling from A to C, then $t_3 = -100$. The flows now are not fractions of water in reservoirs as in the pumps setting, but numbers of cars.)

We assume the "flow conservation" constraints: the net number of cars per hour flowing into each intersection is zero. For example at intersection B, we have the constraint $t_2 - t_1 = 0$. The full set of constraints (one per intersection) is:

$$\begin{cases} t_1 + t_3 = 0 \\ t_2 - t_1 = 0 \\ -t_3 - t_2 = 0 \end{cases}$$

As mentioned earlier, we can place sensors on a road to measure the flow through it, but we have a limited budget, and we would like to determine all of the flows with the smallest possible number of sensors.

Suppose for the network above we have one sensor reading, $t_1 = 10$. Can we figure out the flows along the other roads? (That is, the values of t_2 and t_3). If we can, find the values of t_2 and t_3.

(b) Now suppose we have a larger network, as shown in Figure 2.

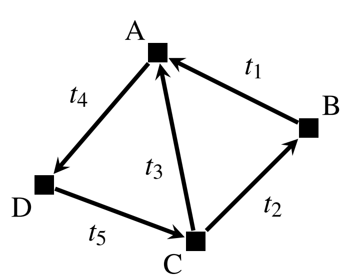


Figure 2: A larger road network.

We would again like to determine the traffic flows on all roads, using measurements from some sensors. A Berkeley student claims that we need two sensors placed on the roads CA (measuring t_3) and DC (measuring t_5). A Stanford student claims that we need two sensors placed on the roads CB (measuring t_2) and BA (measuring t_1). Write out the system of linear equations that represents this flow graph. Is it possible to determine all traffic flows, $\left[t_1, t_2, t_3, t_4, t_5\right]^T$, with the Berkeley student's suggestion? How about the Stanford student's suggestion? *Hint: This can be solved just writing out the relevant equations and reasoning about them.*

(c) We would like a more general way of determining the possible traffic flows in a network. Suppose we write the traffic flow on all roads as a vector $\vec{t} = \begin{bmatrix} t_1 \\ t_2 \\ t_3 \\ t_4 \\ t_5 \end{bmatrix}$. As a first step, let us try to write all the flow conservation constraints (one per intersection) i.e. the system of equations from part (b) as a matrix equation.

Construct a 4×5 matrix $\mathbf{H}$ such that the equation $\mathbf{H}\vec{t} = \vec{0}$:

$$\begin{bmatrix} & & \\ & \mathbf{H} & \\ & & \end{bmatrix} \begin{bmatrix} t_1 \\ t_2 \\ t_3 \\ t_4 \\ t_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

represents the flow conservation constraints for the network in Figure 2.

Hint: You can construct $\boldsymbol{H}$ *using only* $0, 1,$ *and* -1 *entries. Each row represents the inflow/outflow of an intersection. This matrix is called the* ***incidence matrix****.*

(d) Again, suppose we write the traffic flow on all roads as a vector. $\vec{t} = \begin{bmatrix} t_1 \\ t_2 \\ t_3 \\ t_4 \\ t_5 \end{bmatrix}$. Then, write an equation in terms of $\mathbf{H}$, $\vec{t}$ that captures the valid traffic flows in the network. With this equation, we hope you can see that you can find all valid traffic flows by computing the null space of $\mathbf{H}$; however, **you do not need to compute the null space for this problem**.

(e) Notice that we can represent the Berkeley student's measurement as $\mathbf{M}_B\vec{t}$, where:

$$\mathbf{M}_B\vec{t} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \vec{t} = \begin{bmatrix} t_3 \\ t_5 \end{bmatrix}$$

Write a matrix $\mathbf{M}_S$ that can be used to represent the Stanford student's measurement.

(f) **[Challenge, Optional]** Now let us analyze more general road networks. Say there is a road network graph G, with incidence matrix $\mathbf{H}_G$. If $\mathbf{H}_G$ has a k-dimensional null space, does this mean measuring the flows along ***any*** k **roads** is always sufficient to recover all of the true flows? In other words, is there ever a possibility of being unable to recover the true flows depending on which k roads you choose?

If you think measuring the flows along any k roads will always work, then prove it showing various possible scenarios. Otherwise give an example showing a scenario where it does not work (such an example is called a counter example).

Hint: Consider the Stanford student's measurement from part (b).

(g) **[Challenge, Optional]** Assume that $\vec{u}$ and $\vec{t}$ are distinct valid flows, that is $\mathbf{H}_G\vec{u} = \mathbf{H}_G\vec{t} = \vec{0}$. Can you recover all of the network's true flows if $(\vec{u} - \vec{t})$ belongs to the nullspace of $\mathbf{M}_S$?

Clarification: A "valid" flow is one that is possible without violating the constraints on the nodes (so flow in must equal to flow out). There may be many valid flows, but only one "true" flow. The "true flow" is one of many the valid flows, which represents the actual number of cars/ hour on each road.

7. Page Rank

(Contributors: Ava Tan, Avi Pandey, Edward Doyle, Gireeja Ranade, Lydia Lee, Michael Kellman, Panos Zarkos, Raghav Anand, Titan Yuan)

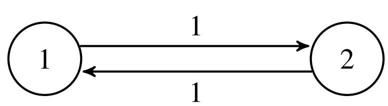
Learning Goal: *This problem highlights the use of transition matrices in modeling dynamical linear systems. Predictions about the steady state of a system can be made using the eigenvalues and eigenvectors of this matrix.*

In homework and discussion, we have discussed the behavior of water flowing in reservoirs and the people flowing in social networks. We now consider the setting of web traffic where the dynamical system can be described with a directed graph, also known as state transition diagram.

As we have seen in lecture and discussion the "transition matrix", $\mathbf{T}$, can be constructed using the state transition diagram, as follows: entries t_{ji}, represent the *proportion* of the people who are at website i that click the link for website j.

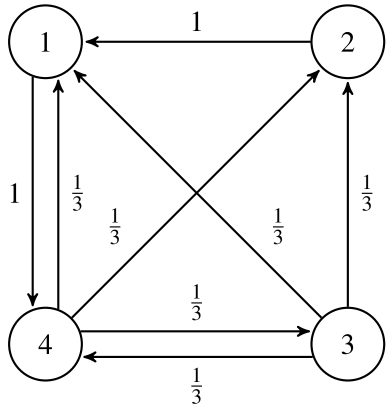
The **steady-state frequency** (i.e. fraction of visitors in steady-state) for a graph of websites is related to the eigenspace associated with eigenvalue 1 for the "transition matrix" of the graph. Once computed, an eigenvector with eigenvalue 1 will have values which correspond to the steady-state frequency for the fraction of people for each webpage. When the elements of this eigenvector are made to **sum to one** (to conserve population), the i^{th} element of the eigenvector will correspond to the fraction of people on the i^{th} website.

(a) For graph A shown below, what are the steady-state frequencies i.e. fraction of visitors in steady-state for the two webpages? Graph A has weights in place to help you construct the transition matrix. Remember to ensure that your steady state-frequencies sum to 1 to maintain conservation.



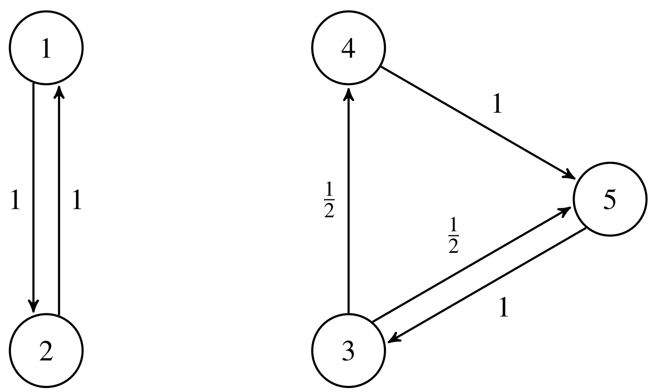
Graph A - weighted

(b) For graph B, what are the steady-state frequencies for the webpages? You may use IPython and the Numpy command `numpy.linalg.eig` for this. It may be helpful to consult the Python documentation for `numpy.linalg.eig` to understand what this function does and what it returns. Graph B is shown below, with weights in place to help you construct the transition matrix.



Graph B - weighted

(c) Find the eigenspace that corresponds to the steady-state for graph C. How many independent systems (disjoint sets of webpages) are there in graph C versus in graph B? What is the dimension of the eigenspace corresponding to the steady-state for graph C? Again, graph C with weights in place is shown below. You may use IPython to compute the eigenvalues and eigenvectors again.



Graph C - weighted

8. Subspaces, Bases and Dimension

(Contributors: Moses Won, Urmita Sikder)

For the set $\mathbb{U}$ (which is a subset of $\mathbb{R}^3$) defined below, state whether it is a subspace of $\mathbb{R}^3$ or not. If $\mathbb{U}$ is a subspace, find a basis for it and state the dimension. You have to show that all three properties of a subspace (as mentioned in Note 8) hold. Even if only one of the properties is violated, it is safe to conclude that the set is not a subspace.

(a) $\mathbb{U} = \left\{ \begin{bmatrix} x \\ y \\ x+1 \end{bmatrix} \mid x, y \in \mathbb{R} \right\}$

9. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

EECS 16A Designing Information Devices and Systems I
Homework 6

Submission Format
Your homework submission should consist of **one** file.

- `hw6.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned)

Submit the file to the appropriate assignment on Gradescope.

1. **Reading Assignment**

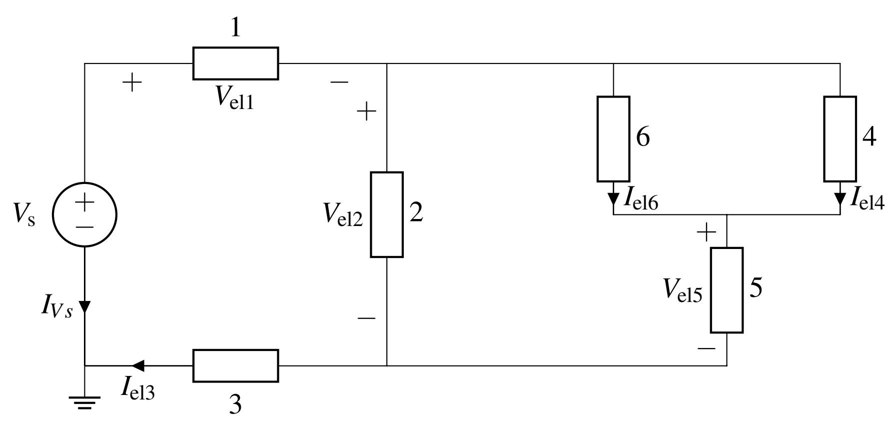
For this homework, please read Notes 11 through 13. Note 11 introduces the basics of circuit analysis. Note 12 will provide an overview of voltage dividers and resistors. Note 13 will refresh you on how simple 1-D resistive touchscreens work, as well as the notion of power in electric circuits.

(a) What is the value of having a systematic procedure for solving circuits?

(b) Describe the key ideas behind how the 1D touchscreen works. In general, why is it useful to be able to convert a "physical" quantity like the position of your finger to an electronic signal (i.e. voltage)?

2. **Intro to Circuits**

(Contributors: Panagiotis Zarkos)



(a) How many nodes does the above circuit have? Label them.
Note: The ground node has been selected for you, so you don't need to label that, but you need to include it in your node count.

(b) Notice that elements 1 - 6 and the voltage source V_s have either the *voltage across* or the *current through* them not labeled. Label the missing *voltages across* or *currents through* for elements 1 - 6, and the voltage source V_s, so that they all follow **passive sign convention**.

(c) Express all element voltages (including the element voltage across the source, V_s) as a function of node voltages. This will depend on the node labeling you chose in part (a).

(d) Write one KCL equation that involves the currents through elements 1 and 2.

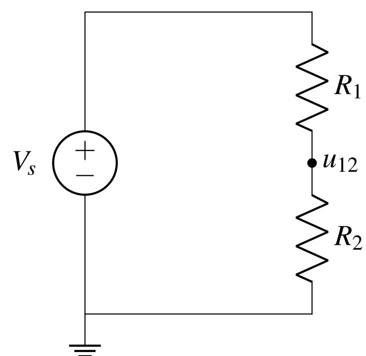
(e) Write a KVL equation for all the loops that contain the voltage source V_s. These equations should be a function of element voltages and the voltage source V_s.

3. Voltage divider

(Contributors: Adhyyan Narang, Panagiotis Zarkos, Sashank Krishnamurthy, Urmita Sikder)

In the following circuit, $V_s = 12\,\text{V}$. Choose resistance values for R_1 and R_2 such that the current through each element is $\leq 0.8\text{A}$ and $u_{12} = 6\,\text{V}$. This is a **design problem**, so there can be more than one set of correct answers to this problem.

Hint: You can refer to Note 11B to solve this question. Other valid circuit solving methods are also acceptable.



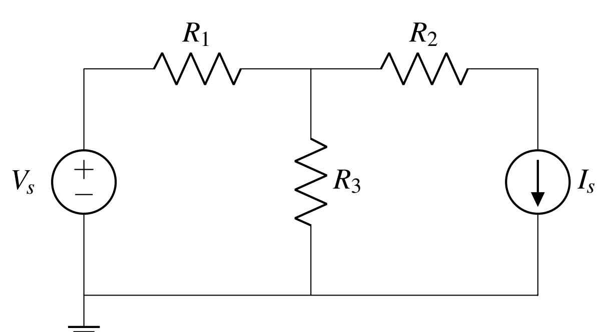
4. Circuit Analysis

(Contributors: Ava Tan, Aviral Pandey, Christos Adamopoulos, Matthew McPhail, Moses Won, Panos Zarkos, Raghav Anand, Titan Yuan, Urmita Sikder)

Learning Goal: *This problem will help you practice circuit analysis using NVA method.*

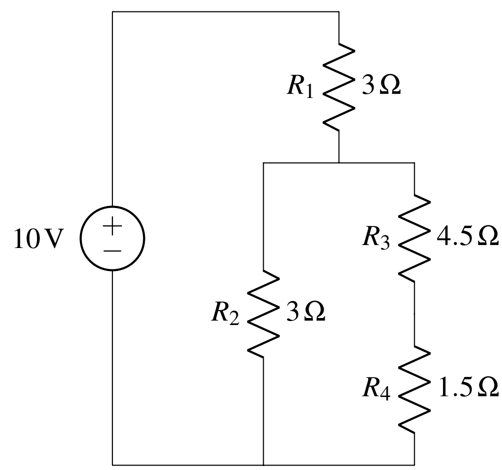
Using the steps outlined in lecture or in Note 11B, analyze the following circuit to calculate the currents through each element and the voltages at each node. Use the ground node labelled for you. You may use a numerical tool such as IPython to solve the final system of linear equations.

$V_s = 5\,\text{V}, I_s = 2\,\text{A}, R_1 = R_2 = 2\,\Omega, R_3 = 4\,\Omega$



5. Mechanical Circuits

Find the voltages across and currents flowing through all of the resistors.



6. It's a Triforce! *(Contributors: Ava Tan, Elena Herbold, Ryan Tsang, Taehwan Kim, Urmita Sikder, Wahid Rahman)*

Learning Goal: *This problem explores passive sign convention and nodal analysis in a slightly more complicated circuit.*

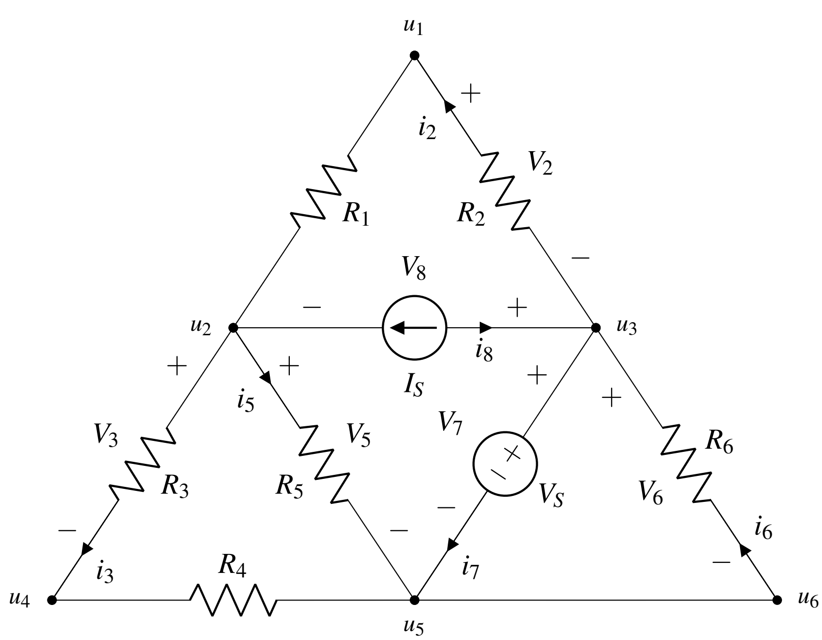


Figure 1: A triangular circuit consisting of a voltage source V_s, current source I_s, and resistors R_1 to R_6.

(a) Which of the elements I_S, V_S, R_2, R_3, R_5, or R_6 in Figure 1 have current-voltage labeling that violates Passive Sign Convention? There could be more than one possible element which violates Passive Sign Convention. Explain your reasoning.

(b) In Figure 1, the nodes are labeled with u_1, u_2, ... etc. There is a subset of u_i's in the given circuit that are redundant, i.e. there might be more than one label for the same node. Which node(s)? Justify your answer.

(c) Redraw the circuit diagram by correctly labeling all the element voltages and element currents according to passive sign convention. (The component labels that were violating Passive Sign Convention in part (a), should be corrected by swapping the element voltage polarity. Also, the elements that have not been labeled yet, should be labeled following Passive Sign Convention.)

(d) Write an equation to describe the current-voltage relationship for element R_4 in terms of the relevant i's, R's, and node voltages in this circuit.

(e) Write the KCL equation for node u_2 in terms of the node voltages and other circuit elements.

7. Fred's Fruits

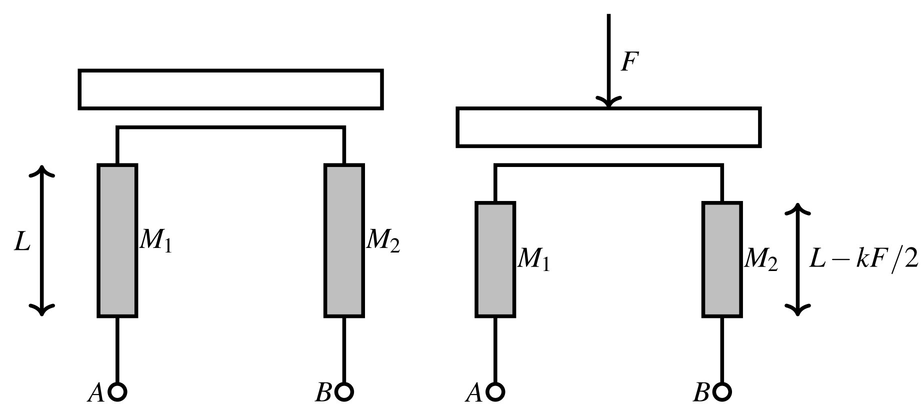
(Contributors: Ava Tan, Christos Adamopoulos, Matthew McPahil, Moses Won, Panos Zarkos, Urmita Sikder, Wahid Rahman)

Learning Goal: *This problem will introduce the process of designing a sensing circuit for the purpose of measuring a physical quantity. This will also help to build your intuition for modeling physical elements.*

Fred just got back from Berkeley Bowl with a bunch of mangoes, pineapples, and coconuts. He wants to sort his mangoes in order of weight, so he decides to use his knowledge from EECS16A to build a scale.

He finds two identical bars of material (M_1 and M_2) of length L (in meters) and a cross-sectional area (i.e. width $\times$ thickness) of A_{ca} (in meters^2). The bars are made of a material with resistivity ρ. He knows that the **length of these bars decreases** by k meters per Newton of force applied, while the **cross-sectional area remains constant**.

He builds his scale as shown below, where the top of the vertical bars are connected with an ideal electrical wire. The left side of the diagram shows the scale at rest (with no object placed on it), and the right side shows it when the applied force is F (Newtons). The force F is equally distributed between two bars, causing the length of each bar to decrease by $kF/2$ meters. Fred's mangoes are not very heavy, so the change in each bar's length is very small compared to the total length (mathematically, we can write this by using the "much smaller than" symbol $\ll$, i.e. $kF/2 \ll L$).



(a) Let R_{AB} be the resistance between nodes A and B with the weights on the scale. Write an expression for R_{AB} as a function of A_{ca}, L, ρ, F, and k. *Hint: You can start by representing each bar as a resistor, then find how they are connected.*

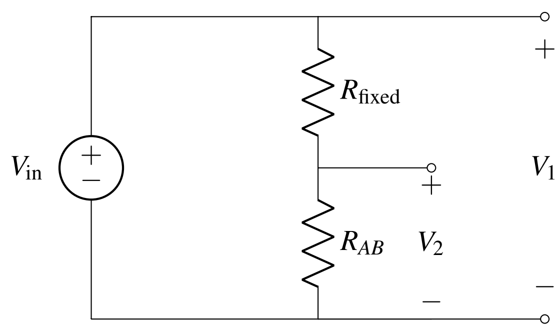
(b) Fred wants to measure a voltage that changes based on how much weight is placed on his scale. He knows that R_{AB} will change with the weight on the scale.

Design a circuit for Fred that outputs a voltage that is some function of the weight F. **Your circuit should include R_{AB}, and you may use any number of voltage sources and resistors in your design.** Be sure to **label** where the voltage should be measured in your circuit.

Also provide an **expression** relating the output voltage of your circuit to the force applied on the scale. This expression can contain any necessary parameters.

Hint: If you connected only a voltage source across A and B and measured the voltage (V_{AB}) between A and B, would V_{AB} change based on the value of R_{AB}? It turns out it wouldn't. Why?

Hint: Consider the following circuit, where R_{fixed} is a value you know and choose. Which voltage would you choose to measure in the circuit to help you determine the weight on the scale: V_1 or V_2?



8. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

EECS 16A Designing Information Devices and Systems I
Homework 7

Submission Format
Your homework submission should consist of **one** file.

- `hw7.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned).

Submit the file to the appropriate assignment on Gradescope.

1. Reading Assignment

For this homework, please review Notes 12 through 14. Note 12 will provide an overview of voltage dividers and resistors. Note 13 will refresh you on how simple 1-D resistive touchscreens work, as well as the notion of power in electric circuits. Note 14 will cover a slightly more complicated 2-D resistive touchscreens and how to analyze them from a circuits perspective.

(a) Describe the significance of positive and negative power. How does this relate to passive sign convention?

(b) For a 2D resistive touchscreen, how would you measure horizontal position measurements and vertical position measurements?

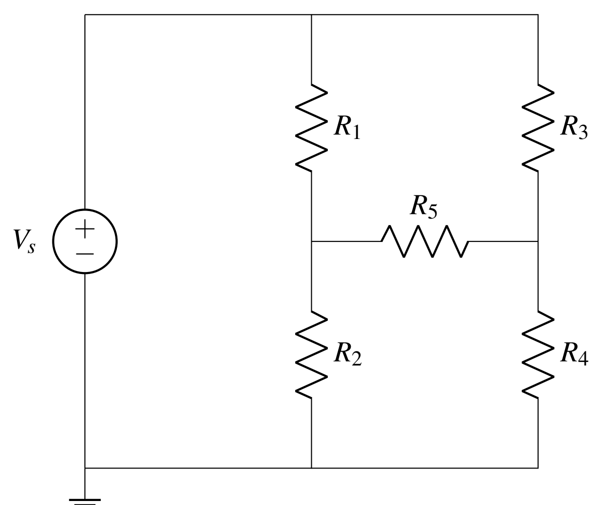
2. Circuit Analysis

(Contributors: Ava Tan, Aviral Pandey, Christos Adamopoulos, Matthew McPhail, Moses Won, Panos Zarkos, Raghav Anand, Titan Yuan, Urmita Sikder)

Learning Goal: *This problem will help you practice circuit analysis using NVA method.*

Using the steps outlined in lecture or in Note 11B, analyze the following circuit to calculate the currents through each element and the voltages at each node. Use the ground node labelled for you. You may use a numerical tool such as IPython to solve the final system of linear equations.

(a) $V_s = 5\,\text{V}, R_1 = 1\,\Omega, R_2 = 2\,\Omega, R_3 = 3\,\Omega, R_4 = 4\,\Omega, R_5 = 5\,\Omega$



3. Resistive Touchscreen

(Contributors: Ava Tan, Ben Osoba, Christos Adamopoulos, Matthew McPhail, Moses Won, Panos Zarkos, Urmita Sikder, Wahid Rahman)

Learning Goal: *The objective of this problem is to provide insight into modeling of resistive elements. This will also help to apply the concepts from 1D resistive touchscreen.*

In this problem, we will investigate how a 1D resistive touchscreen with a defined thickness, width, and length can actually be modeled as a series combination of resistors. As we know the value of a resistor depends on its length.

Figure 1 shows the top view of a resistive touchscreen consisting of a conductive layer with resistivity ρ_1, thickness t, width W, and length L. At the top and bottom it is connected through good conductors ($\rho = 0$) to the rest of the circuit. The touchscreen is wired to voltage source V_s.

Use the following numerical values in your calculations: $W = 50$ mm, $L = 80$ mm, $t = 1$ mm, $\rho_1 = 0.5\,\Omega\,$m, $V_s = 5$V, $x_1 = 20$ mm, $x_2 = 45$ mm, $y_1 = 30$ mm, $y_2 = 60$ mm.

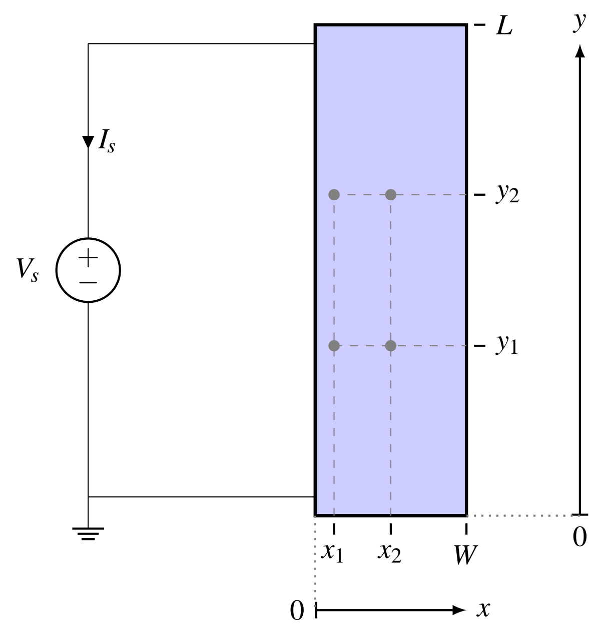


Figure 1: Top view of resistive touchscreen (not to scale). z axis i.e. the thickness not shown (into the page).

(a) Draw a circuit diagram representing Figure 1, where the touchscreen is represented as *a resistor*. **Note that no touch is occurring in this scenario.** Remember that circuit diagrams in general consist of only circuit elements (resistors, sources, etc) represented by symbols, connecting wires, and the reference/ground symbol. Calculate the value of current I_S based on the circuit diagram you drew. Do not forget to specify the correct unit as always.

(b) Let us assume u_{12} is the node voltage at the node represented by coordinates (x_1, y_2) of the touchscreen, as shown in Figure 2. What is the value of u_{12}? You should first draw a circuit diagram representing Figure 2, which includes node u_{12}. Specify all resistance values in the diagram. Does the value of u_{12} change based on the value of the x-coordinate x_1?

Hint: You will need more than one resistor to represent this scenario.

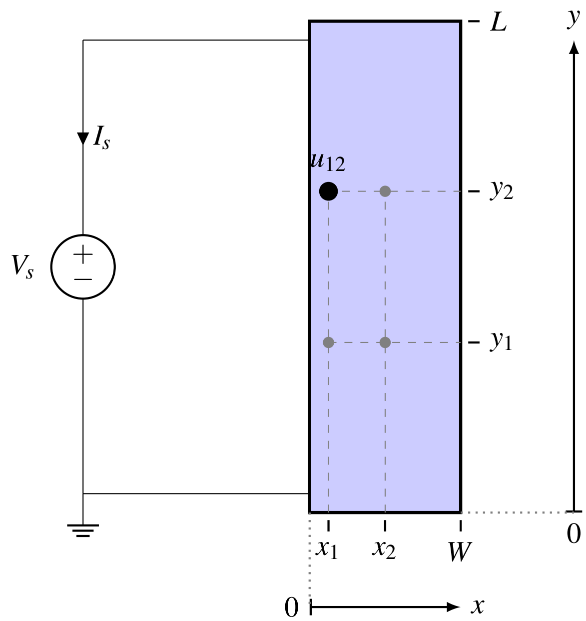


Figure 2: Top view of resistive touchscreen showing node u_{12}.

(c) Assume V_{ab} is the voltage measured between the nodes represented by touchscreen coordinates (x_1, y_1) and coordinates (x_1, y_2), as shown in Figure 3. Calculate the absolute value of V_{ab}. As with the previous part, you should first draw the circuit diagram representing Figure 3, which includes V_{ab}. Calculate all resistor values in the circuit. *Hint: Try representing the segment of the touchscreen between these two coordinates as a separate resistor itself.*

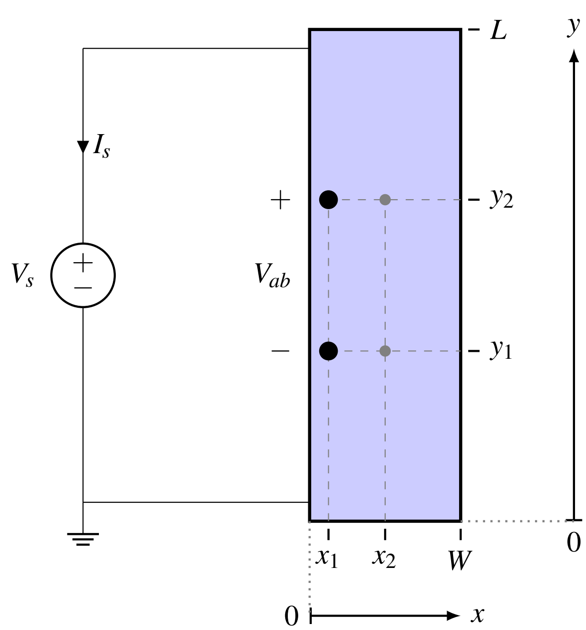


Figure 3: Top view of resistive touchscreen showing voltage V_{ab}.

(d) Calculate (the absolute value of) the voltage between the nodes represented by touchscreen coordinates (x_1, y_1) and coordinates (x_2, y_1).

(e) Calculate (the absolute value of) the voltage between the nodes represented by touchscreen coordinates (x_1, y_1) and coordinates (x_2, y_2).

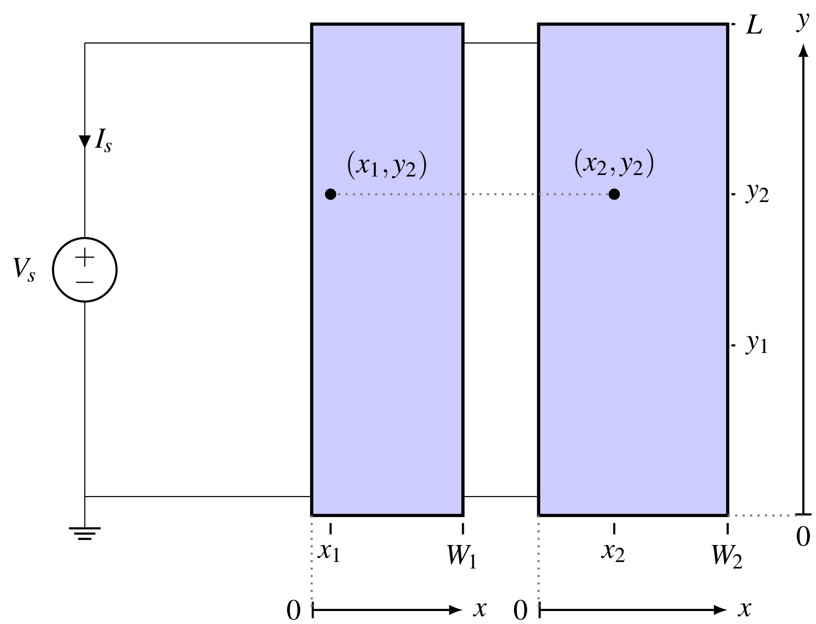


Figure 4: Top view of two touchscreens wired in parallel (not to scale). z axis not shown (into the page).

(f) Figure 4 shows a new arrangement with two touchscreens. The second touchscreen (the one on the right) is identical to the one shown in Figure 1, except for different width, W_2, and resistivity, ρ_2.

Use the following numerical values in your calculations: $W_1 = 50$ mm, $L = 80$ mm, $t = 1$ mm, $\rho_1 = 0.5\,\Omega\,\mathrm{m}$, $V_s = 5\mathrm{V}$, $x_1 = 20$ mm, $x_2 = 45$ mm, $y_1 = 30$ mm, $y_2 = 60$ mm, which are the same values as before. The new touchscreen has the following numerical values which are different: $W_2 = 85$ mm, $\rho_2 = 0.6\,\Omega\,\mathrm{m}$.

Draw a circuit diagram representing Figure 4, where the two touchscreens are represented as *two separate resistors*. **Note that no touch is occurring in this scenario.**

(g) Calculate the value of current I_s for the two touchscreen arrangement based on the circuit diagram you drew in the last part.

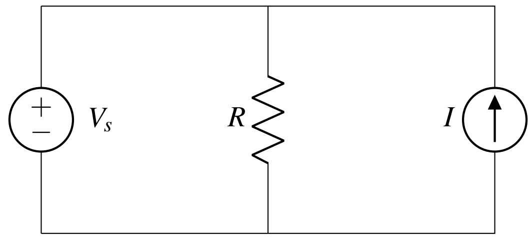
(h) Consider the two points: (x_1, y_2) in the touchscreen on the left, and (x_2, y_2) in the touchscreen on the right in Figure 4. Show that the node voltage at (x_1, y_2) is the same that at (x_2, y_2), i.e. the potential difference between the two points is 0. You can show this without explicitly calculating the node voltages at the two points.

If you were to connect a wire between the two coordinates (x_1, y_2) in the touchscreen on the left, and (x_2, y_2) in the touchscreen on the right, would any current flow through this wire?

4. Power Analysis

(Contributors: Ava Tan, Craig Schindler, Moses Won, Raghav Anand, Urmita Sikder, Wahid Rahman)

Learning Goal: *This problem aims to help you practice calculating power dissipation in different circuit elements. It will also give you insights into how power is conserved in a circuit.*



(a) Find the expressions of power dissipated by each element in the circuit above. Remember to label voltage-current pairs using passive sign convention.

(b) Use $R = 5\text{k}\Omega$, $V_s = 5\text{V}$, and $I = 5\text{mA}$. Calculate the power dissipated by the voltage source (P_{V_s}), the current source (P_I), and the resistor (P_R).

(c) Once again, let $R = 5\text{k}\Omega$, $V_s = 5\text{V}$. What does the value I of the current source have to be such that the current source **dissipates** 40mW? Note that it is possible for a current source to dissipate power, i.e. under passive sign convention, $P_I = 40mW$. For this value of I, compute I, P_{V_s}, P_I, and P_R as well.
As an aside: If the current source were delivering power it would have been $P_I = -40mW$, under passive sign convention, but this is NOT what the question is asking about.

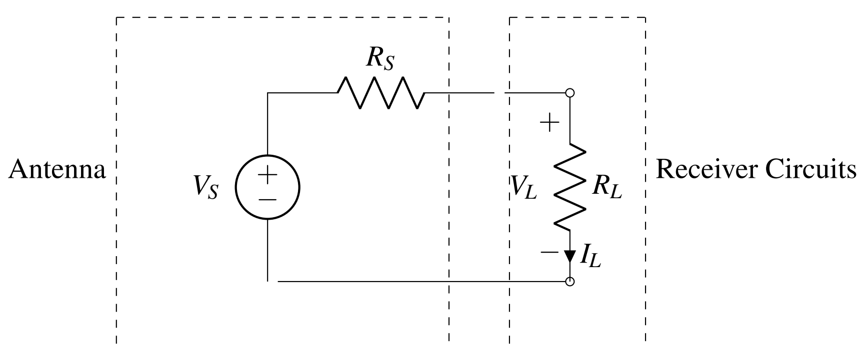
5. Maximum Power Transfer

(Contributors: Craig Schindler, Gireeja Ranade, Panos Zarkos, Urmita Sikder, Sashank Krishnamurthy)

Learning Goal: *This problem shows how the power dissipated in the load depends on the value of the load resistance. It also helps to understand the condition required for maximum power transfer.*

Smartphones use "bars" to indicate strength of the cellular signal. Fewer "bars" translate to slow or no connectivity. But what do these "bars" actually stand for? Voltage, current? Well, not quite. Good radio (a cellular modem is a particular type of radio) reception depends on the **power received at the receiver**. Communication theory tells us that higher received signal power enables higher data rates. To that end, we design a receiver that maximizes the power received, and hence connection speed.

A typical receiver consists of an antenna and receiver circuits. The antenna receives the radio waves propagating in space, and converts it into electrical voltages and currents. A very good abstraction used by circuit designers is to **model the antenna as a voltage source V_S, with a series resistance R_S**. The typical values of V_S in a real cellular receiver are in the range of micro- or milli-volts (10^{-6} and 10^{-3}, respectively) and the typical values of resistance R_S are usually 50Ω or 75Ω, depending on how the antenna is designed. The receiver circuits are quite complex and will be covered in detail in EE142 "Integrated Circuits for Communications". However, a standard abstraction is to **model these receiver circuits as a load resistance R_L** to the antenna, as shown in the figure below.



Models are very important in engineering design for their ability to abstract away details when they are not needed and are the key to successful design of complex systems We will discuss the use and properties of electronic circuit models further in class.

Use the following component values for your calculations: $V_S = 100\mu V$, and $R_S = 50\Omega$.

Note: This homework problem and subsequent class content may require knowledge of calculus. For this problem, please review your understanding of how to compute derivatives.

(a) Consider any value of R_L within the range: $0 \leq R_L \leq \infty$. Find the value of R_L that maximizes the ***voltage*** V_L across resistor R_L. Calculate the values of V_L, I_L, and the power P_L dissipated by resistor R_L for the value you found.

(*Hint: The antenna voltage V_S and the resistance R_S are fixed. Intuitively argue for a particular value for R_L. How does the voltage change as the resistor value changes? Note: To rigorously prove your choice of R_L, you should express the voltage as a function of R_L, and differentiate it with respect to R_L.*)

(b) Consider any value of R_L within the range: $0 \leq R_L \leq \infty$. Find the value of R_L that maximizes the ***current*** I_L through resistor R_L. Calculate the values of V_L, I_L, and the power P_L dissipated by resistor R_L for the value you found.

(*Hint: The antenna voltage V_S and the resistance R_S are fixed. Intuitively argue for a particular value of R_L. How does the current change as the resistor value changes?)*

(c) Find the value of R_L that maximizes the ***power*** P_L delivered to resistor R_L. Calculate the values of V_L, I_L, and the power P_L delivered to resistor R_L. **It is important to note that this value of R_L which maximizes the power delivered to R_L also optimizes cellular connectivity.** (*Hint: The power optimization is best performed algebraically by setting the derivative of P_L with respect to R_L to 0. Alternatively you can do the optimization graphically. Refer to the code notebook provided and complete the code to plot P_L versus R_L and find the maximum.*)

6. Hot Dogs - Fall 2018 Midterm 2

(Contributors: Alice Ye, Michael Kellman, Urmita Sikder)

Learning Goal: *The objective of this problem is to explore how power is dissipated in different resistors in series/parallel combination.*

Michael and Alice are going on a picnic with Daisy the robot dog! But when they arrive at the park, a thunderstorm starts with lots of lightning... However, Daisy still wants to play in the park!

(a) We want to understand what will happen *if* Daisy is struck by lightning. A lightning bolt is represented with a voltage source of 1000 Volts (1 kV) with reference to the Earth. The lightning travels through the ionized air ($R_{Air} = 4\,\text{k}\Omega$), and through Daisy ($R_{Daisy} = 4\,\text{k}\Omega$) before reaching the ground. Daisy can withstand up to 10 Watts (W) of power before breakdown, (i.e. Daisy will breakdown if she dissipates >10 Watts).

Using the circuit diagram in Figure 5, find if it is safe for Daisy to be struck by a lightning bolt. Justify your answer by showing your work!

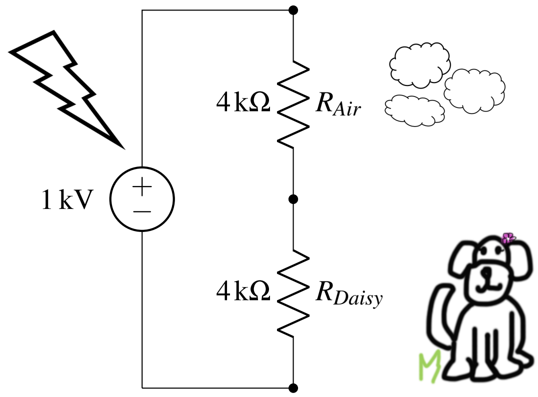


Figure 5: Circuit for Part (a)

(b) Laura also brings along her dog, Elton, for the picnic. Elton has a resistance of $R_{Elton} = 2\,\text{k}\Omega$ and can withstand up to 25 W of power. If Daisy and Elton hide together next to a tree of resistance $R_{Tree} = 4\,\text{k}\Omega$ as shown in the Figure 6, would either of the dogs withstand being struck by lightning? Show your work.

Note: Resistors that have terminals connected to the same two nodes are said to be in parallel. In this case, R_{Tree}, R_{Daisy}, and R_{Elton} are parallel resistors. To find the equivalent resistance (R_{eq}) for n resistors in parallel, where $R_{eq} = R_1||R_2||R_3||...||R_n$, compute the following: $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots \frac{1}{R_n}$

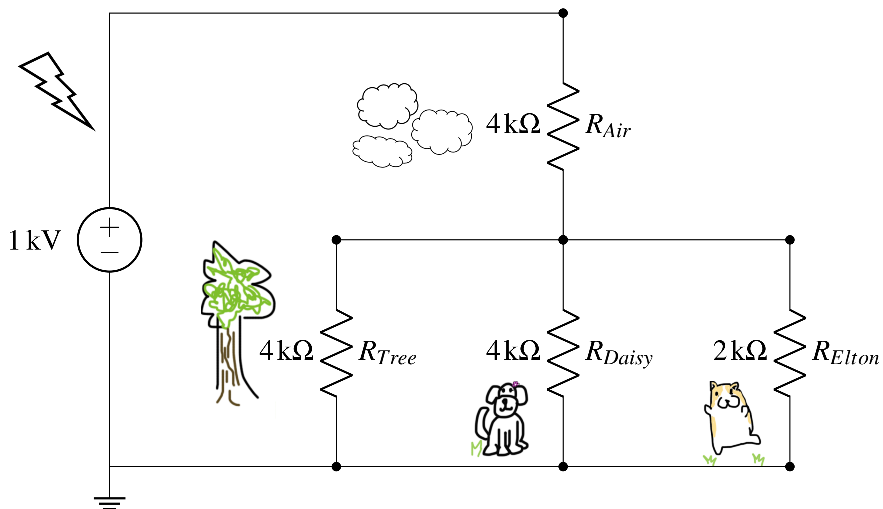


Figure 6: Circuit for Part (b)

7. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

EECS 16A Designing Information Devices and Systems I
Homework 8

Submission Format
Your homework submission should consist of **one** file.

- `hw8.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned).

Submit the file to the appropriate assignment on Gradescope.

1. **Midsemester Survey**

   Please fill out the midsemester survey. The link is given below:

   `https://docs.google.com/forms/d/e/1FAIpQLSchuTtDXMXqy-uwZdmRwh_3a8UbTjViyfXDILSv7g1PWw_pTg/viewform`

   Attach a screenshot of the confirmation page to your submission of this homework.

2. **Reading Assignment**

   For this homework, please read Note 15: Section 15.3 to learn about superposition, a concept that can help to simplify circuit analysis. Also read Note 15: Sections 15.4-15.8, which will explain the ideas of finding the equivalent resistance and Thévenin and Norton equivalent circuits. Please also reread Note 14 to review the idea of the 2D touchscreen. You are always encouraged to read beyond this as well.

   (a) As a part of superposition, you need to zero out independent sources. What circuit elements are equivalent to a zeroed voltage source and zeroed current source, respectively?

   (b) If you connect three resistors (each with value R) in series, what will be the equivalent resistance? What happens if you connect these resistors in parallel?

   (c) What is the difference between the Thévenin and Norton equivalent circuits? In particular, which part of the I–V curve does each equivalent circuit correspond to?

3. **Measuring Voltage and Current**

   (Contributors: Ava Tan, Aviral Pandey, Craig Schindler, Taehwan Kim, Urmita Sikder, Vijay Govindarajan, Wahid Rahman)

   Learning Goal: *The objective of this problem is to provide a deeper understanding of current and voltage measurement processes. It will also help you to understand how the electrical parameters of a measurement tool can affect the measurement precision.*

   In order to measure quantities such as voltage and current, engineers use voltmeters and ammeters. A simple model of a voltmeter is a resistor with a very high resistance, R_{VM}. **The voltmeter measures the voltage across the resistance** R_{VM}. The measured voltage is then relayed to a microprocessor (such as the MSP430 microprocessor, which will be used in lab).

This model of a voltmeter is shown in Figure 1. Let us explore what happens when we connect this voltmeter to various circuits to measure voltages.
Throughout this problem assume $R_{\text{VM}} = 1\text{M}\Omega$. Recall that the SI prefix M or Mega is 10^6.

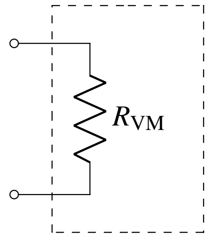
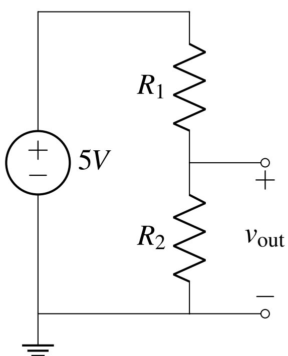


Figure 1: Our model of a voltmeter, $R_{\text{VM}} = 1\text{M}\Omega$

(a) Suppose we wanted to measure the voltage across R_2 (v_{out}) produced by the voltage divider circuit shown in Figure 2 on the left. The circuit on the right in Figure 2 shows how we would connect the voltmeter across R_2. Assume $R_1 = 100\Omega$ and $R_2 = 200\Omega$.
First calculate the value of v_{out}. Then calculate the voltage the voltmeter would measure, i.e. v_{meas}.



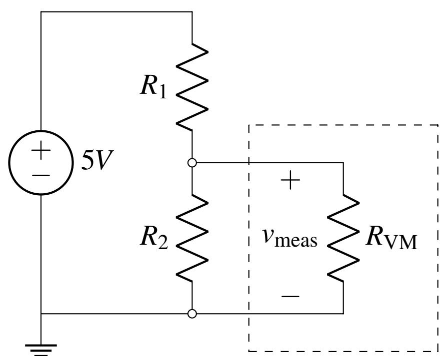


Figure 2: *Left:* Circuit without the voltmeter connected. *Right:* Voltmeter measuring voltage across R_2.

(b) Repeat part (a), but now $R_1 = 10M\Omega$ and $R_2 = 10M\Omega$. Is this particular voltmeter still a good tool to measure the output voltage? Justify why or why not. (Notice that a *good* voltmeter should not significantly affect the value of voltages in a circuit by its presence.)

(c) Now suppose we are working with the same circuit as in part (a), but we know that $R_2 = R_1$. What is the maximum value of R_1 such that $v_{out} - v_{meas} \leq 0.1 \cdot v_{out}$ (i.e. v_{meas} is only 10% smaller than v_{out})?

(d) We can make **an ammeter** and measure the current through an element, using the combination of our voltmeter and an additional resistor R_x. The circuit shown in Figure 3 encompassed by the dashed box can work as an ammeter, where $R_x = 1\,\Omega$. We insert it in the circuit so that the current we want to measure flows through R_x, and then measure the current as $I_{\text{meas}} = \frac{V_{\text{VM}}}{R_x}$ where V_{VM} is the voltage across the voltmeter. $R_{VM} = 1\text{M}\Omega$ is the same as in previous parts.

In Figure 4, the voltmeter-resistor combo is connected so that the current we want to measure flows through resistor $R_1 = 1\text{k}\Omega$. For the circuit on the left, find the current through R_1 without the voltmeter-resistor combo connected (i.e. I_1). Then, for the circuit on the right, find the current measured by the voltmeter-resistor combo when it is connected as an ammeter (i.e. I_{meas}).

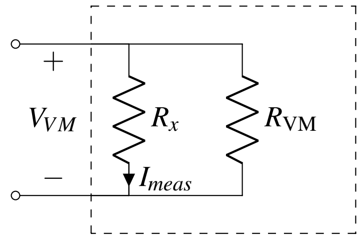


Figure 3: The voltmeter combined with resistor R_x to function as an ammeter (i.e. to measure current), $R_{\text{VM}} = 1M\Omega$.

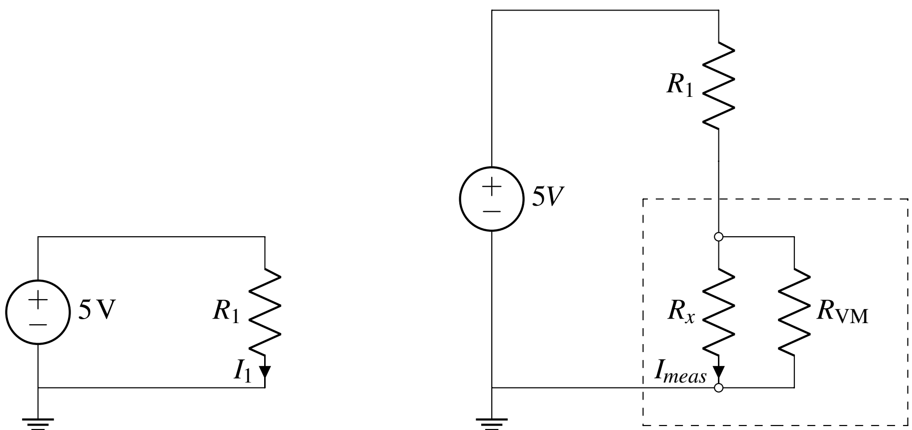


Figure 4: Circuits for part (d). *Left:* Original circuit; *Right:* Circuit with the voltmeter connected as an ammeter.

(e) **(PRACTICE / NOT GRADED)** What is the minimum value of R_1 that ensures the difference between current measurement (I_{meas}) and the the actual value (I_1) is such that $I_1 - I_{\text{meas}} \leq 0.1 \cdot I_1$, i.e. stays within $\pm 10\%$ of I_1? In other words, find the minimum allowable value for R_1 such that $I_1 - I_{\text{meas}} \leq 0.1 \cdot I_1$. **Hint**: You can approximate $R_{\text{VM}}||R_x \approx R_x$ and $\frac{R_{\text{VM}}||R_x}{R_x} \approx 1$.

4. Digital to Analog Converter (DAC)

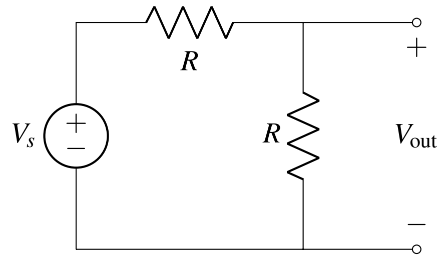
For some outputs, such as audio applications, we need to produce an analog output, or a continuous voltage from 0 to V_s. These analog voltages must be produced from digital voltages, that is sources, that can only be V_s or 0. A circuit that does this is known as a Digital to Analog Converter. It takes a binary representation of a number and turns it into an analog voltage.

The output of a DAC can be represented with the equation shown below:

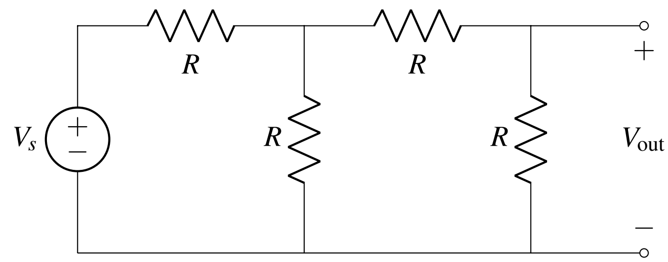
$$V_{\text{out}} = V_s \sum_{n=0}^{N} \frac{1}{2^n} \cdot b_n$$

where each binary digit b_n is multiplied by $\frac{1}{2^n}$.

(a) We know how to take an input voltage and divide it by 2:



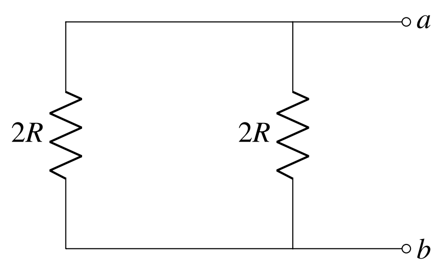
To divide by larger powers of two, we might hope to just "cascade" the above voltage divider. For example, consider:



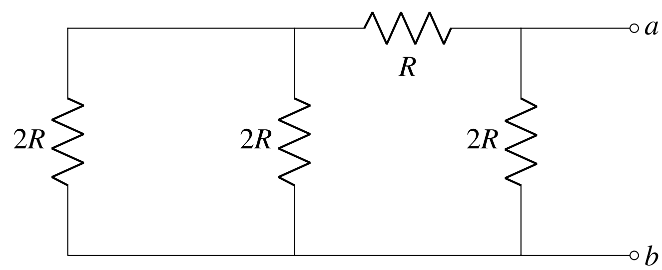
Calculate V_{out} in the above circut. Is $V_{\text{out}} = \frac{1}{4}V_s$?

(b) The R-$2R$ ladder, shown below, has a very nice property. For each of the circuits shown below, find the equivalent resistance looking in from points a and b. Do you see a pattern?

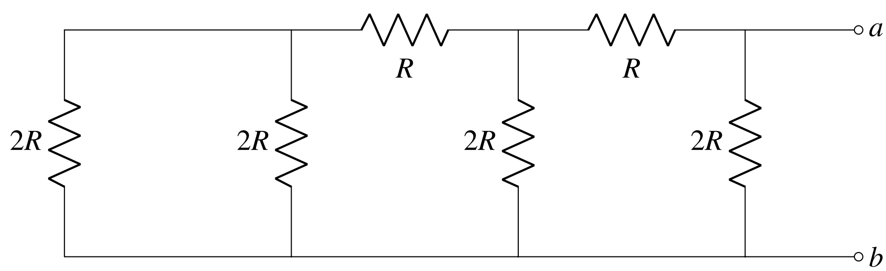
i.



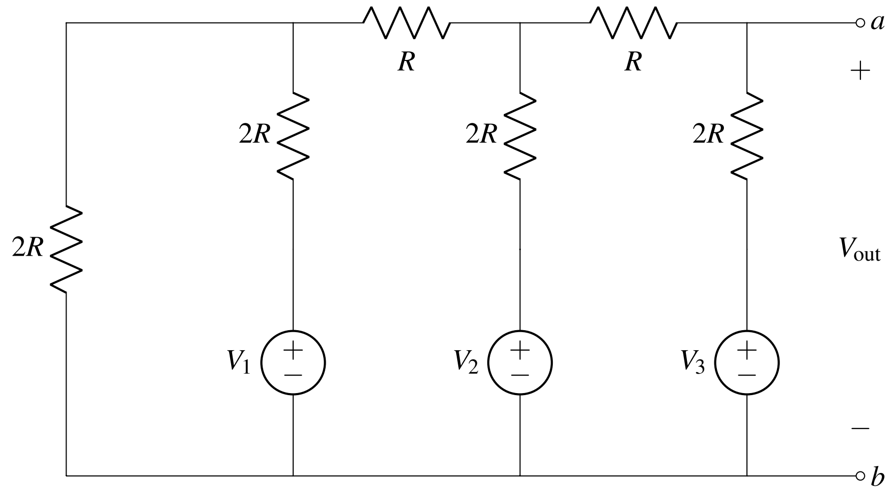
ii.



iii.



(c) The following circuit is an R-$2R$ DAC. To understand its functionality, use superposition to find V_{out} in terms of each V_k in the circuit.



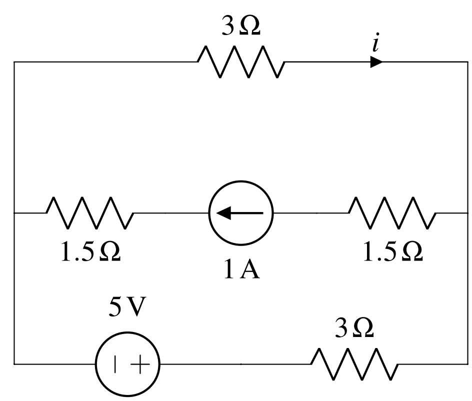
(d) We've now designed a 3-bit R-$2R$ DAC. What is the output voltage V_{out} if $V_2 = 1\,\text{V}$ and $V_1 = V_3 = 0\,\text{V}$?

5. Superposition

(Contributors: Avi Pandey, Gireeja Ranade, Karina Chang, Raghav Anand, Panos Zarkos, Sashank Krishnamurthy, Titan Yuan)

Learning Goal: *The objective of this problem is to help you practice solving circuits using the principles of superposition.*

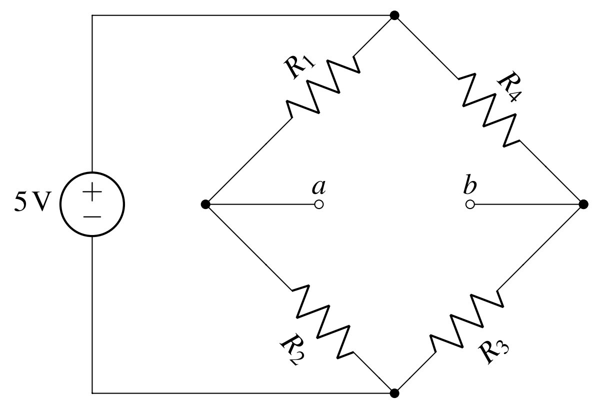
Find the current i indicated in the circuit diagram below using superposition.



6. Wheatstone Bridge

(Contributors: Ava Tan, Panos Zarkos)

A Wheatstone Bridge is a very useful circuit that can be used to help determine unknown resistance values with very high accuracy. This circuit is used in many sensor applications where resistors $R_1 - R_4$ are varying with respect to some external actuation. For example, it can be used to build a strain gauge (`https://en.wikipedia.org/wiki/Strain_gauge`) or a scale (remember Fred's Fruits from HW 7?). In that case the resistors $R_1 - R_4$ would vary with respect to a strain caused by a force, and the Wheatstone Bridge circuit would translate that variation into a voltage difference across the "bridge" terminals a and b.



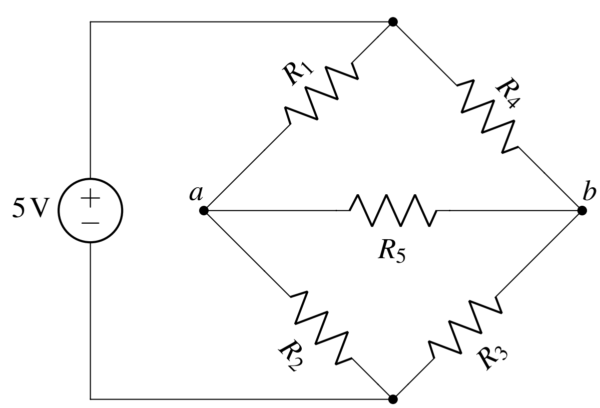
(a) Assume that $R_1 = 2\,\text{k}\Omega$, $R_2 = 2\,\text{k}\Omega$, $R_3 = 1\,\text{k}\Omega$, $R_4 = 4\,\text{k}\Omega$. Calculate the voltage V_{ab} between the two terminals a and b.

Hint: You have analyzed very similar circuits in previous homeworks, in lectures, and in discussions – it may help to redraw the circuit with each branch containing resistors $R_1 - R_4$ using straight vertical wires, rather than diagonal ones.

(b) Now assume that you have added an additional resistor, R_5, between terminals a and b as shown below. Assume that $R_1 = 1\,\text{k}\Omega$, $R_2 = 2\,\text{k}\Omega$, $R_3 = 4\,\text{k}\Omega$, $R_4 =?\ \text{k}\Omega$, $R_5 = 4\,\text{k}\Omega$. In the process of constructing this circuit, you notice that you forgot to write down the value of R_4!

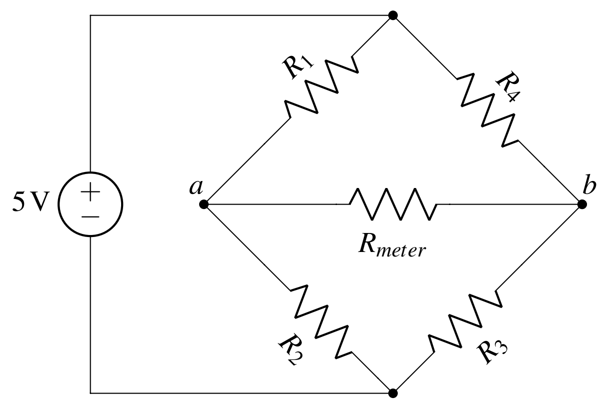
However, you notice something very curious about your Wheatstone Bridge circuit: there is no current flowing through resistor R_5.

Based on this observation, what must the value of resistor R_4 be?



(c) Next, draw the Thévenin equivalent of the Wheatstone bridge circuit between terminals a and b, using resistor values from part (a).

(d) Now assume that you are trying to measure the voltage V_{ab} between nodes a and b using a voltmeter, whose resistance is R_{meter}, so you end up with the circuit below.

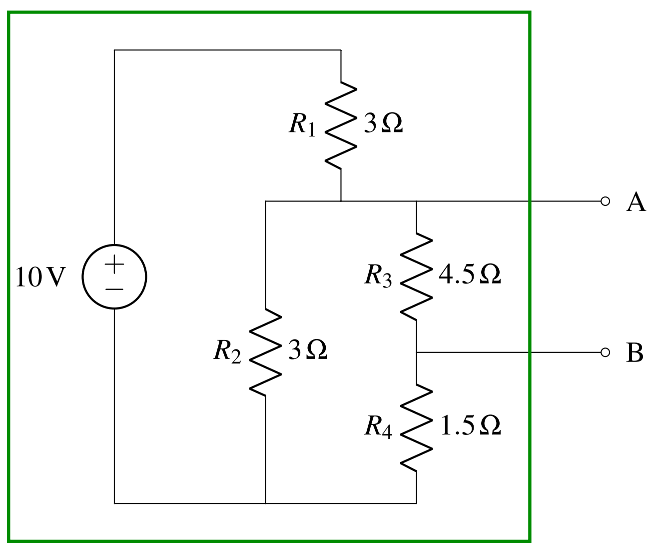


Unfortunately, your voltmeter is far from ideal, so $R_{meter} = 4k\Omega$. Is the voltage V_{ab} you found in part (a) equal to the new voltage $V_{R_{meter}}$ across the voltmeter resistor? Why or why not? Calculate the current $I_{R_{meter}}$ through the voltmeter resistor and the voltage $V_{R_{meter}}$ across the meter resistor. Again, use resistor values from part (a) and $R_{meter} = 4k\Omega$.

Hint: Consider the equivalent circuit you found in part (c).

7. Thévenin and Norton Equivalent Circuits

Find the Thévenin and Norton equivalent circuits seen from terminals A and B.



8. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

EECS 16A Designing Information Devices and Systems I
Homework 9

Submission Format
Your homework submission should consist of **one** file.

- `hw9.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned).

Submit the file to the appropriate assignment on Gradescope.

1. **Reading Assignment**

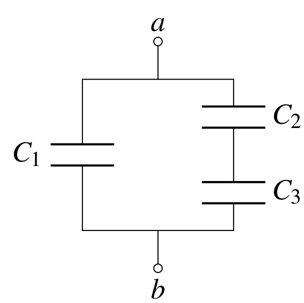
For this homework, please read Notes 16 and 17. Note 16 will provide an introduction to capacitors (a circuit element which stores charge), capacitive equivalence, and the underlying physics behind them. Sections 17.1 - 17.2 in Note 17 will provide an overview of the capacitive touchscreen and how to measure capacitance. 17.3 will introduce you to the op-amp and how it can be used as a comparator. 17.4 will discuss implementing a capacitive touchscreen with an op-amp comparator.

(a) Describe the key ideas behind how a capacitor works. How are capacitor equivalences calculated? Contrast this with how we calculate resistor equivalences.

(b) Consider the capacitive touchscreen. Describe how it works, and compare and contrast it to the resistive touchscreens we have seen in previous lectures and homeworks.

2. **Equivalent Capacitance**

(a) Find the equivalent capacitance between terminals a and b of the following circuit in terms of the given capacitors C_1, C_2, and C_3. Leave your answer in terms of the addition, subtraction, multiplication, and division operators **only**.



(b) Find and draw a capacitive circuit using three capacitors, C_1, C_2, and C_3, that has equivalent capacitance of

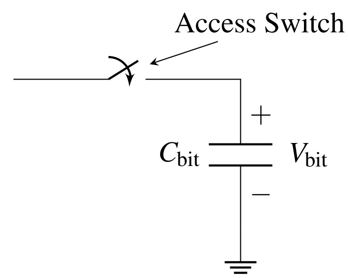
$$\frac{C_1(C_2+C_3)}{C_1+C_2+C_3}$$

3. Dynamic Random Access Memory (DRAM)

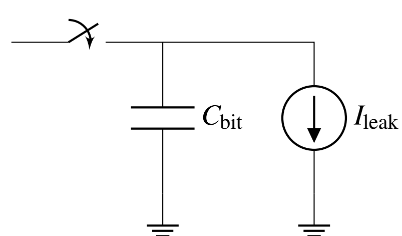
Nearly all devices that include some form of computational capability (phones, tablets, gaming consoles, laptops, ...) use a type of memory known as Dynamic Random Access Memory (DRAM). DRAM is where the "working set" of instructions and data for a processor is typically stored, and the ability to pack an ever increasing number of bits on to a DRAM chip at low cost has been critical to the continued growth in computational capability of our systems. For example, a single DRAM chip today can store > 8 billion bits and is sold for $\approx$ \$3-\$5.

At the most basic level and as shown below, every bit of information that DRAM can store is associated with a capacitor. The amount of charge stored on that capacitor (and correspondingly, the voltage across the capacitor) determines whether a "1" or a "0" is stored in that location.

Single DRAM Bit Cell



In any real capacitor, there is always a path for charge to "leak" off the capacitor and cause it to eventually discharge. In DRAMs, the dominant path for this leakage to happen is through the access switch, which we will **model as a leakage to ground**. The figure below shows **a model of this leakage**:



Fun Fact: This leakage is actually responsible for the "D" in "DRAM" – the memory is "dynamic" because after a cell is "written" by storing some charge onto its capacitor, if you leave the cell alone for too long, the value you wrote in will disappear because the charge on the capacitor leaked away.

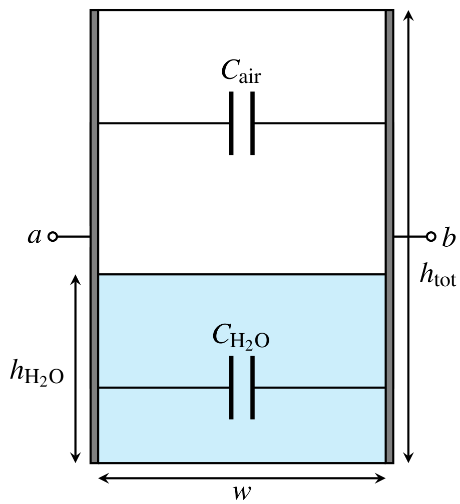
Let's now try to use some representative numbers to compute how long a DRAM cell can hold its value before the information leaks away. Let $C_{\text{bit}} = 28\,\text{fF}$ (note that $1\,\text{fF} = 1 \times 10^{-15}\,\text{F}$) and the capacitor be initially charged to $1.2\,\text{V}$ to store a "1." V_{bit} must be $> 0.9\,\text{V}$ in order for the circuits outside of the column to properly read the bit stored in the cell as a "1."

What is the maximum value of I_{leak} that would allow the DRAM cell retain its value for $> 1\,\text{ms}$?

Hint: Start by writing out the equations you know about charge and current related to the capacitor. Note here that the current source is discharging the capacitor.

4. It's finally raining!

A lettuce farmer in Salinas Valley has grown tired of weather.com's imprecise rain measurements. Therefore, they decided to take matters into their own hands by building a rain sensor. They placed a rectangular tank outside and attached two metal plates to two opposite sides in an effort to make a capacitor whose capacitance varies with the amount of water inside.

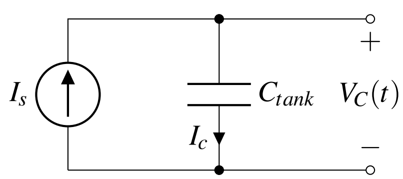


The width and length of the tank are both w (i.e., the base is square) and the height of the tank is h_{tot}.

(a) What is the capacitance between terminals a and b when the tank is full? What about when it is empty? *Note:* the permittivity of air is ε, and the permittivity of rainwater is 81ε.

(b) Suppose the height of the water in the tank is $h_{\text{H}_2\text{O}}$. Modeling the tank as a pair of capacitors in parallel, find the total capacitance between the two plates. Call this capacitance C_{tank}.

(c) After building this capacitor, the farmer consults the internet to assist them with a capacitance-measuring circuit. A fellow internet user recommends the following:



$, V_C(0) = 0\,\text{V}$

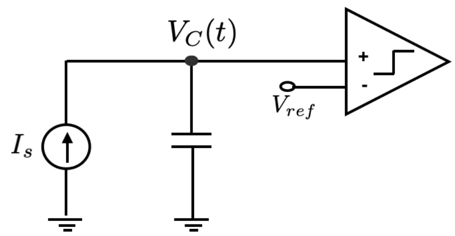
In this circuit, C_{tank} is the total tank capacitance that you calculated earlier. I_s is a known current supplied by a current source.

The suggestion is to measure V_C for a brief interval of time, and then use the difference to determine C_{tank}.

Determine $V_C(t)$, where t is the number of seconds elapsed since the start of the measurement. You should assume that before any measurements are taken, the voltage across C_{tank}, i.e. V_C, is initialized to $0\,\text{V}$, i.e. $V_C(0) = 0$.

(d) Using the equation you derived for $V_C(t)$, describe how you can use this circuit to determine C_{tank} and $h_{\text{H}_2\text{O}}$.

(e) However, after spending some time thinking about different ways of measuring this capacitance you came up with a better idea. You decided to use the circuit proposed in part (c) along with a comparator, as show in the figure below. What you are basically interested in, is the time T_1 needed for V_c to reach V_{ref}. In order to measure time you use a timer. When voltage V_c becomes larger than V_{ref}, the comparator flips its value and you stop the timer. How would you measure in that case the value of the capacitance?



5. Capacitive Touchscreen

(Contributors: Deepshika Dhanasekar, Panos Zarkos, Richard Liou, Wahid Rahman, Urmita Sikder)

The model for a capacitive touchscreen can be seen in Figure 1. See Table 1 for values of the dimensions. The green area represents the contact area of the finger with the top insulator. It has dimensions w_2 x d_1, where w_2 is the horizontal width of the finger contact area and d_1 is the depth (into the page) of the finger contact area. The top metal (red area) has dimensions $w_1 \times d_1$. The bottom metal (grey area) has dimensions $w \times d_2$, where w is larger than both w_1 and w_2.

Table 1: Touchscreen Dimension Values

d_1	10 mm
d_2	1 mm
t_1	2 mm
t_2	4 mm
w_1	1 mm
w_2	2 mm

(a) Draw the equivalent circuit of the touchscreen that contains the nodes F, E_1, and E_2 when: (i) there no finger present; and (ii) when there is a finger present. Express the capacitance values in terms of C_0, C_{F-E1}, and C_{F-E2}.

Hint: Note that node F represents the finger. When there is no touch node F would be non-existent.
Hint: Treat E_1 as the "top node", E_2 as the "bottom node", and the finger F as an intermediate node when present.

(b) What are the values of C_0, C_{E1-F}, and C_{F-E2}? Assume that the insulating material has a permittivity of $\varepsilon = 4.43 \times 10^{-11} F/m$ and that the thickness of the metal layers is small compared to t_1 (so you can ignore the thickness of the metal layers).

(c) What is the difference in effective capacitance between the two metal plates (nodes E_1 and E_2) when a finger is present?

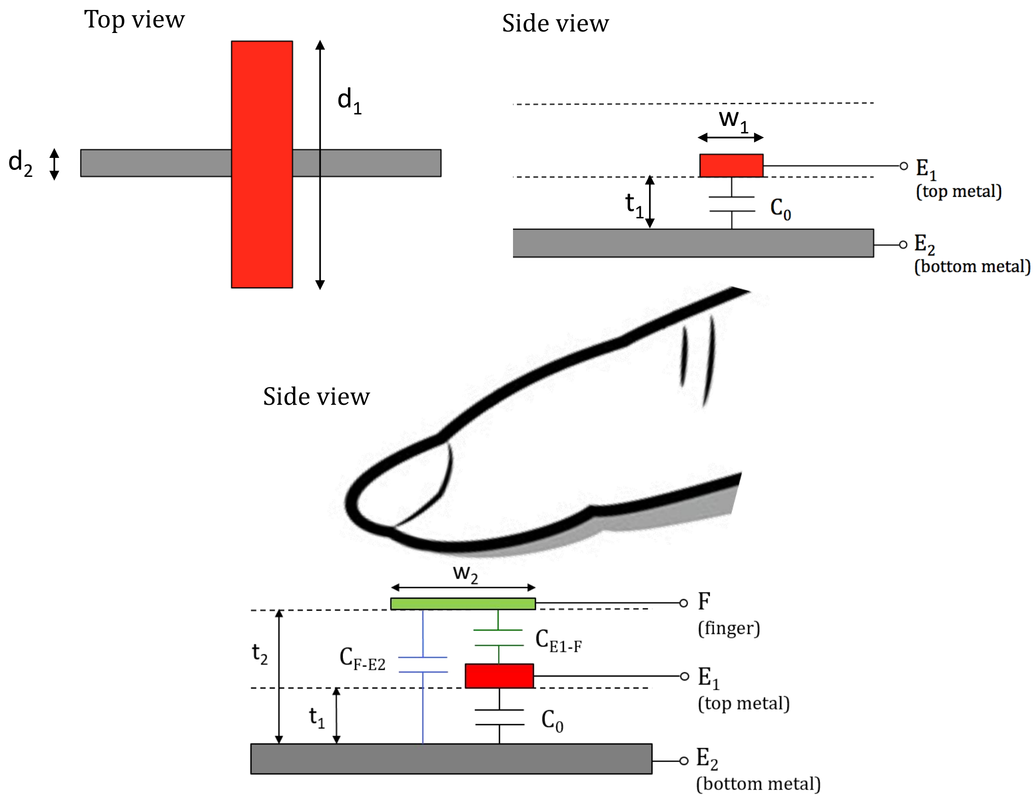


Figure 1: Model of capacitive touchscreen.

6. Super-Capacitors

In order to enable small devices for the "Internet of Things" (IoT), many companies and researchers are currently exploring alternative means of storing and delivering electrical power to the electronics within these devices. One example of these are "super-capacitors" - the devices generally behave just like a "normal" capacitor but have been engineered to have extremely high values of capacitance relative to other devices that fit in to the same physical volume. They can function as a power supply for low power applications such as IoT devices and have the advantage that they can be charged and discharged many times without losing maximum charge capacity. That property makes super-capacitors suitable to store energy from intermittent power sources such as those from energy harvesting. Suppose you are tasked with designing a power supply with a super-capacitor in an IoT device.

(a) Assuming that your electronic device (load) can be modeled as a constant current source with a value of i_{load}, draw circuit models for your device using super-capacitors as the power supply with the following configurations:

- Config 1: a single super-capacitor as the power supply
- Config 2: two super-capacitors stacked in series as the power supply
- Config 3: two super-capacitors connected in parallel as the power supply

(b) If each super-capacitor is charged to an initial voltage v_{init} and has a capacitance of C_{sc}, for each of the three configurations above, write an expression for the voltage supplied to your electronic device as a function of time after the device has been activated (i.e. connected to the super-capacitor(s)).

(c) Now let's assume that your electronic device requires some minimum voltage v_{min} in order to function properly. For each of the three super-capacitor configurations, write an expression for the lifetime of the device.

(d) Assume that a single super-capacitor doesn't provide you sufficient lifetime and so you have to spend the extra money (and device volume) for another super-capacitor. You consider the two following configurations:

- Config 2: two super-capacitors stacked in series
- Config 3: two super-capacitors connected in parallel

When is Config 3 (parallel) better than Config 2 (series)? Your answer should involve conditions on v_{init} and v_{min}.

(e) Calculate the amount of energy delivered by the super-capacitors in Config 3 (parallel) over the device's lifetime.

7. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

EECS 16A Designing Information Devices and Systems I
Homework 10

Submission Format
Your homework submission should consist of **one** file.

- `hw10.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned)

Submit each file to its respective assignment on Gradescope.

1. Reading Assignment

For this homework, please read Notes 18.1-18.6 and Notes 19.1-19.4, as well as Note 19.7. They will provide an overview of operational amplifiers (op-amps), negative feedback, the "golden rules" of op-amps, and some op-amp configurations (inverting and buffers). You are always encouraged to read beyond this as well.

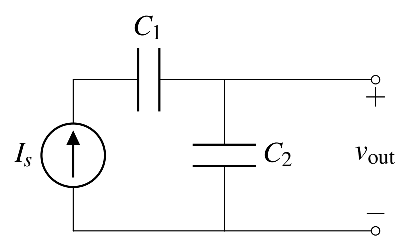
(a) What are the two "golden rules" of ideal op amps? When do these rules hold true?

(b) What is the internal gain of an op-amp, A? What is its value for an ideal op-amp? For non-ideal?

2. More Current Sources And Capacitors

(Contributors: Ava Tan, Panos Zarkos, Wahid Rahman)

(a) For the circuit given below, give an expression for $v_{\text{out}}(t)$ in terms of I_s, C_1, C_2, and time t. Assume that all capacitors are initially uncharged, i.e. the initial voltage across each capacitor is $0\,\text{V}$.

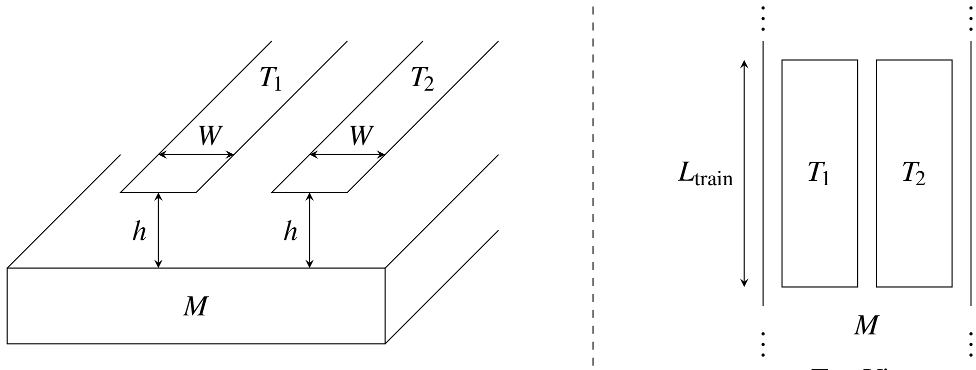


3. Maglev Train Height Control System

(Contributors: Laura Brink, Avi Pandey, Vijay Govindarajan, Titan Yuan, Lam Nguyen, Urmita Sikder)

One of the fastest forms of land transportation are trains that actually travel slightly elevated from the ground using magnetic levitation (or "maglev" for short). Ensuring that the train stays at a relatively constant height above its "tracks" (the tracks in this case are what provide the force to levitate the train and propel it forward) is critical to both the safety and fuel efficiency of the train. In this problem, we'll explore how maglev trains could use capacitors to ensure correct elevation. (Note that real maglev trains may use completely different and much more sophisticated techniques to perform this function, so if you get a contract to build such a train, you'll probably want to do more research on the subject.)

(a) As shown below, we put two parallel strips of metal (T_1, T_2) along the bottom of the train and we have one solid piece of metal (M) on the ground below the train (perhaps as part of the track).



Side View Top View

We assume that the entire train is at a uniform height above the track, so we can use the simple equations developed in lecture to model the capacitance.

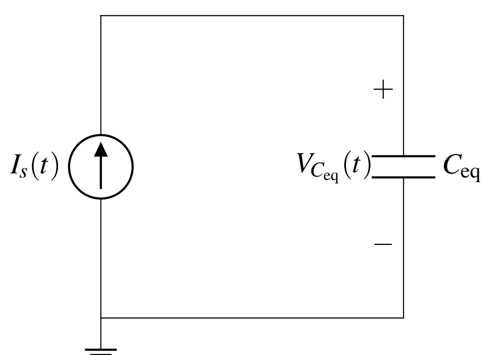
As a function of L_{train} (the length of the train), W (the width of T_1 and T_2), and h (the height of the train above the track), determine the capacitances between T_1 and M and between T_2 and M. *Hint: Note that the area of a capacitor is given by the overlap area between its two plates.*

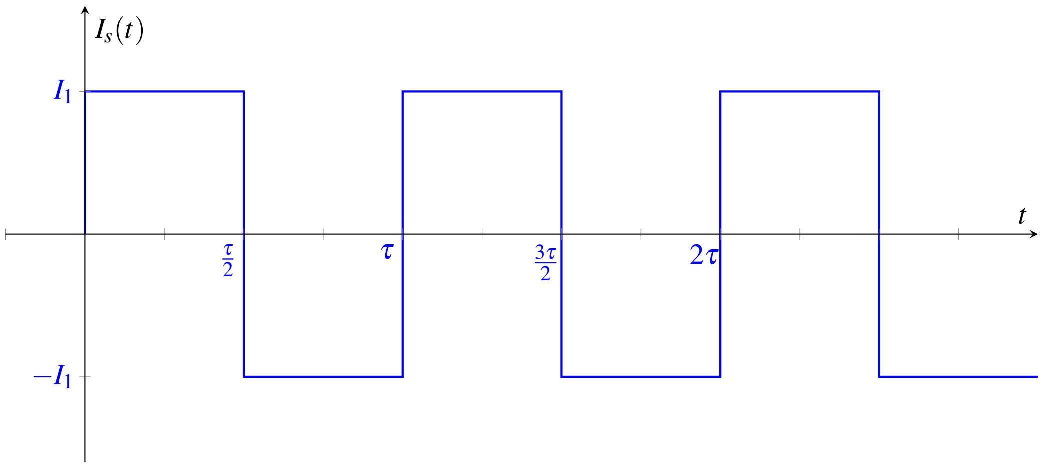
(b) Any circuit on the train can only make direct contact at T_1 and T_2. Thus, you can only measure the equivalent capacitance between T_1 and T_2. Draw a circuit model showing how the capacitors between T_1 and M and between T_2 and M are connected to each other.

(c) Using the same parameters as in part (a), provide an expression for the equivalent capacitance measured between T_1 and T_2.

(d) We want to build a circuit that creates a voltage waveform with an amplitude that changes based on the height of the train. Your colleague recommends you start with the circuit as shown below, where I_s is a periodic current source, and C_{eq} is the equivalent capacitance between T_1 and T_2. The graph below shows $I_s(t)$, a square wave with period τ and amplitude I_1, as a function of time t.

Find an equation for and draw the voltage $V_{C_{\text{eq}}}(t)$ as a function of time t. Assume the capacitor C_{eq} is discharged at time $t = 0$, so $V_{C_{\text{eq}}}(0) = 0\,\text{V}$.

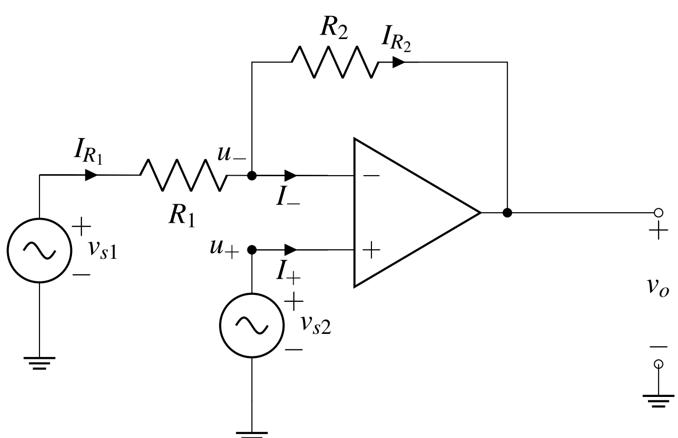




(e) We now want to develop an indicator that alerts us when the train is too high above the tracks. We want to have an output of 5 V (to trigger the alert) when the height of the train h is above 1 cm, and an output of 0 V when h is below 1 cm.
We will assume the train has length $L_{\text{train}} = 100\,\text{m}$ and that the metals, T_1 and T_2, have width $W = 1\,\text{cm}$, and the space between the train and the solid piece of metal M has permitivity $\varepsilon = 8.85 \times 10^{-12}\,\text{F}\,\text{m}^{-1}$.
Design a circuit using **a square wave current source (i.e. I_s in part (d)) with period $\tau = 1\,\mu\text{s}$ and pulses of amplitude $I_1 = 1\,\text{mA}$, a comparator, and any number of voltage sources** to implement this function. *Hint: You should use the circuit you analyzed in part (d).*

4. Amplifier with Multiple Inputs

Consider the circuit below, assuming the op-amp is ideal.



(a) Is this circuit in negative feedback? Why or why not?

(b) What is the current I_- flowing into the negative terminal of the op-amp?

(c) What is the current I_+ flowing into the positive terminal of the op-amp?

(d) What is u_-, the voltage at the negative terminal of the op-amp, in terms of v_o and v_{s1}?

(e) Prove the second golden rule, that $u_+ = u_-$, assuming the op-amp in the circuit above is ideal.
Hint: Recall that $v_o = A(u_+ - u_-)$, where A is the gain of the op-amp.
Hint: Try replacing $\left(\frac{R_2}{R_1+R_2}\right)$ with a constant f.

(f) What is u_- in terms of v_{s2}?

(g) What is I_{R_2}, the current through the resistor R_2, in terms of v_{s1}, v_{s2}, R_1, and R_2? *Hint:* Use KCL at node u_-.

5. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

EECS 16A Designing Information Devices and Systems I
Homework 11

Submission Format
Your homework submission should consist of **one** file.

- `hw11.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned).

Submit the file to the appropriate assignment on Gradescope.

1. **Reading Assignment**

   For this homework, please review Notes 18.5-18.6, Notes 19.5-19.6, and Note 20 (Op-Amp Current Source and Circuit Design).

   (a) Summarize the 4-step design process outlined in the notes, and describe why each step is useful in practice.

2. Op-Amp in Negative Feedback

(Contributors: Adhyyan Narang, Ava Tan, Aviral Pandey, Deepshika Dhanasekar, Lam Nguyen, Panos Zarkos, Titan Yuan, Vijay Govindarajan, Urmita Sikder, Sunash Sharma)

In this question, we analyze op-amp circuits in which the op-amp has a finite gain. Consider the circuit shown in Figure 1. Note that this circuit is in negative feeback. This circuit is known as a non-inverting amplifier circuit.

In parts (a) - (e), we will assume the op-amp in this circuit is ideal (i.e., gain is infinite), and then we will consider the case of finite gain in parts (f) - (h). For all parts, assume $V_{SS} = -V_{DD}$.

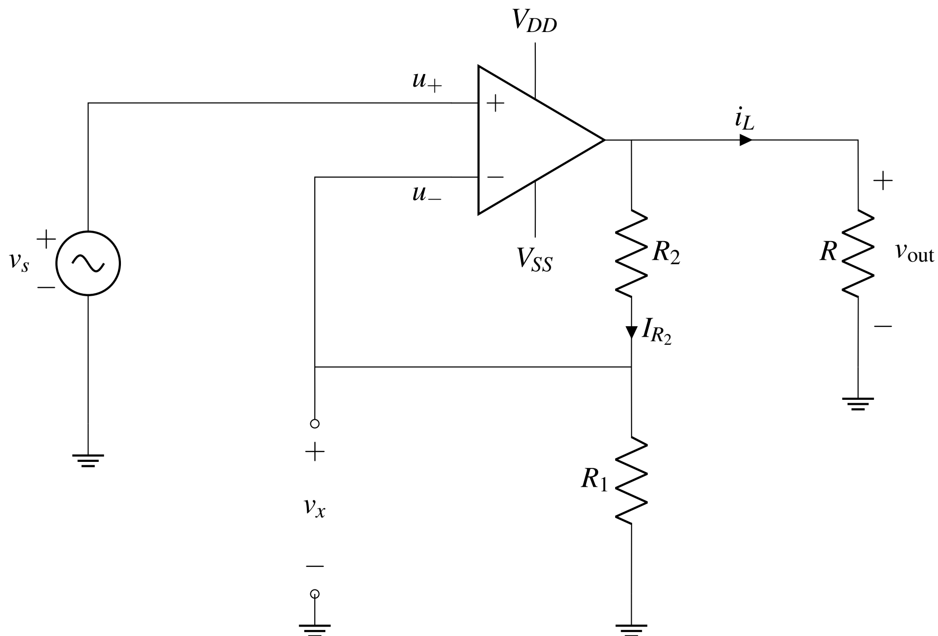


Figure 1: Non-inverting amplifier circuit

(a) Consider the circuit shown in Figure 1. What is $u_+ - u_-$?

(b) Find v_x as a function of v_{out}.

(c) What is I_{R_2}, i.e. the current flowing through R_2 as a function of v_s? *Hint: Find the current through R_1 first.*

(d) Find v_{out} as a function of v_s.

(e) What is the current i_L through the load resistor R? Give your answer in terms of v_{out}.

(f) We will now examine what happens when the op-amp gain is not ∞. To understand what happens in this case, we will consider the equivalent circuit for an op-amp with gain A, shown in Figure 2. We will use this equivalent circuit to analyze the non-inverting amplifier circuit in Figure 1.

First draw an equivalent circuit for Figure 1, **by replacing the ideal op-amp in the non-inverting amplifier in Figure 1 with the op-amp model shown in Figure 2.** You do not need to include V_{DD} and V_{SS} in your equivalent circuit.

Then, using this setup, calculate v_{out} and v_x in terms of A, v_s, R_1, R_2 and R. Assuming A is finite, is the magnitude of v_x larger or smaller than the magnitude of v_s? Do these values depend on R? *Hint: Note that the first golden rule still applies, i.e. the currents through the input terminals are zero.*

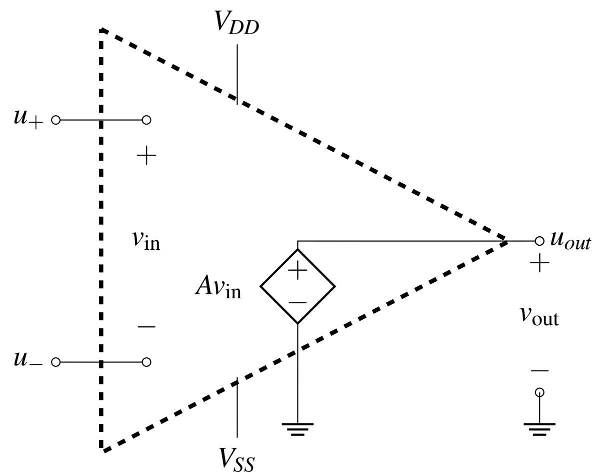


Figure 2: Internal op-amp model

(g) Using your solution to the previous part, calculate the limits of v_{out} and v_x as $A \to \infty$. You should get the same answer as in part (d) for v_{out}.

(h) **[OPTIONAL, CHALLENGE]** Now you want to make a non-inverting amplifier circuit whose gain is nominally $G_{nom} = \frac{v_{\text{out}}}{v_s} = 1 + \frac{R_2}{R_1} = 4$. However, G_{nom} can only be achieved only if the op-amp is ideal, i.e, if its internal gain $A \to \infty$. But, as with most considerations in the physical world, we must account for nonidealities! In reality, because you will be working with an op-amp with finite gain A, your designed circuit gain may come close to but will never quite reach G_{nom} as a result of the real op-amp's finite internal gain A.

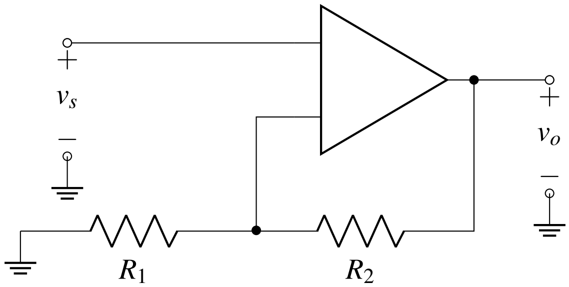
Suppose you would like your real op-amp circuit to have a minimum error of 1% (i.e, a minimum circuit gain of 3.96, i.e. $\frac{v_{\text{out}}}{v_s} \geq 3.96$). Remember that only if your op-amp were ideal, you would have a nominal circuit gain of $G_{nom} = \frac{v_{\text{out}}}{v_s} = 1 + \frac{R_2}{R_1} = 4$.

What is the minimum required value of A, called A_{min}, to achieve that specification? *Hint: Use your expression of v_{out} in part (f) to find an expression for $G_{min} = \frac{v_{\text{out}}}{v_s}$ when $A \not\to \infty$.*

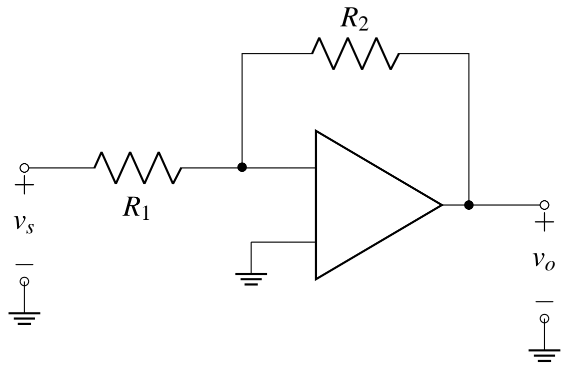
3. Basic Amplifier Building Blocks

(Contributors: Ava Tan, Aviral Pandey, Lam Nguyen, Michael Kellman, Titan Yuan, Vijay Govindarajan, Urmita Sikder)

The following amplifier stages are used often in many circuits and are well known as (a) the non-inverting amplifier and (b) the inverting amplifier.



(a)



(b)

(a) Label the input terminals of the op-amp with $(+)$ and $(-)$ signs in Figure (a), so that it is in negative feedback. Then derive the voltage gain ($G = \frac{v_o}{v_s}$) of the non-inverting amplifier in Figure (a) using the Golden Rules. Why do you think this circuit is called a non-inverting amplifier?

(b) Label the input terminals of the op-amp with $(+)$ and $(-)$ signs in Figure (b), so that it is in negative feedback. Then derive the voltage gain ($G = \frac{v_o}{v_s}$) of the inverting amplifier using the Golden Rules. Can you explain why this circuit is called an inverting amplifier?

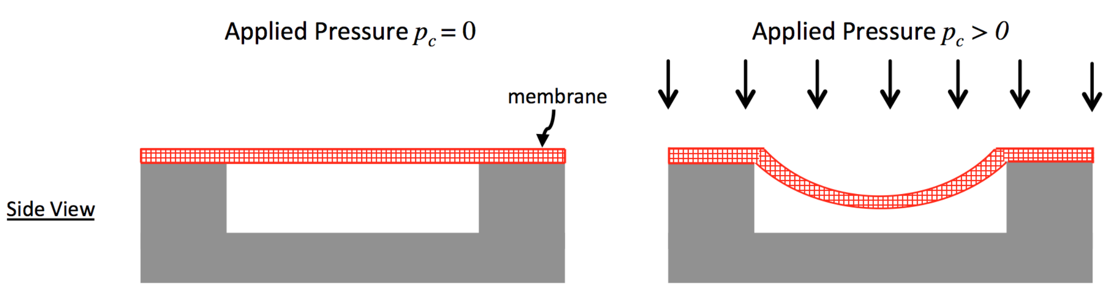
4. Putting on the Pressure: Build Your Own InstantPot

(Contributors: Craig Schindler, Ava Tan, Wahid Rahman, Urmita Sikder)

Prof. Ranade had a great experience with her automatic pressure cooker, so she was inspired to try and build her own. She's enlisting your help! The design of the pressure cooker uses a pressure sensor and a heating element. Whenever the pressure is below a set target value, an electronic circuit turns on the heating element.

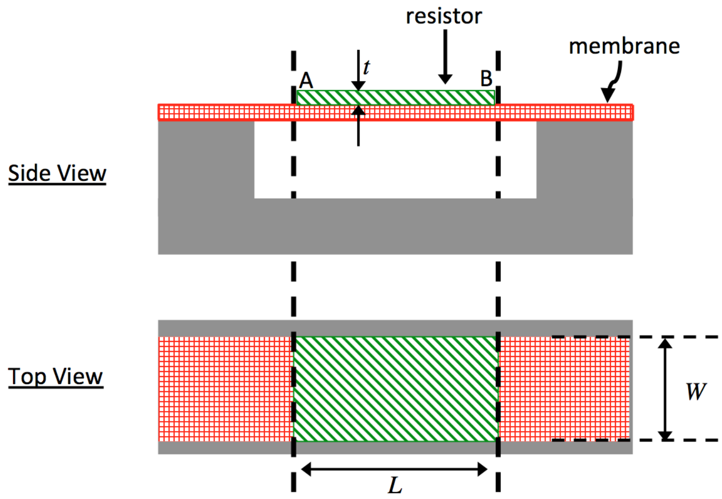
Pressure Sensor Resistance

The first step is designing a pressure sensor. The figure below shows your design. As pressure p_c is applied, the flexible membrane stretches.



(a) You attach a resistor layer R_p with resistivity $\rho = 0.1\Omega\text{m}$, width W, length L, and thickness t to the pressure sensor membrane, as illustrated in the figure below. When the pressure $p_c = 0\,\text{Pa}$ (i.e. there is no applied pressure), $W = 1\,\text{mm}$, $L = L_0 = 1\,\text{cm}$, $t = 100\,\mu\text{m} = 100 \times 10^{-6}$ m.

R_{p0} is the value of R_p when there is no applied pressure. Calculate R_{p0}. Note that direction of current flow in the resistor is from A to B as marked in the diagram.



(b) When pressure is applied, the length of the resistor L changes from L_0 and is a function of applied pressure p_c, and is given by

$$L = L_0 + \beta p_c,$$

where L_0 is the nominal length of the resistor with no pressure applied, and β is a constant with units m/Pa. As a result of the length change, the value of resistance R_p also changes from its nominal value R_{p0} (the value of R_p with no pressure applied).

Derive an expression for R_p as a function of resistivity ρ, width W, thickness t, nominal length L_0, constant β, and applied pressure p_c, when pressure is applied.

Note: The width and thickness of the resistor will also change with applied pressure. However, we ignore this to keep the math simple.

(c) **Pressure Sensor Circuit Design**

For this sub-part and the following sub-parts, we will use a new model for pressure-sensitive resistance R_p. Assume that the resistance R_p is a function of applied pressure p_c according to the relationship $R_p = R_o \times \frac{p_c}{p_{\text{ref}}}$, where $R_o = 1\text{k}\Omega$, and $p_{\text{ref}} = 100\text{kPa}$.

To complete our sensor circuit, we would like to generate a voltage V_p that is a function of the pressure p_c.

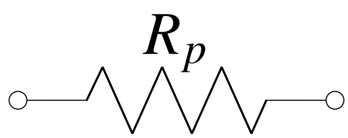
Complete the circuit below so that the output voltage V_p depends on the pressure p_c as:

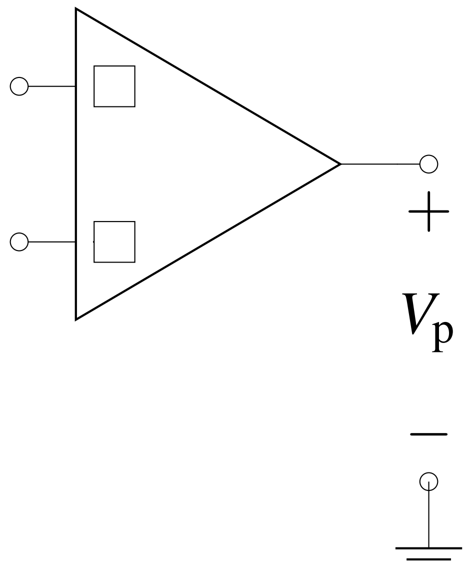
$$V_p = -V_o \times \frac{p_c}{p_{\text{ref}}}, \text{ where } V_o = 1\,\text{V}.$$

Restrictions on your pressure sensor circuit design are as follows:

- **You may add at most one ideal voltage source and one additional resistor to the circuit, but you must calculate their values and mark them in the diagram.**
- **Mark the positive and negative inputs of the operational amplifier with "+" and "-" symbols, respectively, in the boxes provided.**
- **Assume op-amp supply voltages V_{DD} and $V_{SS} = -V_{DD}$ are already provided.**

You may assume that the operational amplifier is ideal.





(d) **Resistive Heating Element**

To heat the pressure cooker, you use a heating element with resistance R_{heat}. Calculate the value of R_{heat} such that the power dissipated is $P_{\text{heat}} = 1000\,\text{W}$ with $V_{\text{heat}} = 100\,\text{V}$ applied across the heating element.

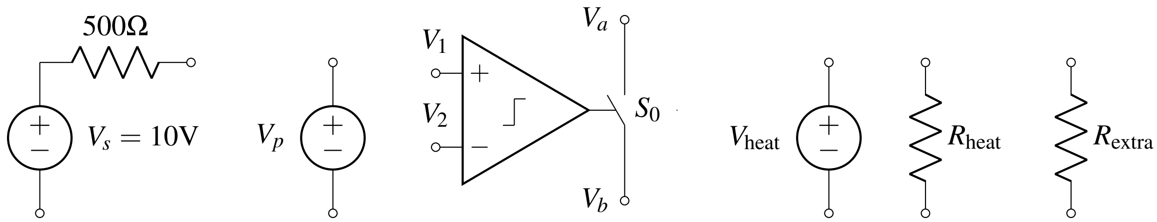
(e) **Pressure Regulation**

You are finally ready to complete the design of your pressure cooker.

Using all of the circuit elements below, make a circuit that will turn the heater on (i.e. will cause a current to flow through R_{heat}) when the pressure is less than 500 kPa, and off (i.e. will cause no current to flow through R_{heat}) when the pressure is greater than 500 kPa.

The elements are:

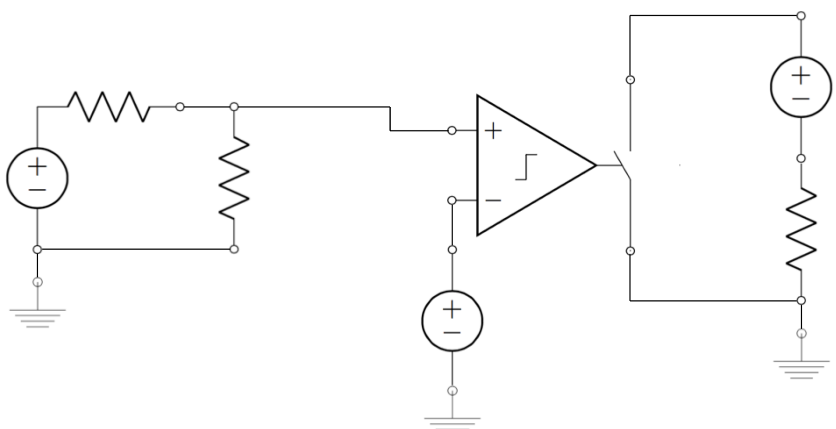
- A voltage source $V_s = 10\text{V}$ in series with a resistance of 500Ω.
- A voltage source $V_p = V_o \times \frac{p_c}{p_{\text{ref}}}$, with V_o = 1V and $p_{\text{ref}} = 100\text{kPa}$. (This is a voltage source whose voltage is a function of pressure p_c, unrelated to any previous parts of the question.)
- A comparator that controls switch S_0. The switch is normally opened (i.e. an open circuit between nodes V_a and V_b), and is closed only when $V_1 > V_2$ (i.e. a short circuit between nodes V_a and V_b).
- The heater supply ($V_{\text{heat}} = 100\text{V}$).
- The heater resistor R_{heat}.
- One additional resistor R_{extra} that can have **any value**.
- You may assume you have access to a ground node.
- Assume comparator supply voltages V_{DD} and $V_{SS} = -V_{DD}$ are already provided.



(i) Since you are looking to *compare* the change in voltage associated with a change in pressure, you decide to assign the **variable voltage** source V_p as one of the inputs to your comparator.
What is the value of the **variable voltage source** V_p for $p_c = 500$kPa?

(ii) For your comparator inputs, you also need to generate a **reference voltage** which can be used to compare against the value of the **variable voltage** source which you calculated above.
Combine the voltage source $V_s = 10$V with an associated resistance of 500Ω with an an additional resistor R_{extra} to **generate a reference voltage equal to the voltage V_p you calculated above.** What would you choose the value of R_{extra} to be?

(iii) Label the circuit elements in the schematic below with the **circuit elements presented above and your calculated R_{extra} value** to turn the heater on when the pressure is less than 500 kPa.



5. Island Karaoke Machine

(Contributors: Aviral Pandey, Lam Nguyen, Pangiotis Zarkos, Sashank Krishnamurthy, Titan Yuan, Urmita Sikder, Vijay Govindarajan)

Learning Goal: *The objective of this problem is design a circuit that calculates the difference between two signals and amplifies the result.*

You're stuck on a desert island and everyone is bored out of their minds. Fortunately, you have your EECS16A lab kit with op-amps, wires, resistors, and your handy breadboard. You decide to build a karaoke machine. You recover one speaker from the crash remains and use your iPhone as your source. You know that many songs put instruments on either the "left" or the "right" channel, but the vocals are usually present on both channels with equal strength.

In our case, the vocals are present on both left and right channels, but the instruments are only present on the right channel, i.e.

$$v_{\text{left}} = v_{\text{vocals}}$$
$$v_{\text{right}} = v_{\text{vocals}} + v_{\text{instrument}},$$

where the voltage source v_{vocals} can have values anywhere in the range of $\pm 120\,\text{mV}$ and $v_{\text{instrument}}$ can have values anywhere in the range of $\pm 50\,\text{mV}$.

What is the goal of a karaoke machine? **The ultimate goal is to *remove* the vocals from the audio output.** We're going to do this by first building a circuit that takes the left and right audio outputs of the smartphone and then calculates its **difference**. Let's see what happens.

The equivalent circuit model of the iPhone audio jack and speaker is shown in Figure 3. We model the **audio signals and jack** as v_{left} and v_{right} with **equivalent source resistance** of the left/right audio channels of $R_{\text{left}} = R_{\text{right}} = 3\,\Omega$. The **speaker** has an equivalent resistance of $R_{\text{speaker}} = 4\,\Omega$.

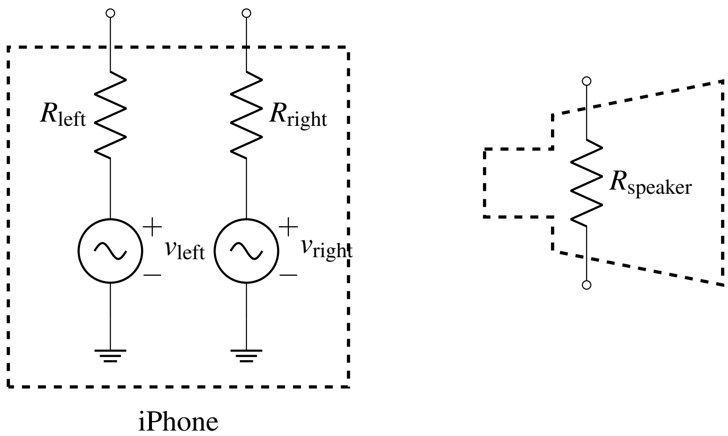


Figure 3: Audio jack and speaker of an iPhone.

(a) One of your island survivors suggests the circuit in Figure 4 to do this. **Find the expression for the voltage across the speaker R_{speaker} as a function of v_{vocals} and $v_{\text{instruments}}$.**
Does the voltage across the speaker depend on v_{vocals}? In other words, what do you think the islanders will hear – vocals, instruments, or both?

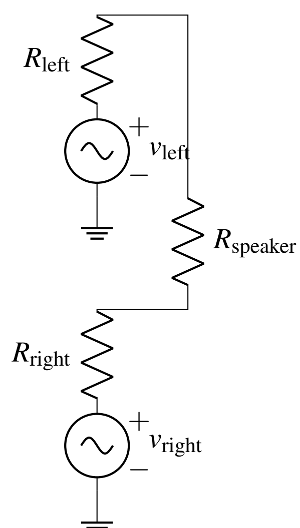


Figure 4: Circuit for part (a).

(b) We need to boost the sound level to get the party going. To this end, we want a range of $\pm 2\,\mathrm{V}$ across the speaker. Design a circuit by completing the Figure 5 below that takes in $\{v_{\text{left}}, R_{\text{left}}\}$ and $\{v_{\text{right}}, R_{\text{right}}\}$ combos as inputs and outputs an **amplified version of $v_{\text{instrument}}$ across R_{speaker}**. Consider all op-amps to be **ideal** for this problem.

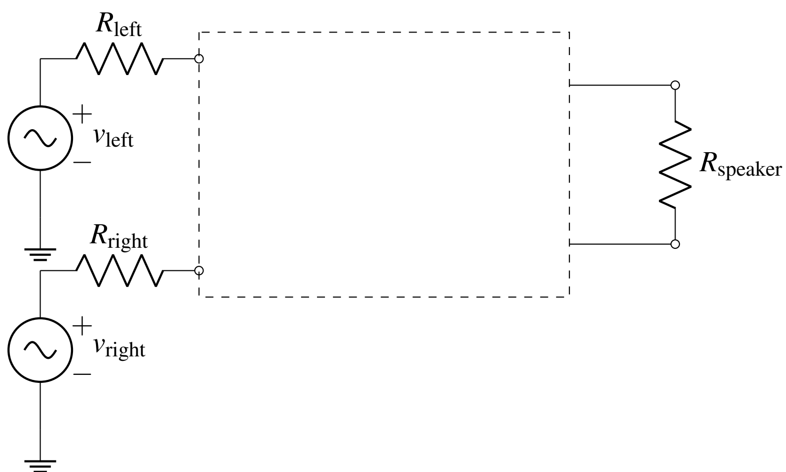


Figure 5: Circuit for part (b).

Hint 1: *We need to get an output voltage with the range of* $\pm 2\,\mathrm{V}$*. The input voltage* $v_{instrument}$ *can have values anywhere in the range of* $\pm 50\,\mathrm{mV}$*. What gain is needed from the op-amp based amplifier circuits?*

Hint 2: *Use two op-amps in the non-inverting configuration. The non-ideal voltage source* $\{v_{left}, R_{left}\}$ *must be the input to one non-inverting amplifier and the non-ideal voltage source* $\{v_{right}, R_{right}\}$ *must be the input to the other non-inverting amplifier.*

Hint 3: *Connect the speaker* $R_{speaker}$ *across the outputs of those two op-amps.*

(c) The trouble with the approach in part (b) is that two op-amps are required. Let's say you only have **one op-amp** with you. What would you do? One night in your dreams, you have an inspiration. Why not combine an inverting and non-inverting amplifier into one, as shown in Figure 6!

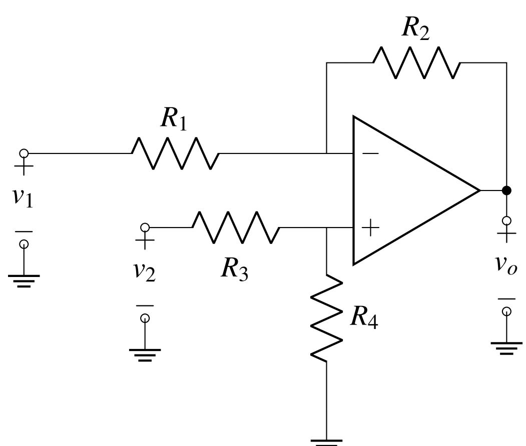


Figure 6: The new amplifier for part (c).

If we set $v_2 = 0\,\text{V}$, what is the output v_o in terms of v_1? (This is the inverting path.)

(d) Consider the circuit in Figure 6 again. If we set $v_1 = 0\,\text{V}$, what is the output v_o in terms of v_2? (This is the non-inverting path.)

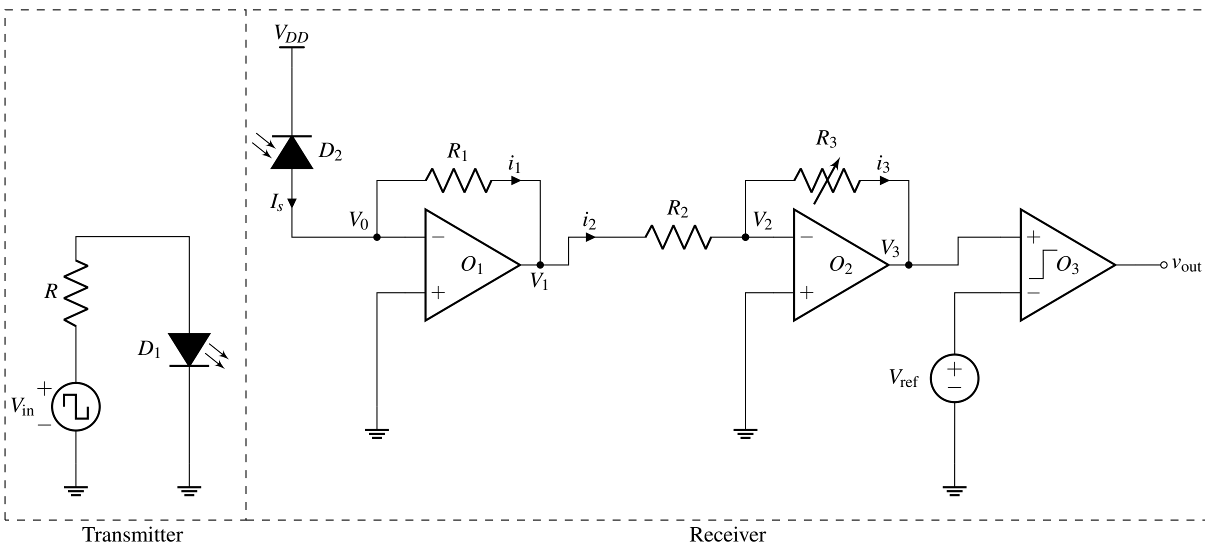
(e) Now, determine v_o in terms of v_1 and v_2 by superposing your results from the last two parts. Choose values for R_1, R_2, R_3 and R_4, such that the speaker has $v_o = \pm 2\,\text{V}$ across it for $v_2 - v_1 = \pm 50\,\text{mV}$.

6. Wireless Communication With An LED

(Contributors: Aviral Pandey, Lam Nguyen, Pangiotis Zarkos, Sashank Krishnamurthy, Titan Yuan, Urmita Sikder)

Learning Goal: *This problem is designed to find the response of a complex circuit by breaking it into smaller circuit blocks and analyzing each block individually.*

In this question, we are going to analyze the system shown in the figure below. It shows a circuit that can be used as a wireless communication system using visible light (or infrared, very similar to remote controls).



The element D_1 in the transmitter is a ***light-emitting diode (LED).*** An LED is an element that emits light and whose brightness is controlled by the current flowing through it. You can recall controlling the light emitted by an LED in various labs during the course. In our circuit, the current through the LED, hence its brightness, can be controlled by choosing the applied voltage V_{in} and the value of the resistor R.

The light from the element D_1 in the transmitter is received by the element D_2 in the receiver. In this circuit, the LED D_1 is used as a means for **transmitting information with light**, and the element D_2 is used as **a light receiver to see if anything was transmitted**.

In the receiver, the element labeled as D_2 behaves like a ***"reverse biased" solar cell***, which means that when it receives light, it generates a current. We will denote the current generated by the solar cell by I_S.

Remark: *Non-idealities, such as background light, affect the performance of a system that does light measurements. In this question, we will assume ideal conditions, that is, there is no source of light around except for the LED.*

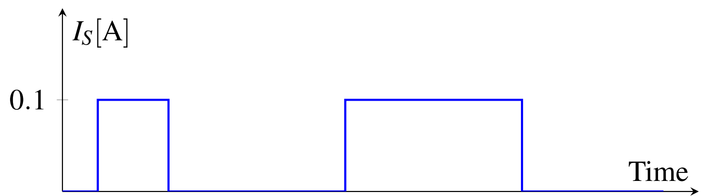
In our system, we define two states for the transmitter: (i) the transmitter is sending something when **they turn on the LED** and (ii) the transmitter is not sending anything when **they turn off the LED**. On the receiver side, the goal is to **convert the current I_S generated by the solar cell into a voltage and amplify it**, so that we can read the output voltage V_{out} to see if the transmitter was sending something or not. The circuit implements this operation through a series of op-amps and a comparator. *It might look look complicated at first glance, but we can analyze it one section at a time.*

(a) Currents i_1, i_2 and i_3 are labeled on the diagram. Assuming *op-amps are ideal* and the Golden Rules hold, is $I_S = i_1$? $i_1 = i_2$? $i_2 = i_3$? Treat the solar cell as an ideal current source supplying I_s.

(b) Use the Golden Rules to find V_0, V_1, V_2 and V_3 in terms of I_S, R_1, R_2 and R_3.

Hint: Solve for V_0 first, then use V_0 to find V_1. Afterwards, find V_2 and V_3 in a similar manner.

(c) Now, assume that the transmitter has chosen the values of V_{in} and R to control the intensity of light emitted by LED, such that when the **transmitter is sending something**, I_S is equal to 0.1 A and when the **transmitter is not sending anything**, I_S is equal to 0 A. The following figure shows a visual example of how this current I_S might look like as time changes.



For the receiver, suppose $V_{ref} = 2\,V$, $R_1 = 10\,\Omega$, $R_2 = 1000\,\Omega$, and the supply voltages of the comparator are $V_{DD} = 5\,V$ and $V_{SS} = -5\,V$. Pick a value of R_3 such that **V_{out} is V_{DD} when the transmitter is sending something (i.e. $I_S = 0.1\,A$) and V_{SS} when the transmitter is not sending anything (i.e. $I_S = 0$).**

7. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

EECS 16A Designing Information Devices and Systems I
Homework 12

Submission Format
Your homework submission should consist of **one** file.

- `hw12.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned) as well as your IPython notebook saved as a PDF.

  If you do not attach a PDF "printout" of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

Submit the file to the appropriate assignment on Gradescope.

1. Reading Assignment

For this homework, please read Notes 21, 22, and 23 to learn about inner products, norms, trilateration, correlation, and least squares. It may also be helpful to recap Note 19 about op-amps, negative feedback, and common amplifiers designs. You are always encouraged to read beyond this as well.

(a) What does it mean for two vectors $\vec{x}$ and $\vec{y}$ to be orthogonal, in terms of their inner product?

(b) In trilateration, the distances between the beacons and the unknown location $\vec{x}$ involve quadratic terms of $\vec{x}$. What trick can we use to get a system of linear equations in $\vec{x}$?

(c) Suppose the signal $x[n]$ is only defined for timesteps $0, 1, \cdots, 5$. For the purpose of computing linear cross-correlation, what value of $x[n]$ do we assume when n is a timestep out of the range: $0 \leq n \leq 5$ (e.g. $n = 6$ or $n = -1$)?

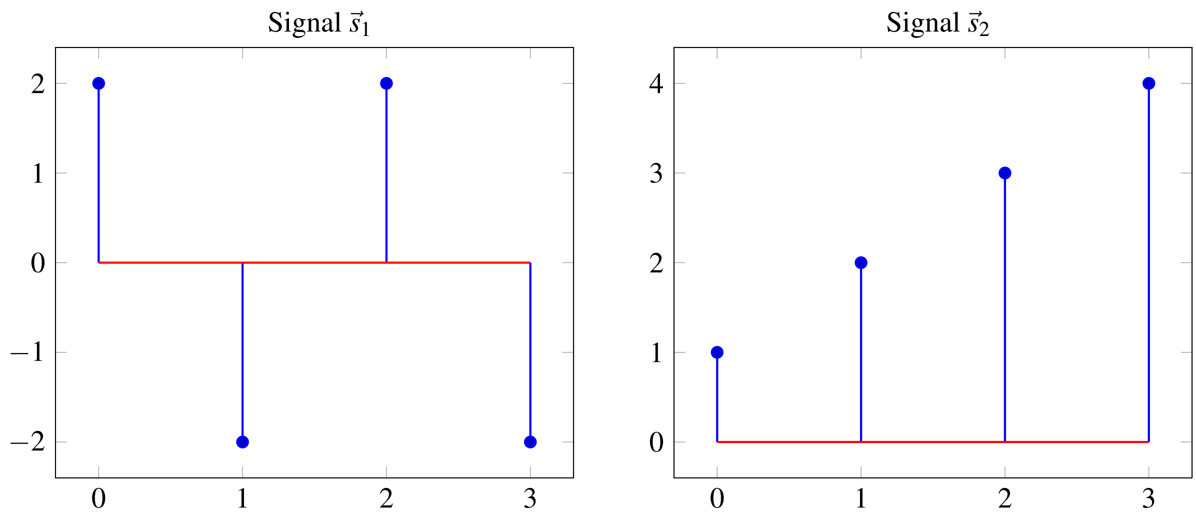
2. Mechanical Linear Correlation

(Contributors: Michael Kellman, Avi Pandey, Linda Liu, Moses Won, Amanda Jackson, Gireeja Ranade, Lam Nguyen, Laura Brink, Sang Min Han, Spencer Kent, Urmita Sikder, Vijay Govindarajan, Raghav Anand)

Learning Goal: *The objective of this problem is to understand how to compute the linear correlation between signals.*

We recall that the linear correlation of signal $\vec{y}$ with signal $\vec{x}$ is given as:

$$\text{corr}_{\vec{x}}(\vec{y})[k] = \sum_{n=-\infty}^{\infty} x[n]y[n-k]$$



Assume that both of the above signals extend to $\pm\infty$, and are 0 everywhere outside of the region shown in the above graphs. First, we will demonstrate the procedure for linear correlation by computing the linear correlation between signal $\vec{s}_1$ with itself (*i.e.* $\text{corr}_{\vec{s}_1}(\vec{s}_1)[k]$). This is referred to as the linear autocorrelation. This can be computed by evaluating the inner product between the signal and the shifted version of the signal (outlined in the below tables). Here, we compute this quantity for shifts between -3 and 3. For all shifts outside this range, the inner product is zero. Finally, we plot the non-zero values of the linear autocorrelation.

$\vec{s}_1[n]$	0	0	0	2	-2	2	-2	0	0	0	
$\vec{s}_1[n+3]$	2	-2	2	-2	0	0	0	0	0	0	
$\langle \vec{s}_1[n], \vec{s}_1[n+3] \rangle$	0 +	0 +	0 +	-4 +	0 +	0 +	0 +	0 +	0 +	0	= -4

$\vec{s}_1[n]$	0	0	0	2	-2	2	-2	0	0	0	
$\vec{s}_1[n+2]$	0	2	-2	2	-2	0	0	0	0	0	
$\langle \vec{s}_1[n], \vec{s}_1[n+2] \rangle$	0 +	0 +	0 +	4 +	4 +	0 +	0 +	0 +	0 +	0	= 8

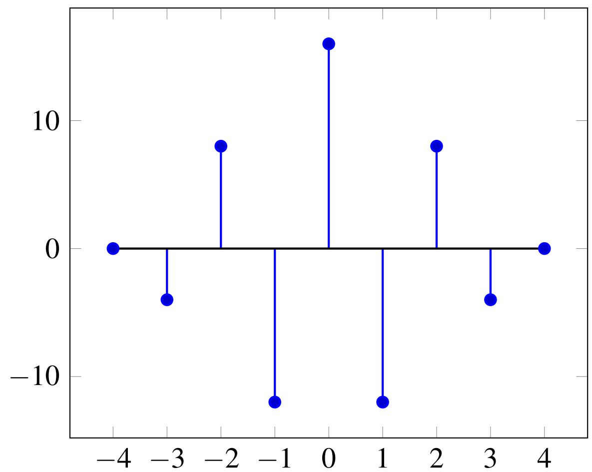
$\vec{s}_1[n]$	0	0	0	2	-2	2	-2	0	0	0	
$\vec{s}_1[n+1]$	0	0	2	-2	2	-2	0	0	0	0	
$\langle \vec{s}_1[n], \vec{s}_1[n+1] \rangle$	0 +	0 +	0 +	-4 +	-4 +	-4 +	0 +	0 +	0 +	0	= -12

$\vec{s}_1[n]$	0	0	0	2	-2	2	-2	0	0	0	
$\vec{s}_1[n+0]$	0	0	0	2	-2	2	-2	0	0	0	
$\langle \vec{s}_1[n], \vec{s}_1[n+0] \rangle$	0 +	0 +	0 +	4 +	4 +	4 +	4 +	0 +	0 +	0	= 16

$\vec{s}_1[n]$	0	0	0	2	-2	2	-2	0	0	0	
$\vec{s}_1[n-1]$	0	0	0	0	2	-2	2	-2	0	0	
$\langle \vec{s}_1[n], \vec{s}_1[n-1] \rangle$	0 +	0 +	0 +	0 +	-4 +	-4 +	-4 +	0 +	0 +	0	= -12

$\vec{s}_1[n]$	0	0	0	2	-2	2	-2	0	0	0	
$\vec{s}_1[n-2]$	0	0	0	0	0	2	-2	2	-2	0	
$\langle \vec{s}_1[n], \vec{s}_1[n-2] \rangle$	0 +	0 +	0 +	0 +	0 +	4 +	4 +	0 +	0 +	0	= 8

$\vec{s}_1[n]$	0		0		0		2		-2		2		-2		0		0		0	
$\vec{s}_1[n-3]$	0		0		0		0		0		0		2		-2		2		-2	
$\langle \vec{s}_1[n], \vec{s}_1[n-3] \rangle$	0	+	0	+	0	+	0	+	0	+	0	+	-4	+	0	+	0	+	0	= -4



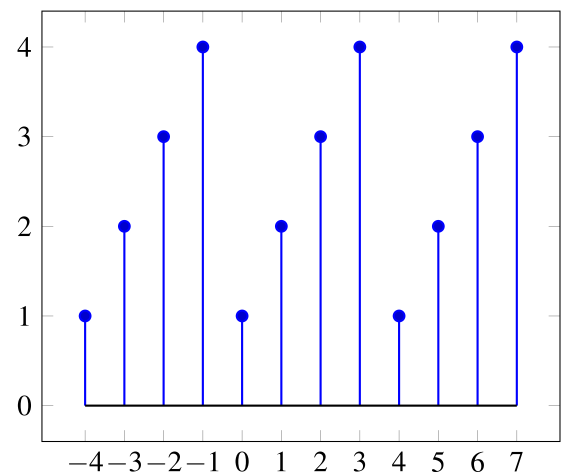
(a) Using the procedure demonstrated above, compute $\text{corr}_{\vec{s}_1}(\vec{s}_2)[k]$, the linear cross-correlation of $\vec{s}_2$ with $\vec{s}_1$. Like the example, use tables like the one given below for $k = -3$ and plot the resulting correlation.

$\vec{s}_1[n]$	0	0	0	2	-2	2	-2	0	0	0
$\vec{s}_2[n+3]$	1	2	3	4	0	0	0	0	0	0
$\langle \vec{s}_1[n], \vec{s}_2[n+3] \rangle$										

(b) Will the linear cross-correlation of $\vec{s}_2$ with $\vec{s}_1$ ($\text{corr}_{\vec{s}_1}(\vec{s}_2)[k]$) be the same as the cross-correlation of $\vec{s}_1$ with $\vec{s}_2$ ($\text{corr}_{\vec{s}_2}(\vec{s}_1)[k]$)? You can use the iPython notebook **prob12.ipynb** to figure this out. How are they related to each other?

Now, we will review the procedure to perform linear cross-correlation between one signal that is periodic with a period of 4 and another that is finite length and extended with zeros as in the previous parts. As an example, we will compute the linear correlation $\text{corr}_{\vec{p}_2}(\vec{s}_1)[k]$ between the periodic signal $\vec{p}_2$ (with period 4), formed by repeating $\vec{s}_2$, and the finite length signal $\vec{s}_1$ extended with zeros. The result will be a periodic signal with period 4.

The periodic signal, $\vec{p}_2$, formed by repeating $\vec{s}_2$ is plotted below for indices -4 to 7. It is defined and non-zero for all indices from $-\infty$ to $+\infty$.



We compute one period of the result of the cross-correlation by starting at a shift of $k=-1$ and ending at a shift of $k=2$.

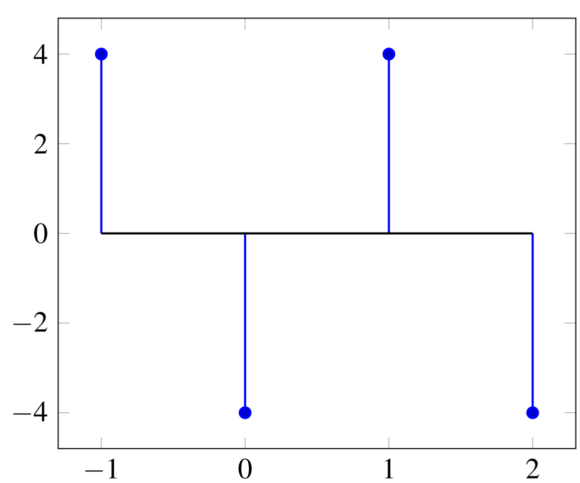
$\vec{p}_2[n]$	2	3	4	1	2	3	4	1	2	3	
$\vec{s}_1[n+1]$	0	0	2	-2	2	-2	0	0	0	0	
$\langle \vec{p}_2[n], \vec{s}_1[n+1] \rangle$	0 +	0 +	8 +	-2 +	4 +	-6 +	0 +	0 +	0 +	0	= 4

$\vec{p}_2[n]$	2	3	4	1	2	3	4	1	2	3	
$\vec{s}_1[n+0]$	0	0	0	2	-2	2	-2	0	0	0	
$\langle \vec{p}_2[n], \vec{s}_1[n+0] \rangle$	0 +	0 +	0 +	2 +	-4 +	6 +	-8 +	0 +	0 +	0	= -4

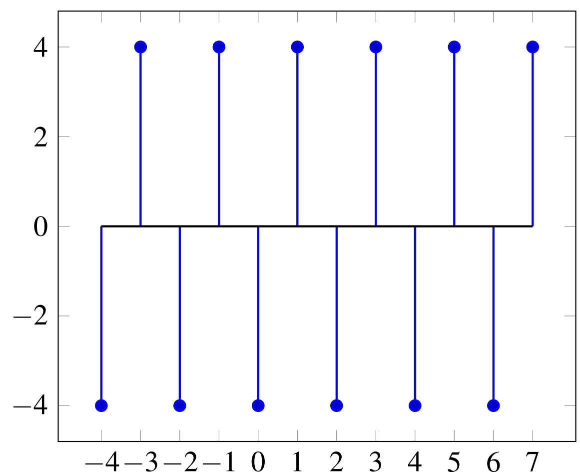
$\vec{p}_2[n]$	2	3	4	1	2	3	4	1	2	3	
$\vec{s}_1[n-1]$	0	0	0	0	2	-2	2	-2	0	0	
$\langle \vec{p}_2[n], \vec{s}_1[n-1] \rangle$	0 +	0 +	0 +	0 +	4 +	-6 +	8 +	-2 +	0 +	0	= 4

$\vec{p}_2[n]$	2	3	4	1	2	3	4	1	2	3	
$\vec{s}_1[n-2]$	0	0	0	0	0	2	-2	2	-2	0	
$\langle \vec{p}_2[n], \vec{s}_1[n-2] \rangle$	0 +	0 +	0 +	0 +	0 +	6 +	-8 +	2 +	-4 +	0	= -4

The computed single period of the resulting linear cross correlation is plotted below.



The resulting linear cross correlation for shifts from $k=-4$ to $k=7$ is plotted below.



(c) Repeat the procedure described above to compute the correlation $\text{corr}_{\vec{p}_1}(\vec{s}_1)[k]$ between a periodic signal $\vec{p}_1$ (with period 4), formed by repeating $\vec{s}_1$, and the finite-length signal $\vec{s}_1$ extended with zeros. Like the example, evaluate tables like the one below for $k = -3$ for different shifts and plot a single period of the result.

$\vec{p}_1[n]$	-2	2	-2	2	-2	2	-2	2	-2	2
$\vec{s}_1[n+3]$	2	-2	2	-2	0	0	0	0	0	0
$\langle\vec{p}_1[n], \vec{s}_1[n+3]\rangle$										

3. Cauchy-Schwarz Inequality

(Contributors: Amanda Jackson, Avi Pandey, Gireeja Ranade, Michael Kellman, Panos Zarkos, Richard Liou, Vijay Govindarajan, Titan Yuan)

Learning Goal: *The objective of this problem is to understand and prove the Cauchy-Schwarz inequality for real-valued vectors.*

The Cauchy-Schwarz inequality states that for two vectors $\vec{v}, \vec{w} \in \mathbb{R}^n$:

$$|\langle\vec{v}, \vec{w}\rangle| = |\vec{v}^T\vec{w}| \leq \|\vec{v}\| \cdot \|\vec{w}\|$$

In this problem we will prove the Cauchy-Schwarz inequality for vectors in $\mathbb{R}^2$.

Take two vectors: $\vec{v} = r\begin{bmatrix}\cos\theta \\ \sin\theta\end{bmatrix}$ and $\vec{w} = t\begin{bmatrix}\cos\phi \\ \sin\phi\end{bmatrix}$, where $r > 0$, $t > 0$, θ, and ϕ are scalars. Make sure you understand why any vector in $\mathbb{R}^2$ can be expressed this way and why it is acceptable to restrict $r, t > 0$.

You may find it helpful to refer to trigonometric identities in your analysis.

(a) In terms of some or all of the variables r, t, θ, and ϕ, what are $\|\vec{v}\|$ and $\|\vec{w}\|$?

(b) In terms of some or all of the variables r, t, θ, and ϕ, what is $\langle\vec{v}, \vec{w}\rangle$?

(c) Show that the Cauchy-Schwarz inequality holds for any two vectors in $\mathbb{R}^2$. *Hint: consider your results from part (b). Also recall* $-1 \leq \cos x \leq 1$ *and use both inequalities.*

(d) Note that the inequality states that the inner product of two vectors must be less than *or equal to* the product of their magnitudes. What conditions must the vectors satisfy for the equality to hold? In other words, when is $\langle\vec{v}, \vec{w}\rangle = \|\vec{v}\| \cdot \|\vec{w}\|$?

4. GPS Receivers

(Contributors: Ava Tan, Aviral Pandey, Craig Schindler, Lam Nguyen, Michael Kellman, Moses Won, Terry Chern, Titan Yuan, Urmita Sikder, Vijay Govindarajan, Vasuki Narasimha Swamy)

Learning Goal: *This problem will help to understand how GPS satellites transmit encoded signals to GPS receivers and how a receiver decodes the received signals and calculates the distance to the satellites using the signal propagation delays. It also shows how the GPS system is designed to be immune to noise.*

The Global Positioning System (GPS) is a space-based satellite navigation system that provides location and time information in all weather conditions, anywhere on or near the Earth where there is an unobstructed line of sight to four or more GPS satellites. In this problem, we will understand how a receiver (e.g. your cellphone) can disambiguate signals from the different GPS satellites that are simultaneously received.

GPS satellites employ "spread-spectrum" technology (very similar to Code-division multiple access, CDMA, which is commonly used in cellphone transmissions) and a special coding scheme where each transmitter is assigned a code that serves as its "signature".

Each GPS satellite uses a unique 1023 element long sequence as its "signature." These codes used by the satellites are called "Gold codes," and they have some special properties:

- The auto-correlation of a Gold code (cross-correlation with itself) is very **high** at the 0^{th} shift and very **low** at all other shifts.
- The cross-correlation between different Gold codes is very **low** at all shifts, i.e. different Gold codes are almost orthogonal to each other.

Gold codes are generated using a linear feedback shift register (LFSR). Understanding how this works is out of scope for the class, but you can read more about LFSR and CDMA if you are interested.

The important thing to know is that the Gold codes are 1023 element vectors where each element is either $+1$ or -1, and that any Gold code is "almost orthogonal" to any other Gold code.

A receiver listening for signature transmissions from a satellite has copies of all of the different GPS satellites' Gold codes. The receiver can determine how long it took for a particular GPS satellite's signal to reach it by **taking the cross-correlation of the received signal with a satellite's Gold code. (The Gold code is the signal which is shifted during cross-correlation — as discussed in lecture.)** The shift value (delay) that corresponds to distinct peaks (positive/negative) in the correlation determines the "propagation delay" between when the GPS satellite transmitted it's signal and when the receiver received it. This time delay can then be converted into a distance (in the case of GPS, electromagnetic waves are used for transmissions, distance is equal to the speed of light multiplied by the time delay).

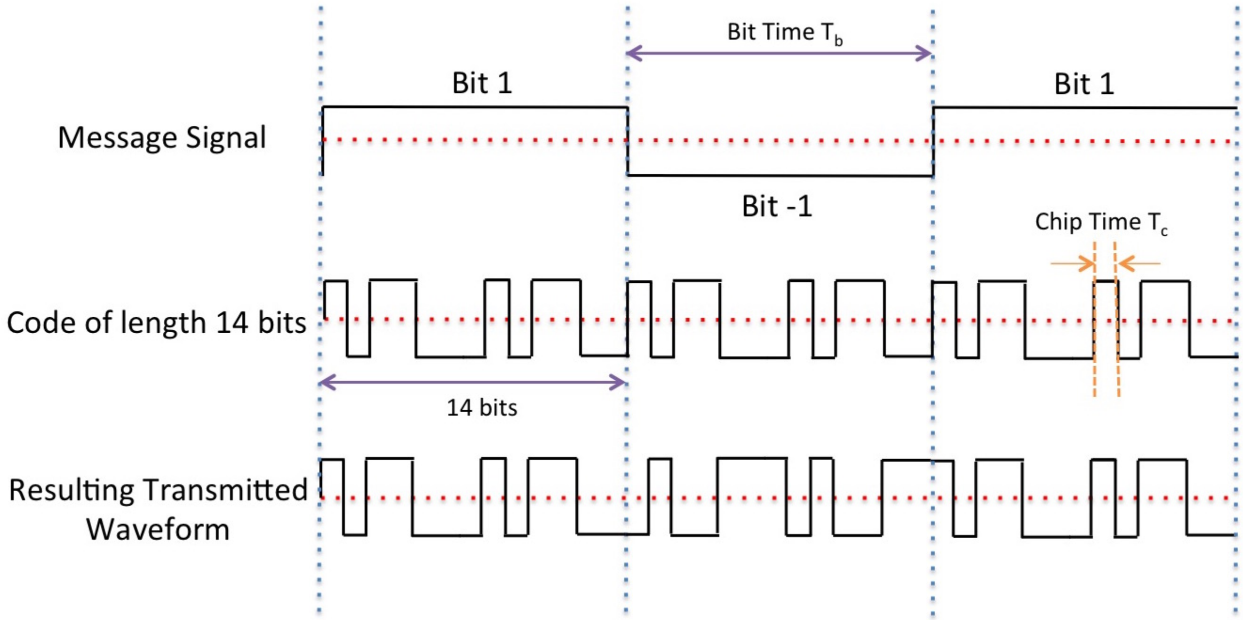
The GPS satellite is constantly transmitting its signature. In addition to identifying itself through its signature, it can also **"modulate" the signature** to communicate more information. Modulating a signature means multiplying the entire signature block by $+1$ or -1, as shown in the figure.

In the figure below, the signature is of length 14, i.e. it is made of 14 ± 1 symbols. T_c is the duration of one ± 1 symbol, and T_b is the duration of a whole signature. The figure shows 3 blocks of length 14 being transmitted. The message signal (made of $+1$ and -1 as well) multiplies the entire block of the signature, to give the resulting transformed waveform at the bottom of the figure. The message being transmitted in the figure is $\begin{bmatrix}1 & -1 & 1\end{bmatrix}$. So to send these three symbols of message, we need to send 14×3 symbols of the gold code.

The waveform that is actually transmitted is the multiplication of the message signal with the signature signal. Now, when a receiver receives a signal, in addition to finding the time delay between transmission and reception, the receiver will be able to decode the message by noting a very high correlation if the message bit is equal to 1, and a very negative correlation if the message bit is equal to -1.

For the problem you will now do, $T_b = 1023T_c$. (In reality, transmissions can be much faster and have $T_b = 20 \times 1023 \times T_c$.)

You will use the ideas of linear correlation to figure out which of the satellites are transmitting.



For the purpose of this question we only consider 24 GPS satellites. Download the IPython notebook and the corresponding data files for the following questions:

Note: this code is calculation-heavy, and can take up to a few mins to run for each code block. Be patient!

The iPython code required for part (a) is already given for you. You only need to write codes for parts (b)-(g).

(a) Auto-correlate (i.e. cross-correlate with itself) the Gold code of satellite 10 and plot it. What do you observe? Use the helper function `array_correlation` in the notebook to perform correlation for all sub-parts of this problem. Try to understand what the helper function `array_correlation` is doing.

(b) Cross-correlate the Gold code of satellite 10 with satellite 13 and plot it. What do you observe?

(c) Consider a random signal, i.e. a signal that is not generated due to a specific code but is a random ± 1 sequence. A helper function `integernoise_generator` in the notebook will generate this for you. Cross-correlate it with the Gold code of satellite 10. What do you observe? What does this mean about our ability to identify satellites in the presence of random ± 1 noise?

(d) The signal actually received by a receiver will be the satellites' transmissions plus additive noise, and this need not be just noise that takes values ± 1. Use the helper function `gaussiannoise_generator` in the notebook to generate a random noise sequence of length 1023, and compute the cross-correlation of this sequence with the Gold Code of satellite 10. What does this mean about our ability to identify satellites in the presence of real-valued noise?

For the next subparts of this problem, the received signals are corrupted by real-valued noise. Use the observation from this subpart for solving the rest of the question.

(e) The receiver may receive signals from multiple satellites simultaneously, in which case the signals will all be added together. In addition, noise might be added to the signal. What are the satellites present in the received signal from `data1.npy`?

Use helper function `find_peak` in your code to help you figure out whether peak correlation values of magnitude greater than a pre-specified threshold value is present. Note that the threshold is 800 for this problem.

(f) Let's assume that you can hear only one satellite, Satellite X, at the location you are in (though this never happens in reality). Let's also assume that this satellite is transmitting an unknown sequence of $+1$ and -1 of length 5 (after encoding it with the 1023 bit Gold code corresponding to Satellite X).

First, find out from `data2.npy` which satellite is transmitting using the same procedure that you used in part (e).

Next, find the 5 element sequence of ±1's that is being transmitted. To do this, you can observe the cross-correlation of the received signal from `data2.npy` with the Gold code of Satellite X and then visually find the peaks (positive /negative) and use these to understand the message.

(g) **[OPTIONAL / NOT GRADED]**

Signals from different transmitters arrive at the receiver with different delays. We use these delays to figure out the distance between the satellite and receiver.

The signals from different satellites are superimposed on each other with different offsets at the start.

What satellites are you able to see in `data3.npy`? Assume that all satellites begin transmission at time 0. What are the delays of all the satellites that are present? Assume you are told that all the satellites have the same message signal given by $\begin{bmatrix}1 & 1 & -1 & -1 & -1\end{bmatrix}$.

5. Mechanical Trilateration

Trilateration is the problem of finding one's coordinates given distances from known location coordinates. For each of the following trilateration problems, you are given 3 positions and the corresponding distance from each position to your location. Find your location or possible locations. If a solution does not exist, state that it does not.

(a) $\vec{s}_1 = \begin{bmatrix}4\\5\end{bmatrix}$, $d_1 = 5$, $\vec{s}_2 = \begin{bmatrix}1\\-1\end{bmatrix}$, $d_2 = 2$, $\vec{s}_3 = \begin{bmatrix}-11\\6\end{bmatrix}$, $d_3 = 13$.

(b) $\vec{s}_1 = \begin{bmatrix}0\\0\end{bmatrix}$, $d_1 = 5\sqrt{2}$, $\vec{s}_2 = \begin{bmatrix}10\\0\end{bmatrix}$, $d_2 = 5\sqrt{2}$, $\vec{s}_3 = \begin{bmatrix}5\\0\end{bmatrix}$, $d_3 = 5$.

(c) $\vec{s}_1 = \begin{bmatrix}3\\4\end{bmatrix}$, $d_1 = 5$, $\vec{s}_2 = \begin{bmatrix}0\\-2\end{bmatrix}$, $d_2 = 2$, $\vec{s}_3 = \begin{bmatrix}-12\\5\end{bmatrix}$, $d_3 = 12$.

6. Mechanical Projections

(Contributors: Michael Kellman)

Learning Goal: *The objective of this problem is to practice calculating projection of a vector and the corresponding squared error.*

(a) Find the projection of $\vec{b} = \begin{bmatrix}3\\2\\-1\end{bmatrix}$ onto $\vec{a} = \begin{bmatrix}1\\0\\1\end{bmatrix}$. What is the squared error between the projection and $\vec{b}$, i.e. $\|e\|^2 = \|\text{proj}_{\vec{a}}(\vec{b}) - \vec{b}\|^2$?

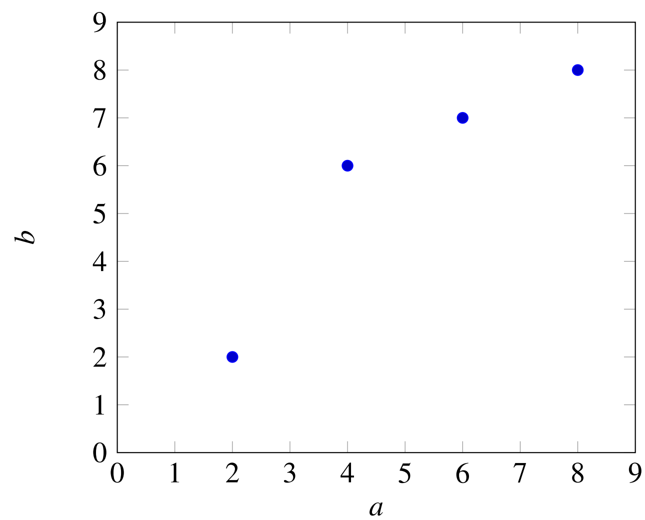
(b) Find the projection of $\vec{b} = \begin{bmatrix} 1 \\ 4 \\ -5 \end{bmatrix}$ onto the subspace spanned by the vectors $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$. What is the projection's squared error with $\vec{b}$, i.e. $\|e\|^2 = \|\text{proj}_{\vec{a}}(\vec{b}) - \vec{b}\|^2$?

7. Mechanical Least Squares

(Contributors: Adhyyan Narang, Aviral Pandey, Jiazhen Chen, Michael Kellman, Wahid Rahman)

Learning Goal:

The goal of this problem is to use least squares to fit different models (i.e. equations) to a data set. Depending on the model's number of parameters, the model will fit the data better or worse. A better model results in a lower squared error than a worse one. In part (a), we consider a linear model that contains a single slope parameter and intercepts the vertical axis at zero. In part (b), we consider a linear model with a possibly non-zero vertical axis intercept parameter, also known as an affine model.



a	2	4	6	8
b	2	6	7	8

(a) Consider the above data points. Find the linear model of the form

$$\vec{a}x = \vec{b}$$

that best fits the data, where x is a scalar that minimizes the squared error

$$\|\vec{e}\|^2 = \left\| \begin{bmatrix} a_1 \\ \vdots \\ a_4 \end{bmatrix} x - \begin{bmatrix} b_1 \\ \vdots \\ b_4 \end{bmatrix} \right\|^2 = \|\vec{a}x - \vec{b}\|^2. \tag{1}$$

Note: By using this linear model, we are implicitly forcing the line to go through the origin.

You may use a calculator but show your work. Do not directly plug your numbers into IPython. Note that we can model this linear model as a generic system $\mathbf{A}\vec{x} = \vec{b}$ where $\mathbf{A} = [\vec{a}]$ and $\vec{x} = x$. Once you've computed the optimal solution $\hat{\vec{x}}$, compute the squared error between your model's prediction and the actual b values as shown in Equation 1. **Optional but recommended:** Plot the best fit line along with the data points to examine the quality of the fit. You may plot however you wish - one option is to use the helper code in the IPython notebook provided.

(b) Now, let us consider an affine model for the same data, i.e. one with a non-zero vertical b-intercept. We believe we can get a better fit for the data by assuming an affine model of the form

$$ax_1 + x_2 = b,$$

for each point. Note that x_1 is the slope and x_2 is the b-intercept here. We can write this equation jointly for all the points using the vector notation below:

$$\vec{a}x_1 + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} x_2 = \vec{b}.$$

Make sure you understand why a column vector of 1's is required above. In order to do this, we need to augment our $\mathbf{A}$ matrix from the previous part for the least squares calculation with a column of 1's (do you see why?), so that it has the form

$$\mathbf{A} = \begin{bmatrix} a_1 & 1 \\ \vdots & \vdots \\ a_4 & 1 \end{bmatrix}.$$

Set up a least squares problem to find the optimal x_1 and x_2 and compute the squared error between your model's prediction and the actual $\vec{b}$ values. Is it a better fit for the data? Provide a quantitative, numerical justification. **Optional:** Plot your affine model and examine qualitatively how close the best fit line is to the data points compared to part (a). You may plot however you wish - one option is to use the helper code in the IPython notebook provided.

(c) **Prove the following theorem.**
Let $\mathbf{A} \in \mathbb{R}^{m,n}$. If $\hat{\vec{x}}$ is the solution to the least squares problem

$$\min_{\vec{x}} \left\| \mathbf{A}\vec{x} - \vec{b} \right\|^2,$$

then the error vector $\mathbf{A}\hat{\vec{x}} - \vec{b}$ is orthogonal to the columns of $\mathbf{A}$, i.e. show that: $\mathbf{A}^T(\mathbf{A}\hat{\vec{x}} - \vec{b}) = \vec{0}$.
Note: Trying to show individually that $\langle \vec{a}_i, \mathbf{A}\hat{\vec{x}} - \vec{b} \rangle = \vec{a}_i^T(\mathbf{A}\hat{\vec{x}} - \vec{b}) = 0$ for $i = 0, \ldots, n$, where $\vec{a}_i$ is the ith column of $\mathbf{A}$ can be a bit tricky, however, stacking all of the $\vec{a}_i^T$ on top of each other and showing that $\mathbf{A}^T(\mathbf{A}\hat{\vec{x}} - \vec{b}) = \vec{0}$ is easier.
For this question, it is sufficient to prove that $\mathbf{A}^T(\mathbf{A}\hat{\vec{x}} - \vec{b}) = \vec{0}$.
Hint: Can you substitute $\hat{\vec{x}}$ using the least-squares formula?

8. Image Analysis

(Contributors: Amanda Jackson, Ava Tan, Aviral Pandey, Lydia Lee, Michael Kellman, Panagiotis Zarkos, Spencer Kent, Titan Yuan, Urmita Sikder)

Learning Goal: *This problem introduces a method of fitting a non-linear model through a set of measured data points using the least squares method.*

Applications in medical imaging often require an analysis of images based on the image's pixels. For instance, we might want to count the number of cells in a given biological sample. One way to do this is to take a picture of the cells and use the pixels to determine their locations and how many there are.

Automatic detection of shape is useful in image classification as well (e.g. consider a robot trying to find out autonomously where a mug is in its field of vision).

Let us focus back on the medical imaging scenario. You are interested in finding the exact position and shape of a cell in an image. You will do this by finding the **equation of the circle or ellipse** that bounds the cell relative to a given coordinate system in the image. Your collaborator uses edge detection techniques to find **a bunch of points that are approximately along the edge of the cell**. We assume that the origin of the coordinate system is in the center of the image with standard axes (x,y) and your collaborator gives you the following points that approximately bound the cell:

$(0.3,-0.69)$, $(0.5,0.87)$, $(0.9,-0.86)$, $(1,0.88)$, $(1.2,-0.82)$, $(1.5,0.64)$, $(1.8,0)$.

Recall that an equation of the form

$$a_1x^2+b_1xy+c_1y^2+d_1x+e_1y=1$$

can be used to represent an ellipse (if $b_1^2-4a_1c_1<0$), and an equation of the form

$$a_1(x^2+y^2)+d_1x+e_1y=1$$

is a circle if $d_1^2+e_1^2+4a_1>0$. Notice that the circle has fewer parameters.

(a) How can you find the equation of a *circle* that surrounds the cell by fitting the data points? First, provide a setup and formulate a minimization problem to do this, i.e. a least squares problem minimizing the squared error $\left\|\mathbf{A}\vec{v}-\vec{b}\right\|^2$, where you attempt to find the **unknown coefficients** a_1**,** d_1**, and** e_1 from your data points. Here your unknown vector $\vec{v}=\begin{bmatrix}a_1\\d_1\\e_1\end{bmatrix}$. *Hint: The quantities* (x^2+y^2)*, x, and y can be thought of as known values calculated from your data points.*

You do not need to simplify the numerical values for the matrix elements; just writing out the matrix with numerical expressions will suffice.

(b) How can you find the equation of an ellipse (instead of a circle) that surrounds the cell? Provide a setup and formulate a minimization problem similar to that in part (a). Now the unknown vector $\vec{v}$ will be different from the earlier parts.

You do not need to simplify the numerical values for the matrix elements; just writing out the numerical expressions will suffice.

(c) In the IPython notebook, write a short program that uses least-squares to fit a circle to the given points. A helper function `plot_circle` is provided. What is $\frac{\|\vec{e}\|}{N}$, where $\vec{e}=\mathbf{A}\vec{v}-\vec{b}$ and N is the number of data points? Plot your points and the best fit circle in IPython.

(d) In the IPython notebook, write a short program that uses least-squares to fit an ellipse to the given points. A helper function `plot_ellipse` is provided. What is $\frac{\|\vec{e}\|}{N}$, where $\vec{e}=\mathbf{A}\vec{v}-\vec{b}$ and N is the number of data points? Now the unknown vector $\vec{v}$ will be different from the earlier parts. Plot your points and the best fit ellipse in IPython. How does this error compare to the one in the previous subpart? Which technique is better, and why?

9. Study Group Survey

We believe working collaboratively is one of the keys to success in this course! We want to learn more about your study group experience over the past semester, and ask you to fill out the following feedback form to help us better understand what collaboration looked like for you in EECS16A:

https://forms.gle/pKXwv3wZfij64ui77
To complete this part of the HW, you should fill out the survey and attach a screenshot indicating completion. Nothing else is required.

10. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

EECS 16A Designing Information Devices and Systems I
Homework 13

Submission Format
Your homework submission should consist of **one** file.

- `hw13.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned) as well as your IPython notebook saved as a PDF.

  If you do not attach a PDF "printout" of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

Submit the file to the appropriate assignment on Gradescope.

1. Audio File Matching

(Contributors: Aviral Pandey, Michael Kellman, Moses Won, Spencer Kent, Titan Yuan, Urmita Sikder, Vijay Govindarajan)

Learning Goal: *This problem motivates the application of correlation for pattern matching applications such as Shazam.*

Many audio processing applications rely on representing audio files as vectors, referred to as audio *signals*. Every component of the vector determines the sound we hear at a given time. We can use inner products to determine if a particular audio clip is part of a longer song, similar to an application like *Shazam*.

Let us consider a very simplified model for an audio signal, $\vec{x}$. At each timestep k, the audio signal can be either $x[k] = -1$ or $x[k] = 1$.

(a) Say we want to compare two audio files of the same length N to decide how similar they are. First, consider two vectors that are exactly identical, namely $\vec{x}_1 = \begin{bmatrix}1 & 1 & \cdots & 1\end{bmatrix}^T$ and $\vec{x}_2 = \begin{bmatrix}1 & 1 & \cdots & 1\end{bmatrix}^T$. What is the inner product of these two vectors? What if $\vec{x}_1 = \begin{bmatrix}1 & 1 & \cdots & 1\end{bmatrix}^T$ but $\vec{x}_2$ oscillates between 1 and -1? Assume that N, the length of the two vectors, is an even number.
Use this to suggest a method for comparing the similarity between a generic pair of length-N vectors.

(b) Next, suppose we want to find a short audio clip in a longer one. We might want to do this for an application like *Shazam*, which is able to identify a song from a short clip. Consider the vector of length 8, $\vec{x} = \begin{bmatrix}-1 & 1 & 1 & -1 & 1 & 1 & -1 & 1\end{bmatrix}^T$.
We want to find the short segment $\vec{y} := \begin{bmatrix}y[0] & y[1] & y[2]\end{bmatrix}^T = \begin{bmatrix}1 & 1 & -1\end{bmatrix}^T$ in the longer vector. To do this, perform the linear cross correlation between these two finite length sequences and identify at what shift(s) the linear cross correlation is maximized. Apply the same technique to identify what shift(s) gives the best match for $\vec{y} = \begin{bmatrix}1 & 1 & 1\end{bmatrix}^T$.
(If you wish, you may use iPython to do this part of the question, but you do not have to.)

(c) Now suppose our audio vector is represented using integers beyond simply just 1 and -1. Find the short audio clip $\vec{y} = \begin{bmatrix}1 & 2 & 3\end{bmatrix}^T$ in the song given by $\vec{x} = \begin{bmatrix}1 & 2 & 3 & 1 & 2 & 2 & 3 & 10\end{bmatrix}^T$. Where do you expect to see the peak in the correlation of the two signals? Is the peak where you want it to be, i.e. does it pull out the clip of the song that you intended? Why?
(If you wish, you may use iPython to do this part of the question, but you do not have to.)

(d) Let us think about how to get around the issue in the previous part. We applied cross-correlation to compare segments of $\vec{x}$ of length 3 (which is the length of $\vec{y}$) with $\vec{y}$. Instead of directly taking the cross correlation, we want to normalize each inner product computed at each shift by the magnitudes of both segments, i.e. we want to consider $\frac{<\vec{x}_k,\vec{y}>}{\|\vec{x}_k\|\|\vec{y}\|}$, where $\vec{x}_k$ is the length 3 segment starting from the k-th index of $\vec{x}$. This is referred to as normalized cross correlation. Using this procedure, now which segment matches the short audio clip best?

(e) We can use this on a more 'realistic' audio signal – refer to the IPython notebook, where we use normalized cross-correlation on a real song. Run the cells to listen to the song we are searching through, and add a simple comparison function `vector_compare` to find where in the song the clip comes from. Running this may take a couple minutes on your machine, but note that this computation can be highly optimized and run super fast in the real world! Also note that this is not exactly how Shazam works, but it draws heavily on some of these basic ideas.

2. Constrained Least Squares Optimization

In this problem, we will guide you through solving the following optimization problem:

Consider a matrix $\mathbf{A} \in \mathbb{R}^{M \times N}$ where $M > N$ and all N columns are linearly independent. Determine a unit vector $\hat{\vec{x}}$ that minimizes $\|\mathbf{A}\vec{x}\|$, where $\|\cdot\|$ denotes the norm—that is,

$$\|\mathbf{A}\vec{x}\|^2 \triangleq \langle \mathbf{A}\vec{x}, \mathbf{A}\vec{x} \rangle = (\mathbf{A}\vec{x})^T \mathbf{A}\vec{x} = \vec{x}^T \mathbf{A}^T \mathbf{A}\vec{x}.$$

This is equivalent to solving the following optimization problem:

$$\min_{\vec{x}} \|\mathbf{A}\vec{x}\|^2 \quad \text{subject to the constraint} \quad \|\vec{x}\|^2 = 1.$$

This task may *seem* like solving a standard least squares problem $\mathbf{A}\vec{x} = \vec{b}$ where $\vec{b} = \vec{0}$, but it is different. As an example, notice $\vec{x} = \vec{0}$ is *not* a valid solution to our problem because the norm of the zero vector does not equal one. Our optimization problem is a least squares problem with a constraint—hence the term *constrained least squares optimization*. The constraint can be visualized as limiting the vector $\vec{x}$ to lie on a unit circle (radius of the circle is one) if $N = 2$ and on a unit sphere if $N = 3$.

Let $(\lambda_1, \vec{v}_1), \ldots, (\lambda_N, \vec{v}_N)$ denote the eigenpairs (i.e., eigenvalue/eigenvector pairs) of $\mathbf{A}^T\mathbf{A}$. Assume that the eigenvalues are all real, distinct and indexed in an descending fashion—that is,

$$\lambda_1 > \cdots > \lambda_N.$$

Assume, too, that each eigenvector has been normalized to have unit length—that is, $\|\vec{v}_k\| = 1$ for all $k \in \{1, \ldots, N\}$.

(a) Show that $\lambda_N > 0$, i.e. all the eigenvalues are strictly positive.
Hint: Consider $\|\mathbf{A}\vec{v}\|^2$, where $\vec{v}$ is an eigenvector of $\mathbf{A}^T\mathbf{A}$ with eigenvalue λ.

(b) Consider two eigenpairs $(\lambda_k, \vec{v}_k)$ and $(\lambda_\ell, \vec{v}_\ell)$ corresponding to distinct eigenvalues of $\mathbf{A}^T\mathbf{A}$—that is, $\lambda_k \neq \lambda_\ell$. Prove that the corresponding eigenvectors $\vec{v}_k$ and $\vec{v}_\ell$ are orthogonal: $\vec{v}_k \perp \vec{v}_\ell$.
To help you get started, consider the two equations

$$\mathbf{A}^T\mathbf{A}\,\vec{v}_k = \lambda_k\,\vec{v}_k \tag{1}$$

and

$$\vec{v}_\ell^T\,\mathbf{A}^T\mathbf{A} = \lambda_\ell\,\vec{v}_\ell^T. \tag{2}$$

The second equation can be derived by taking the transpose of both sides in the eigenvalue equation $\mathbf{A}^T\mathbf{A}\vec{v}_\ell = \lambda_\ell \vec{v}_\ell$. Premultiply Equation 1 with $\vec{v}_\ell^T$, postmultiply Equation 2 with $\vec{v}_k$, compare the two, and explain how one may then infer that $\vec{v}_k$ and $\vec{v}_\ell$ are orthogonal, i.e. $\langle \vec{v}_k, \vec{v}_\ell \rangle = 0$.

Premultiplication by a vector $\vec{u}$ means multiplying an expression by $\vec{u}$ on the left. For example, premultiplying the matrix $\mathbf{A}$ by $\vec{u}$ gives $\vec{u}\mathbf{A}$. Postmultiplication means multiplying on the right. Remember that in general matrix-vector multiplication is not commutative, so those operations are not identical.

(c) The results of part (b) imply that the N eigenvectors of $\mathbf{A}^T\mathbf{A}$ are mutually orthogonal. A basis formed by vectors that are both (1) mutually orthogonal and (2) have unit length is called an orthonormal basis. Since the eigenvalues of $\mathbf{A}^T\mathbf{A}$ are distinct, and the corresponding eigenvectors are scaled to have a norm of one, the scaled eigenvectors form an orthonormal basis in $\mathbb{R}^N$. This means that we can express an arbitrary vector $\vec{x} \in \mathbb{R}^N$ as a linear combination of the eigenvectors $\vec{v}_1, \ldots, \vec{v}_N$, as follows:

$$\vec{x} = \sum_{n=1}^{N} \alpha_n \vec{v}_n.$$

i. Determine the n^{th} coefficient α_n in terms of $\vec{x}$ and one or more of the eigenvectors $\vec{v}_1, \ldots, \vec{v}_N$.

ii. Suppose $\vec{x}$ is a unit-length vector (i.e., a unit vector) in $\mathbb{R}^N$. Show that

$$\sum_{n=1}^{N} \alpha_n^2 = 1$$

where the α_n's are the coefficients of $\vec{x}$ in the basis defined earlier.
Hint: We know that $\|\vec{x}\|^2 = 1$. *Use the previous part to expand this expression.*

(d) Now express $\|\mathbf{A}\vec{x}\|^2$ in terms of $\{\alpha_1, \alpha_2 \ldots \alpha_N\}$, $\{\lambda_1, \lambda_2 \ldots \lambda_N\}$, and $\{\vec{v}_1, \vec{v}_2 \ldots \vec{v}_N\}$, and find an expression for $\vec{\hat{x}}$ such that $\|\mathbf{A}\vec{x}\|^2$ is minimized. Do *not* use any tool from calculus to solve this problem, so avoid differentiation.

Hint 1: Start with expressing $\mathbf{A}^T\mathbf{A}\vec{x}$ *in terms of* $\{\alpha_1, \alpha_2 \ldots \alpha_N\}$, $\{\lambda_1, \lambda_2 \ldots \lambda_N\}$, *and* $\{\vec{v}_1, \vec{v}_2 \ldots \vec{v}_N\}$, *then extend this to* $\|\mathbf{A}\vec{x}\|^2$.

Hint 2: After expressing $\|\mathbf{A}\vec{x}\|^2$ *in terms of* $\{\alpha_1, \alpha_2 \ldots \alpha_N\}$, $\{\lambda_1, \lambda_2 \ldots \lambda_N\}$, *and* $\{\vec{v}_1, \vec{v}_2 \ldots \vec{v}_N\}$, *which variable are you minimizing over?*

3. Noise Canceling Headphones

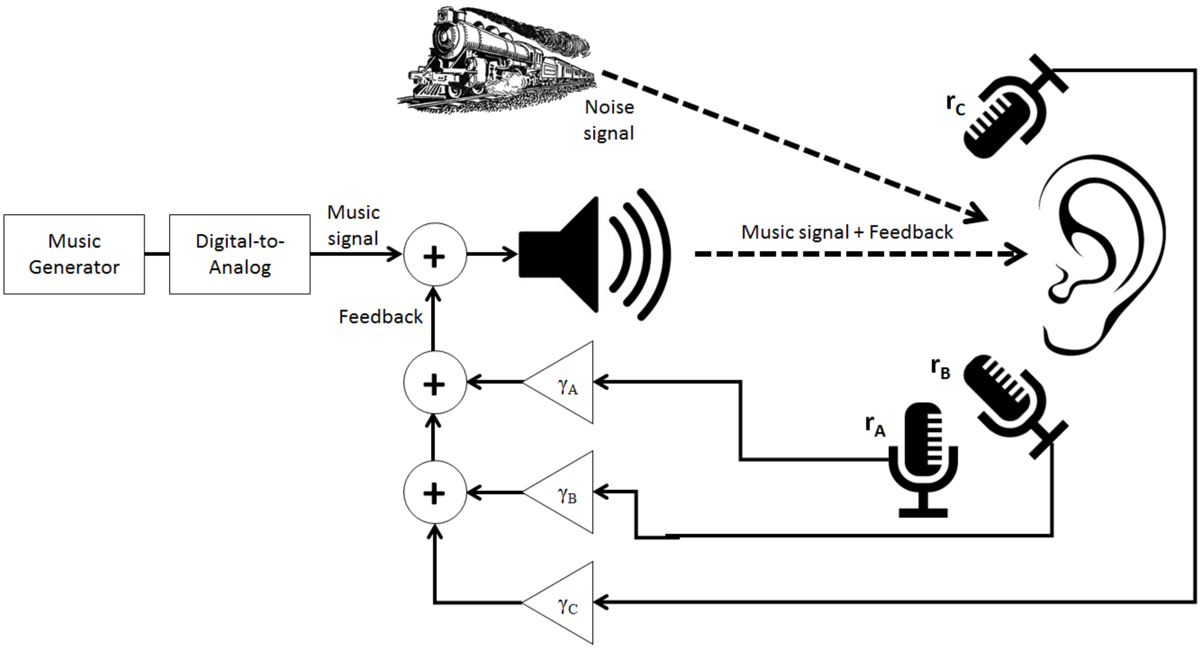
(Contributors: Aviral Pandey, Lydia Lee, Michael Kellman, Moses Won, Urmita Sikder, Wahid Rahman)

Learning Goal: *This problem will employ least squares method to minimize the effect of additive noise through headphones.*

In this problem, we will explore a common design for noise cancellation using noise-canceling headphones as an example application. We will work with the model shown in the figure below.

A music signal is generated at a speaker and transmitted to the listener's ear. If there is noise in the environment (*e.g.* other people's voices, a train going by), this noise signal will be superimposed (i.e. added) on the music signal and the listener will hear both. In order to cancel the noise, we will try **to record the noise and subtract it directly from the transmitted signal** with the hope that we can achieve perfect cancellation of everything but the music. Since our system is imperfect, we'll have to solve a **least squares problem**.

The gain blocks marked by γ **(Greek "gamma") represent scalar multiplication**, and we will assume that they can take on any real number, positive or negative.



Assume we have an **additive noise signal** noted by $\vec{n}$, which we want to cancel. This represents the extra noise coming from the train.

$$\vec{n} = \begin{bmatrix} n_1 \\ n_2 \\ n_3 \\ n_4 \\ n_5 \end{bmatrix}.$$

Let us assume the music signal is represented by

$$\vec{m} = \begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ m_5 \end{bmatrix}.$$

In absence of any noise cancellation, the user will hear:

$$\vec{s} = \vec{m} + \vec{n}.$$

However, we want to cancel this noise. For this we use three microphones to record this noise, Mic A, Mic B, and Mic C. However, they cannot perfectly record the noise $\vec{n}$ and have erroneous recordings. Let $\vec{r}_A$, $\vec{r}_B$, and $\vec{r}_C$ represent the **noise that each microphone picks up**:

$$\vec{r}_A = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix}, \vec{r}_B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix}, \vec{r}_C = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{bmatrix}.$$

We can arrange the recordings into a matrix $\mathbf{R}$ and the microphone gains, γ, into a vector $\vec{\gamma}$ to get:

$$\mathbf{R} = \begin{bmatrix} \vec{r}_A & \vec{r}_B & \vec{r}_C \end{bmatrix} = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \\ a_4 & b_4 & c_4 \\ a_5 & b_5 & c_5 \end{bmatrix}, \vec{\gamma} = \begin{bmatrix} \gamma_A \\ \gamma_B \\ \gamma_C \end{bmatrix}.$$

We want to choose a $\vec{\gamma}$ such that we minimize the impact of the noise.

The listener's ear will hear a total signal of:

$$\vec{s} = \vec{m} + \mathbf{R}\vec{\gamma} + \vec{n},$$

$$\begin{bmatrix} s_1 \\ s_2 \\ s_3 \\ s_4 \\ s_5 \end{bmatrix} = \begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ m_5 \end{bmatrix} + \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \\ a_4 & b_4 & c_4 \\ a_5 & b_5 & c_5 \end{bmatrix} \begin{bmatrix} \gamma_A \\ \gamma_B \\ \gamma_C \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \\ n_3 \\ n_4 \\ n_5 \end{bmatrix}.$$

This problem focuses on how to choose $\vec{\gamma}$ so as to have the signal $\vec{s}$ be as close to $\vec{m}$ as possible.

(a) Ideally, we would want to have a signal at the ear that matches the original music signal $\vec{m}$ perfectly. In reality, this is not possible, so we will aim to minimize the effect of the noise. What quantity would we need to minimize to make sure this happens? Write your answer in terms of the matrix $\mathbf{R}$, the vector of mic gains $\vec{\gamma}$, and the noise vector $\vec{n}$. *Hint: Your answer should be the norm of something.*

(b) We can solve the noise minimization problem by the least squares method. In effect, if we have a problem, $\min_{\vec{x}} \left\| \mathbf{A}\vec{x} - \vec{b} \right\|$, then the $\vec{x}$ that solves this problem is,

$$\vec{\hat{x}} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\vec{b}. \tag{3}$$

Implement this least squares solution in the IPython Notebook helper function `doLeastSquares`. Remember that $\min_{\vec{x}} \left\| \mathbf{A}\vec{x} + \vec{b} \right\| = \min_{\vec{x}} \left\| \mathbf{A}\vec{x} - (-\vec{b}) \right\|$.

(c) For the given $\vec{n}$ and the recordings, $\vec{r}_A$, $\vec{r}_B$, $\vec{r}_C$, below, find the γ's that minimize the effect of noise. Use the helper function `doLeastSquares` you made in the last part to solve this.

$$\vec{n} = \begin{bmatrix} 0.10 \\ 0.37 \\ -0.45 \\ 0.068 \\ 0.036 \end{bmatrix}, \vec{r}_A = \begin{bmatrix} 0 \\ 0.11 \\ -0.31 \\ -0.012 \\ -0.018 \end{bmatrix}, \vec{r}_B = \begin{bmatrix} 0 \\ 0.22 \\ -0.20 \\ 0.080 \\ 0.056 \end{bmatrix}, \vec{r}_C = \begin{bmatrix} 0 \\ 0.37 \\ -0.44 \\ 0.065 \\ 0.038 \end{bmatrix}.$$

The next few questions can be answered in the IPython notebook by running the associated cells.

(d) Follow the instructions in the IPython notebook to load a **music signal** and some **noise signals**. Listen to the music signal `music_y` and the two noise signals `noise1_y` and `noise2_y`. Which ones are full of static and which ones are not?

Also listen to the **noisy music signal** `noisyMusic`, generated by adding the music signal `music_y` and the first noise signal `noise1_y`. What do you hear?

(e) For this part we assume the noise signal $\vec{n}$ is given by `noise1_y` only. $\mathbf{R}$, i.e. the matrix of recorded noise from 3 microphones, is simulated with the `recordAmbientNoise` function. Calculate a vector $\vec{\gamma}$ that minimizes the effect of noise using `doLeastSquares`.

(f) Use you answers from the previous part to find the **noise-cancellation signal $\mathbf{R}\vec{\gamma}$**. Add this noise-cancellation signal ($\mathbf{R}\vec{\gamma}$), the music signal $\vec{m}$ (`music_y`), and the noise signal $\vec{n}$ (`noise1_y`), in order to generate a **noise-cancelled signal**. Play the noisy signal from part (e) and the noise-cancelled signal. Can you hear a difference?

(g) **[Optional]** Try adding the other noise signal `noise2_y` to the music signal to generate a new noisy signal. Don't re-calculate new values for $\vec{\gamma}$ i.e. don't solve the least squares problem again. Recalculate $\mathbf{R}$ and find the new noise-cancellation signal $\mathbf{R}\vec{\gamma}$ using $\vec{\gamma}$ from part (f). Add this noise-cancellation signal to your noisy signal. Comment on the quality of the resulting noise-cancelled signal. Is it perfect or are there artifacts?

Note: We want to find out how the $\vec{\gamma}$ we calculated work for any type of noise.

4. Course Evaluations [Optional]

Please take a moment to fill out course evaluations if you have not done so already! It really helps us improve the course for future semesters! Links to this should have already been sent to your email!